

SA-AI (Spalart–Allmaras with Autogenous Inception): Technical Summary

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1 Model

1.1 Transport equation and blended sources

$$\frac{D\tilde{\nu}}{Dt} = P - D + \frac{1}{\sigma} \nabla \cdot [(c_{\nu, \text{ai}} \nu + \tilde{\nu}) \nabla \tilde{\nu}] + \frac{c_{b2}}{\sigma} |\nabla \tilde{\nu}|^2, \quad (1)$$

$$P = \max[(1 - \sigma_P) a S \omega \tilde{\nu}, \sigma_P P_{\text{SA}}], \quad D = \sigma_D D_{\text{SA}}, \quad (2)$$

$\sigma_P \equiv 1$ (and with it $\sigma_D = 1$), $c_{\nu, \text{ai}} = 1$, and $b \equiv 0$ recover baseline SA exactly.

$$\begin{aligned} P_{\text{SA}} &= c_{b1} \tilde{S} \tilde{\nu}, & D_{\text{SA}} &= c_{w1} f_w \left(\frac{\tilde{\nu}}{d} \right)^2, & \tilde{S} &= \omega + \frac{\tilde{\nu}}{\kappa^2 d^2} f_{v2}, & f_{v2} &= 1 - \frac{\chi}{1 + \chi f_{v1}}, \\ f_{v1} &= \frac{\chi^3}{\chi^3 + c_{v1}^3}, & f_w &= g \left(\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6} \right)^{1/6}, & g &= r + c_{w2}(r^6 - r), & r &= \min\left(\frac{\tilde{\nu}}{\tilde{S} \kappa^2 d^2}, 10\right), \end{aligned} \quad (3)$$

The non-negative $\tilde{S}$ modification and the negative- $\tilde{\nu}$ safeguards of Allmaras et al. (2012) are retained, with one consistency requirement: the negative-branch diffusion factor f_n is formed with the scaled coefficient $c_{\nu, \text{ai}} \nu$, which keeps the total diffusivity positive, $c_{\nu, \text{ai}} \nu + f_n \tilde{\nu} > 0$.

$$\sigma_P = \sigma_t(\chi; \tau), \quad \sigma_t(\chi; \tau) = \max\left[1 - e^{-(\chi-1)/\tau}, 0\right], \quad \sigma_D = 1 - \frac{c_{b1}}{\kappa^2 c_{w1}} (1 - \sigma_P). \quad (4)$$

1.2 Local indicators

In a thin quasi-two-dimensional layer the profile is captured to second order about any wall-normal point by three velocity scales formed from the wall distance d : the velocity u , the shear du' , and the curvature $\frac{1}{2}d^2u''$. Normalized by their common magnitude $R = \sqrt{u^2 + (du')^2 + (\frac{1}{2}d^2u'')^2}$, they place the local state on the unit sphere of directions

$$(X, Y, Z) = (u, du', \frac{1}{2}d^2u'')/R, \quad (5)$$

$$\hat{\Omega} = \frac{Y}{\sqrt{X^2 + Y^2}} \quad (\text{vorticity fraction}), \quad \hat{I} = Y - X - Z \quad (\text{accumulated inflection}). \quad (6)$$

A near-wall parabola $u = By + Cy^2$ satisfies $\hat{I} = 0$ identically.

$$I(d) \equiv R \hat{I} = d u' - u - \frac{1}{2} d^2 u'' = - \int_0^d \frac{1}{2} s^2 u'''(s) ds, \quad (7)$$

Every profile leaves the wall with $\hat{I}$ third-order small ($u_w''' = 0$ on a steady wall). The vorticity Reynolds number $Re_\Omega = d^2 \omega / \nu$ is reserved for the onset gate. Fig. 1.

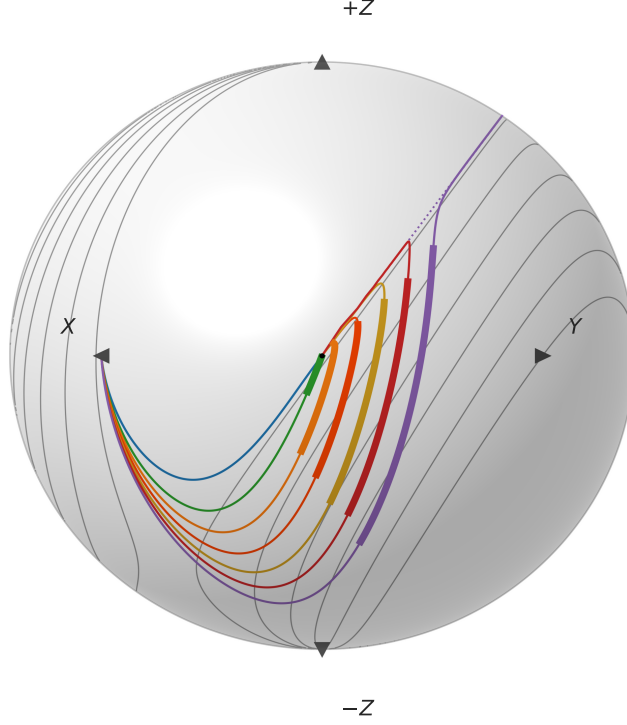


Figure 1: Indicator sphere. Falkner–Skan profiles $\beta = +0.30, 0, -0.10, -0.16, -0.19, -0.1988$; separated Stewartson $\beta = -0.19, H \approx 4.9$. Attached profiles at $Re_\theta = 500$, reversed-flow at $Re_\theta = 200$; $\hat{\Omega}\hat{I}$ contours $\{0.025, 0.2, 0.4, 0.6, 0.8, 1.0\}$.

In the solver the triple is assembled from four local vector-calculus objects: the velocity $\mathbf{u}$, the vorticity $\boldsymbol{\omega}$, the wall-distance vector $\mathbf{d} = d \nabla d$, and the curvature vector $\boldsymbol{\ell} = \frac{1}{2} \langle \mathbf{d}, \mathbf{d} \rangle \nabla^2 \mathbf{u}$, with $\nabla^2 \mathbf{u} = -\nabla \times \boldsymbol{\omega}$ supplying the second derivative.

$$(X, Y, Z) = \left(\langle \mathbf{u}, \mathbf{u} \rangle, \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle \langle \mathbf{d}, \mathbf{d} \rangle \langle \boldsymbol{\omega}, \boldsymbol{\omega} \rangle}, \langle \boldsymbol{\ell}, \mathbf{u} \rangle \right), \quad \hat{\Omega} = \frac{Y}{\sqrt{X^2 + Y^2}}, \quad \hat{I} = \frac{Y - X - Z}{\sqrt{X^2 + Y^2 + Z^2}}, \quad (8)$$

The solver’s equivalent ratio realization ($|\mathbf{u}|, d|\boldsymbol{\omega}|, \frac{1}{2}d^2(\nabla^2 \mathbf{u}) \cdot \hat{\mathbf{u}}$) coincides with it wherever $\mathbf{u} \neq 0$.

1.3 Amplification rate and onset threshold

The complete amplification source is

$$P_{\text{AI}} = a S \omega \tilde{\nu}, \quad a = a_{\text{max}} \text{clip} \langle \hat{\Omega} \hat{I} \rangle_0^1, \quad S = S(Re_\Omega / Re_\Omega^c(\hat{\Omega} \hat{I})), \quad S(\zeta) = \frac{1}{2} [1 + \tanh((\zeta - 1)/w)], \quad (9)$$

$$Re_\Omega^c(\hat{\Omega} \hat{I}) = k \text{softmin}_2 \left(C, A + B \langle \hat{\Omega} \hat{I} \rangle_+^{-2} \right), \quad \text{softmin}_2(x_1, x_2) = \frac{x_1 x_2}{\sqrt{x_1^2 + x_2^2}}, \quad \langle x \rangle_+ = \max(x, 0). \quad (10)$$

Fig. 2: graze ratios 0.93–1.14, root-mean-square $\sim 6\%$, from incipient separation through $\beta = +0.15$. The marched Blasius $N=9$ crossing lands at $Re_\theta = 1181$, 7% past Drela’s 1108; across the attached family the $N=1$ stations land on the Drela–Giles stations to 6% root-mean-square (adverse wedges within $\pm 4\%$, the favorable pair 11–12% early). By $\beta = +0.35$ the rate is $\approx 0.1 \times$ Drela and the onset $\approx 3 \times$ its critical Re_θ ; by $\beta \approx 0.45$ transition has retreated beyond any station of practical relevance, and stronger favorable wedges are suppressed outright (Figs. 3–4).

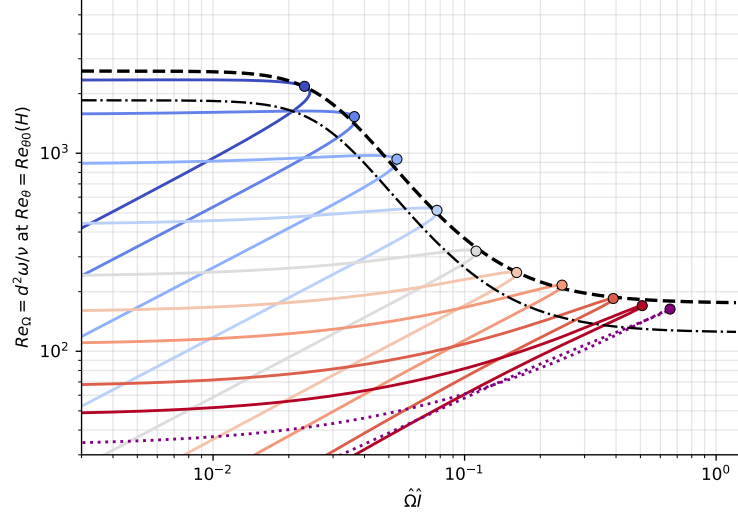


Figure 2: Onset-threshold graze. Attached family $\beta = +0.15, +0.10, +0.05, 0, -0.05, -0.10, -0.15, -0.19, -0.1988$, each at its Drela–Giles $Re_{\theta 0}(H)$; separated Stewartson $\beta = -0.19$ ($H \approx 4.9$, $Re_{\theta 0} \approx 26$). $\max_y \hat{\Omega} \hat{I} = 0.162/0.078/0.037$ at $\beta = -0.10/0/+0.10$. Envelope $k = 1$; model threshold $k = 0.712$.

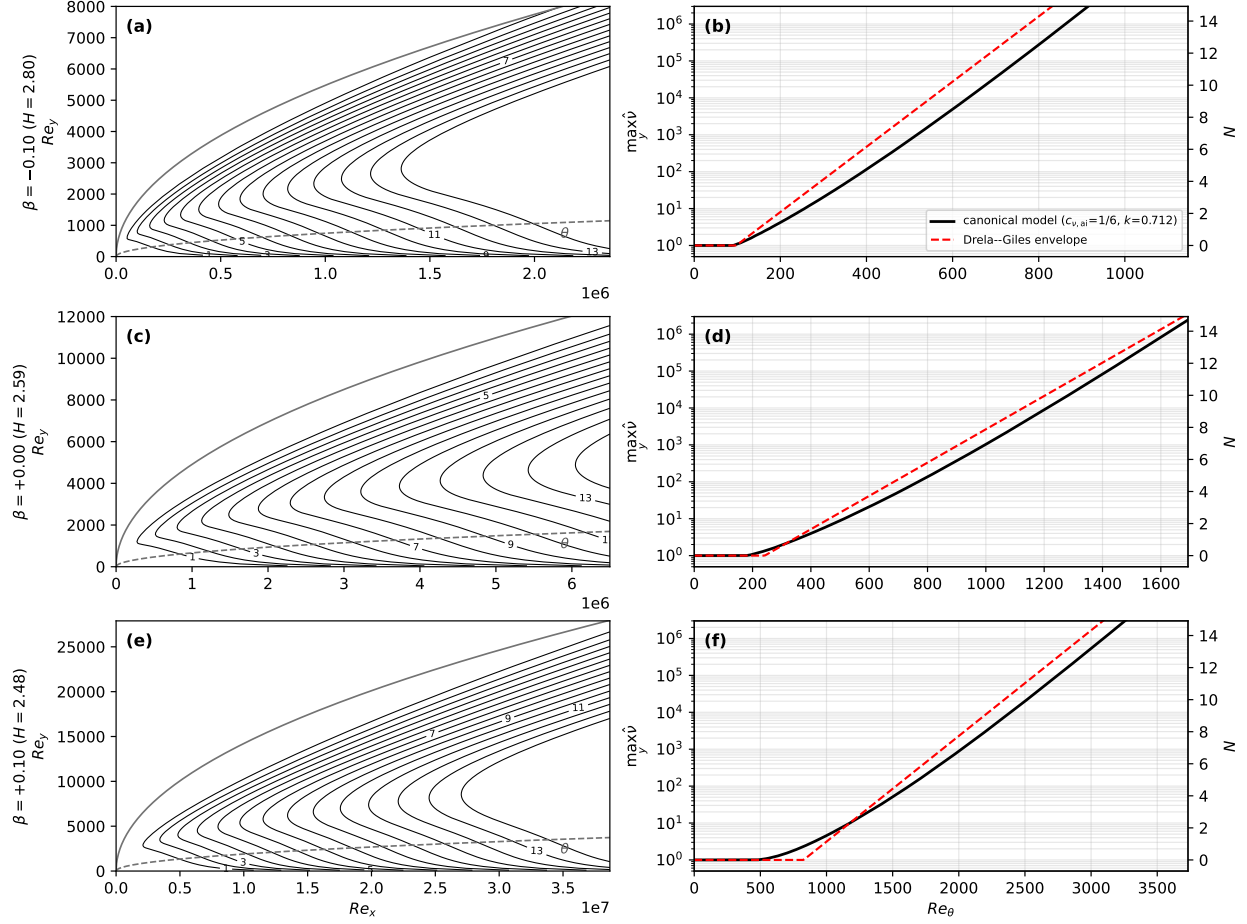


Figure 3: Disturbance transport, three wedges. Adverse $\beta = -0.10$ ($\max_y \hat{\Omega} \hat{I} = 0.162$), Blasius $\beta = 0$ (0.078), favorable $\beta = +0.10$ (0.037); seeded $\dot{\nu}_\infty = 1$, $a_{\max} = 0.19$, $(c_{\nu, ai}, k) = (\frac{1}{6}, 0.712)$. Anchor $N = 9$ crossing 7% past Drela; favorable $N = 1$ station $\sim 11\%$ early.

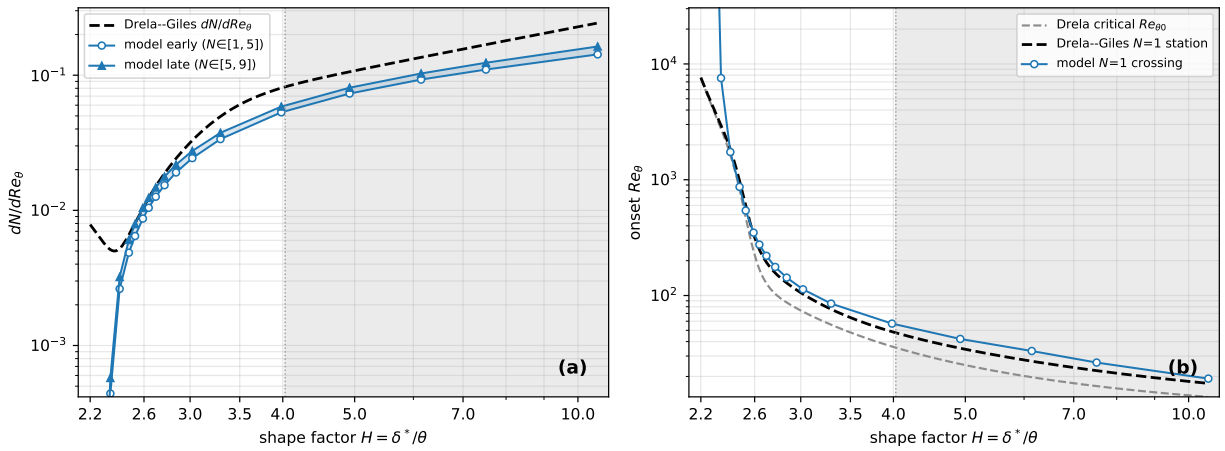


Figure 4: Calibrated model vs Drela-Giles. Stagnation $H \approx 2.2$, Blasius $H \approx 2.59$, separation limit $H = 3.98$, reversed-flow Stewartson $H \approx 10.6$; $a_{\max} = 0.19$, $c_{\nu, ai} = 1/6$. (b) onset tracks the Drela-Giles $N = 1$ stations within $\sim \pm 12\%$ from $H = 2.44$ ($\beta = +0.15$) to $H = 3.98$.

1.4 Reduced laminar diffusion

The molecular coefficient in the diffusion term of Eq. (1) only is scaled by $c_{\nu,\text{ai}} < 1$, leaving the mean-flow viscosity, the SA production and destruction, and $\chi = \tilde{\nu}/\nu$ (formed with the true ν) untouched. The frozen-profile growth eigenvalue on Blasius at $Re_\theta = 400$ delivers only 0.49 of the Drela–Giles rate, and the marched inception fails outright (the Blasius $N = 1$ crossing sits at $Re_\theta = 392$ even with the threshold removed entirely). Down the ladder $c_{\nu,\text{ai}} = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{6}, \frac{1}{12}$ that station’s eigenvalue ratio climbs 0.49, 0.75, 0.87, 1.02, 1.13: the value

$$c_{\nu,\text{ai}} = \frac{1}{6}$$

puts the just-past-critical stations on the correlation (the separation branch holds 0.82–0.91). The equilibrium disturbance-band thickness scales as $\sqrt{c_{\nu,\text{ai}}}$, so every halving thins the band the grid must resolve by $\sqrt{2}$.

1.5 Handover blend and destruction tie

The destruction is not gated off but tied: σ_D tracks σ_P with slope $c_{b1}/(\kappa^2 c_{w1}) \approx 0.249$ from the floor $(1 + c_{b2})/(\sigma c_{w1}) \approx 0.751$, the two summing to one by the definition of c_{w1} . The tie is derived from the constant-stress wall layer, on which SA’s solution is the linear profile $\tilde{\nu} = \kappa u_\tau y$ with height-independent terms $P_{\text{SA}} = c_{b1} u_\tau^2$, $D_{\text{SA}} = c_{w1} \kappa^2 u_\tau^2$, and self-diffusion $(1 + c_{b2}) \kappa^2 u_\tau^2 / \sigma$; requiring the gated balance to hold pointwise gives Eq. (4), under which the linear profile is an exact solution of the gated equation at every height and for any τ —continuously and discretely (verified to below 10^{-10} in profile and log-law intercept with two unrelated discretizations). In the laminar range the destruction floor is inert at the disturbance’s quadratic scale: its ratio to the amplification production stays below 10^{-3} up to the $\chi = 1$ crossing for every wedge and seed tested, and at $\chi = c_{v1}$ is at most 1.4% of the σ_P -weighted SA production it is tied against; marching the calibration instrument with the floor included moves the anchor and prediction stations by less than 1%.

1.6 Eddy-viscosity assembly in lifted transitional layers

The near-wall damping f_{v1} enforces the attached-wall-layer identity $\chi = \kappa y^+$. Writing $y^+ := \chi/\kappa$, the reconstruction

$$q = \frac{|\mathbf{u}| d/\nu}{y^+ U^+(y^+)}, \quad U^+(y^+) = \frac{1}{\kappa} \ln(1 + \kappa y^+) + 7.8(1 - e^{-y^+/11} - \frac{y^+}{11} e^{-y^+/3}) \quad (\text{Reichardt}), \quad (11)$$

The assembly is blended,

$$\mu_t = [(1 - b) f_{v1}(\chi) + b] \rho \tilde{\nu}, \quad b = s(\chi) G(q), \quad s = \text{clip}(\chi - 1, 0, 1), \quad G = \text{clip}((q - 2)/2, 0, 1), \quad (12)$$

inert for $\chi < 1$ (negligible μ_t either way), for $\chi \gtrsim 30$ ($f_{v1} \rightarrow 1$), and in the attached inner layer ($q = 1$, G shut). Controlled on/off batteries bound the net effect on attached-layer outer edges below 1.5 drag counts (flat plate and NLF at their cold-budget protocol).

1.7 Freestream seed

A case seeded with χ_∞ reaches the handover where its envelope crosses $N_{\text{crit}} = \ln(c_{v1}/\chi_\infty)$.

$$N_{\text{crit}}(Tu) = -8.43 - 2.4 \ln(Tu_{\text{frac}}), \quad \chi_\infty = c_{v1} e^{-N_{\text{crit}}}, \quad (13)$$

One e -fold in χ_∞ is one unit of N_{crit} , which on the Blasius envelope ($dN/dRe_\theta \approx 10^{-2}$) shifts onset by $\Delta Re_\theta \approx 100$. Strongly bypass-dominated conditions ($Tu \gtrsim 1\%$) are out of scope: at $Tu = 1\%$ the map leaves only $N_{\text{crit}} \approx 2.6$, and by $Tu \approx 3\%$ the seed exceeds c_{v1} outright.

1.8 Constants

Baseline SA, unchanged:

$$c_{b1} = 0.1355, \quad \sigma = \frac{2}{3}, \quad c_{b2} = 0.622, \quad \kappa = 0.41, \quad c_{w1} = \frac{c_{b1}}{\kappa^2} + \frac{1 + c_{b2}}{\sigma}, \quad c_{w2} = 0.3, \quad c_{w3} = 2, \quad c_{v1} = 7.1.$$

SA-AI:

constant	value	determination
$a_{\max}$	0.19	tanh-layer Kelvin–Helmholtz temporal eigenvalue, $\omega_{i,\max}/\omega_{\text{peak}} = 0.1897$
(C, A, B)	(2600, 175, 2)	graze envelope of the attached Falkner–Skan family at its critical Re_θ (rms $\sim 6\%$)
k	0.712	Blasius march crosses $N = 1$ at the Drela–Giles station $Re_\theta = 338$ at $c_{\nu,\text{ai}} = \frac{1}{6}$
w	0.35	onset ramp width; $N = 1$ stations move $< 1\%$ halved, $< 2.5\%$ doubled
$c_{\nu,\text{ai}}$	1/6	frozen-profile growth eigenvalue reaches the Drela–Giles rate just past critical (1.02 at $Re_\theta = 400$)
τ	4	numerics bracket: handover 78% complete at $\chi = c_{v1}$, 97% at $\chi \approx 14.8$
σ_D tie	Eq. (4)	derived: constant-stress wall-layer balance; linear wall solution exact
bypass (s, G)	Eq. (12)	parameter-free law-of-the-wall discriminator; G ramps over $q \in (2, 4)$
$(A, B)_{\text{Mack}}$	(−8.43, 2.4)	Mack 1977 e^N correlation, wholesale

1.9 Running the model: branch selection and seed handling

On the frozen Hiemenz stagnation field with the freestream seed exactly zero, standard SA sustains the wedge above a critical nose Reynolds number bisected to $Re_r = 4.66\text{--}4.74 \times 10^5$ (Fig. 5).

Table 1: Attachment-anchored branch: steady $\max \chi$ versus nose Reynolds number $Re_r = L^2$ on the frozen Hiemenz wedge, standard SA, $\chi_\infty = 0$. Below the bisected critical band the layer collapses to $\chi = 0$.

$L = \sqrt{Re_r}$	Re_r	steady $\max \chi$
≤ 683	$\leq 4.66 \times 10^5$	$< 10^{-3}$ (collapses to 0)
683–689	$4.66\text{--}4.74 \times 10^5$	critical band (bisected)
689	4.74×10^5	2.19
700	4.90×10^5	6.15
1000	1.00×10^6	24.1
3000	9.00×10^6	75.7

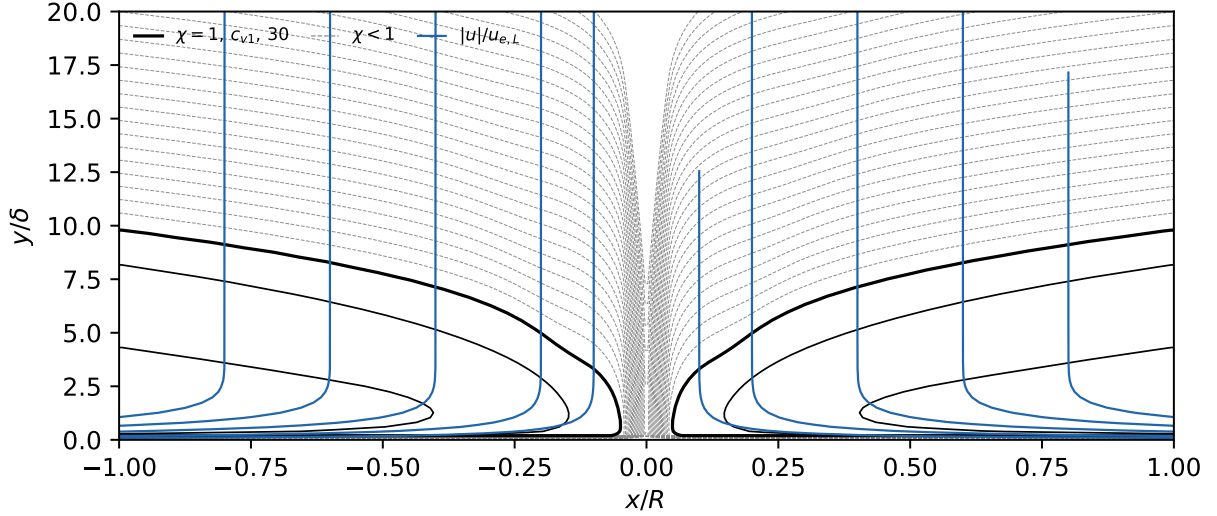


Figure 5: Attachment-anchored turbulent branch, $L = 3000$ ($Re_r = 9 \times 10^6$), $\max \chi = 76$. Standard SA, frozen Hiemenz, $\chi_\infty = 0$. Abscissa x/R (arc length in nose radii, $R/\delta = L$); ordinate y/δ . χ contours dashed per decade for $\chi < 1$ (10^{-24} – 10^{-1} ; the zero freestream leaves no floor, so χ decays smoothly over decades), solid at 1, c_{v1} , 30; velocity magnitude $|u|/u_{e,L}$ (blue, normalized by the $x=L$ edge speed) at 0.1, 0.2, 0.4, 0.6, 0.8. Branch data in Table 1.

The protocol used for every result in this paper: the $\tilde{\nu}$ -equation residual is scaled by f^{1-g} , where g rises smoothly from 0 (laminar) to 1 (turbulent) with χ and $f = 10^{-2}$ (the solver uses $g = 1 - e^{-(\chi - c_{v1})/4}$ above $\chi = c_{v1}$, zero below). Because the freestream boundary condition flows through the same scaling, the freestream viscosity ratio supplied to the solver is $\chi_\infty f$; the seeds quoted throughout are the physical χ_∞ , recovered at the boundary-layer edge. (In the reference implementation the scaling is `AI_LAMINAR_SLOWDOWN` = f , and the case files carry the pre-multiplied boundary seed $\chi_\infty f$; running a committed case at $f = 1$ without rescaling the seed mis-seeds it by $1/f$.)

2 Results

Coupled RANS: Flow360, unstructured node-centered second-order finite volume, Roe flux, $M = 0.1$. Airfoil cases: $\chi_\infty = c_{v1}e^{-9} \approx 8.76 \times 10^{-4}$ (the $N_{\text{crit}} = 9$ anchor; via Mack’s map, $Tu \approx 0.07\%$); two independently generated mesh families—all-triangular unstructured cavity meshes, constrained-Delaunay through the boundary layer, and all-quadrilateral structured O-grids—at three refinement levels, 1.5×10^4 to 2.6×10^5 nodes. Transition locations are the solver’s near-wall $\chi = 1$ crossing (the $\chi = c_{v1}$ crossing sits up to $0.05c$ aft where the front grows slowly); bubble stations are signed- C_f zero crossings. The cross-solver replication is a cell-centered, pressure-based, incompressible OpenFOAM implementation of the model on the same flat-plate grid specification and the same structured airfoil grids (the port omits the f_{v1} bypass); thirty airfoil cases plus the flat-plate sweep.

2.1 Flat plate, $Tu = 0.04$ – 0.60%

Structured 320×80 grid, $Re_x \leq 6 \times 10^6$, $y^+ \lesssim 0.3$; five seeds from Mack’s map ($\chi_\infty \approx 2.3 \times 10^{-4}$, 1.2×10^{-3} , 6.3×10^{-3} , 2.9×10^{-2} , 1.5×10^{-1}). Figure 6: at the $\chi = c_{v1}$ crossing the onset brackets the Abu-Ghannam & Shaw correlation within 10% across the range. Figure 7: $\chi = 1$ crossings within 7% in Re_θ . Table 2: both solvers’ crossings.

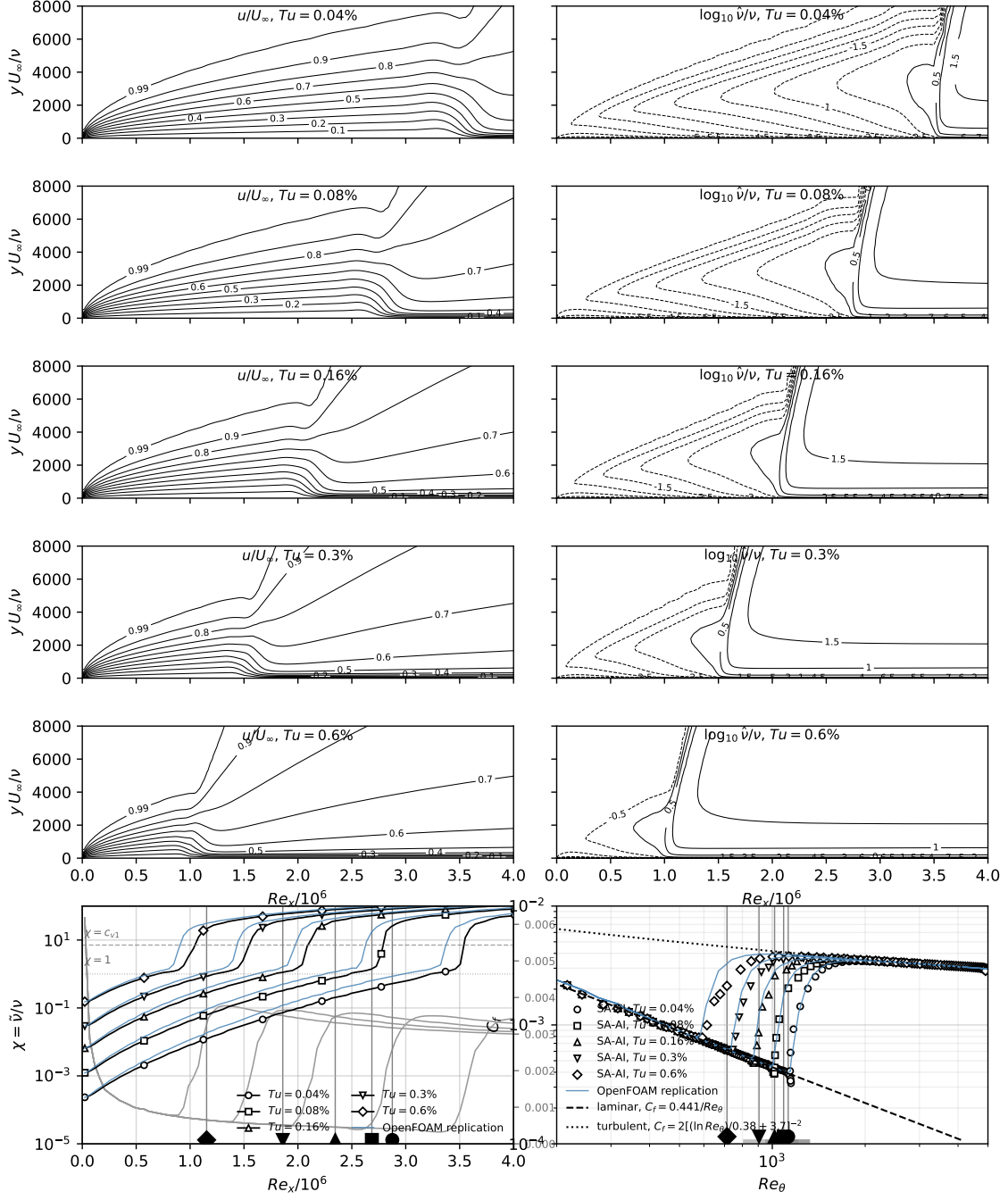


Figure 6: Flat plate, Flow360. Five levels $Tu = 0.04\text{--}0.60\%$; references $\chi = 1$ and $\chi = c_{v1} = 7.1$; laminar $C_f = 0.441/Re_\theta$, turbulent $C_f = 2[(\ln Re_\theta)/0.38 + 3.7]^{-2}$; $\chi = c_{v1}$ onset brackets Abu-Ghannam & Shaw within 10%.

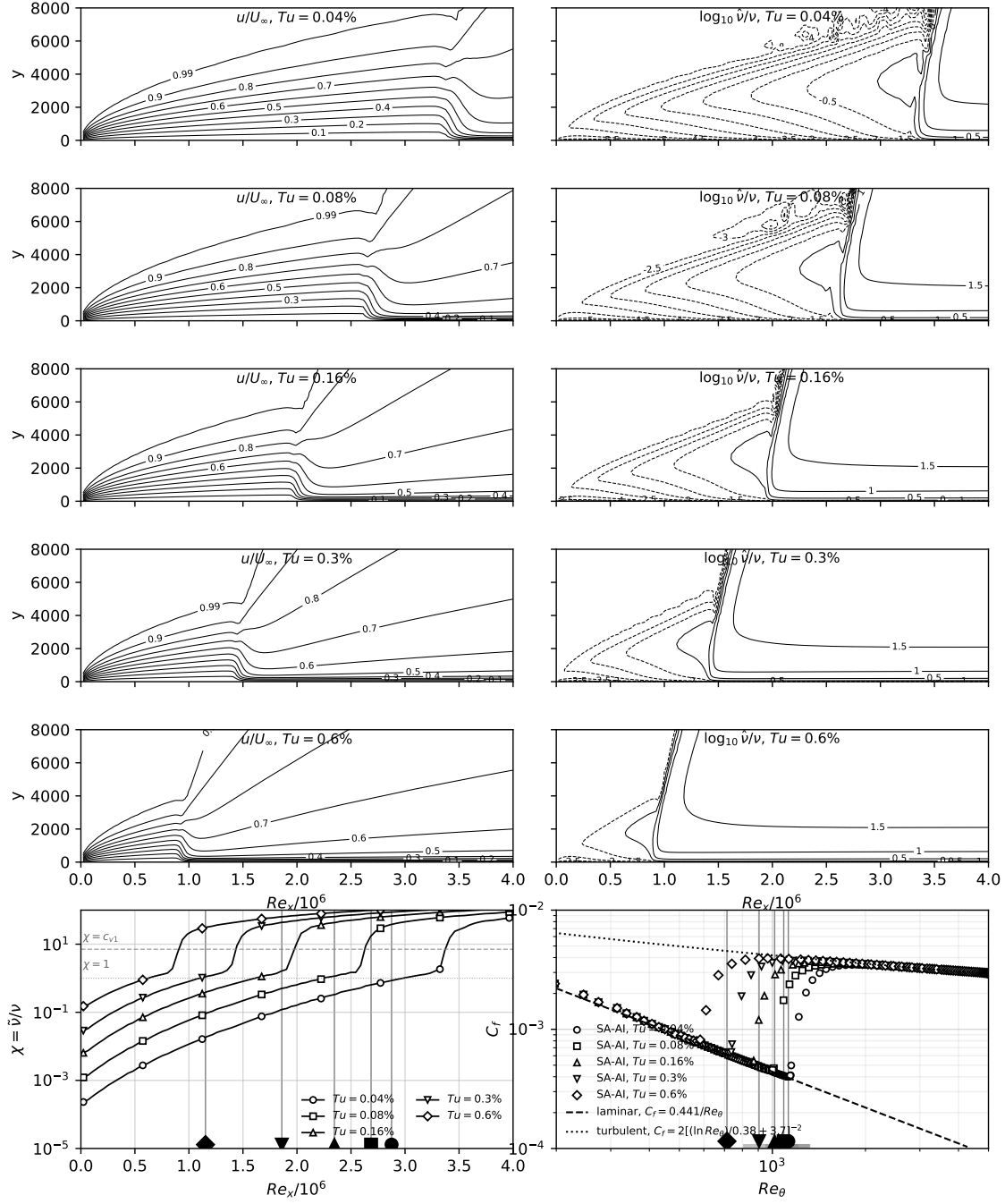


Figure 7: Flat plate, OpenFOAM. Same sweep as Fig. 6, five levels $Tu = 0.04\text{--}0.60\%$.

Table 2: Flat-plate transition-onset Re_θ (integrated from the computed profiles) at the $\chi = 1$ and $\chi = c_{v1}$ crossings, Flow360 and the OpenFOAM replication (OF), against the Abu-Ghannam & Shaw correlation.

Tu [%]	AGS Re_θ	$\chi = 1$	$\chi = c_{v1}$	OF $\chi = 1$	OF $\chi = c_{v1}$
0.04	1126	1149	1203	1098	1154
0.08	1088	1002	1069	949	1004
0.16	1017	845	934	798	887
0.3	905	700	815	656	739
0.6	713	524	685	487	589

2.2 NLF(1)-0416 at $Re = 4 \times 10^6$

$\alpha \in \{0^\circ, 4^\circ, 9^\circ, 15^\circ\}$ on both families at L0–L2 plus $\alpha = -8^\circ, -4^\circ$ on all six grids (thirty-six cases). Figure 8: transition fronts against the LTPT microphone measurements, Coder’s AFT (OVERFLOW) at $N_{crit} = 10.07$ and recalibrated 7.18, the fourteen First AIAA Transition Modeling Workshop submittals, SA-LM2015, Langtry–Menter, and XFOIL. Figure 9: drag polar against Somers TP-1861, the e^9 reference, and fully turbulent SA. Figures 10–12: surface distributions. Table 3: forces and fronts. Table 4: at L2 every front within $0.016c$ and the lift within 0.9% of the SA-AI values.

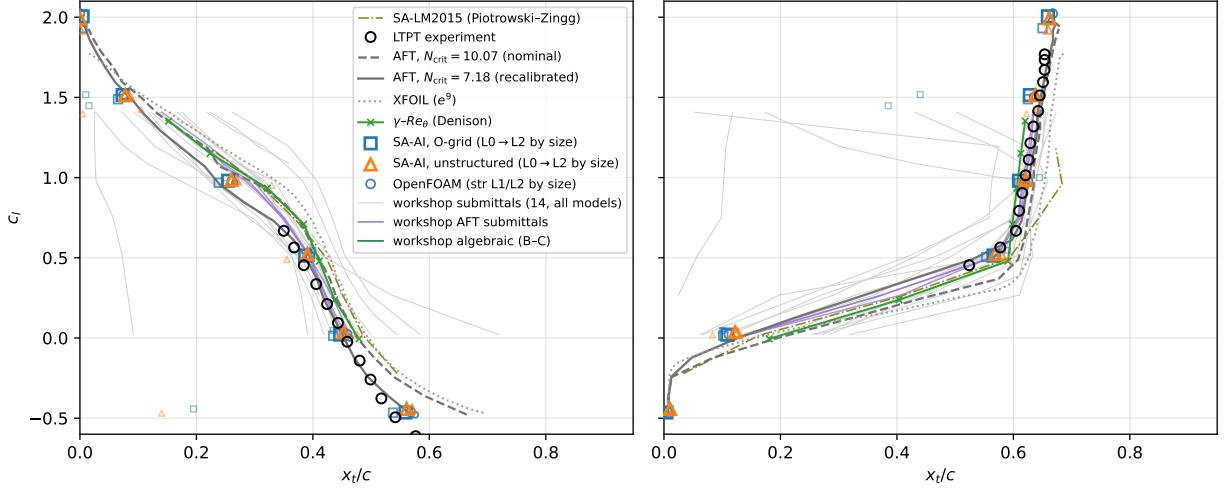


Figure 8: NLF(1)-0416 transition fronts, $Re = 4 \times 10^6$; near-wall $\chi = 1$ convention. SA-AI at $\alpha = 0^\circ, 4^\circ, 9^\circ, 15^\circ, -8^\circ, -4^\circ$; workshop sweep $\alpha = -4^\circ$ – 8° .

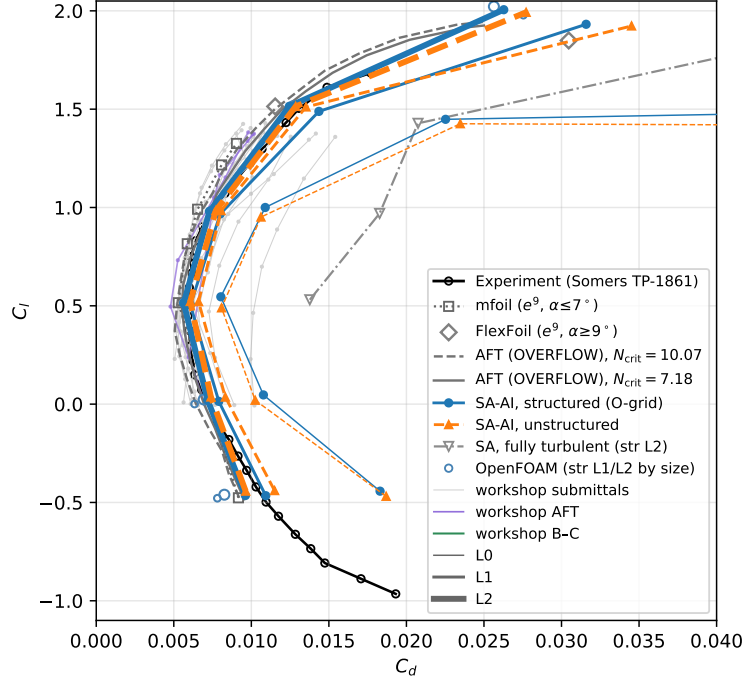


Figure 9: NLF(1)-0416 drag polar, $Re = 4 \times 10^6$. Fully turbulent SA ($\chi_\infty = 3$): 1.6–2.5 $\times$ above the SA-AI drag, +144/+150% in the bucket at $\alpha = 0^\circ/4^\circ$, +67/+74% at $9^\circ/15^\circ$.

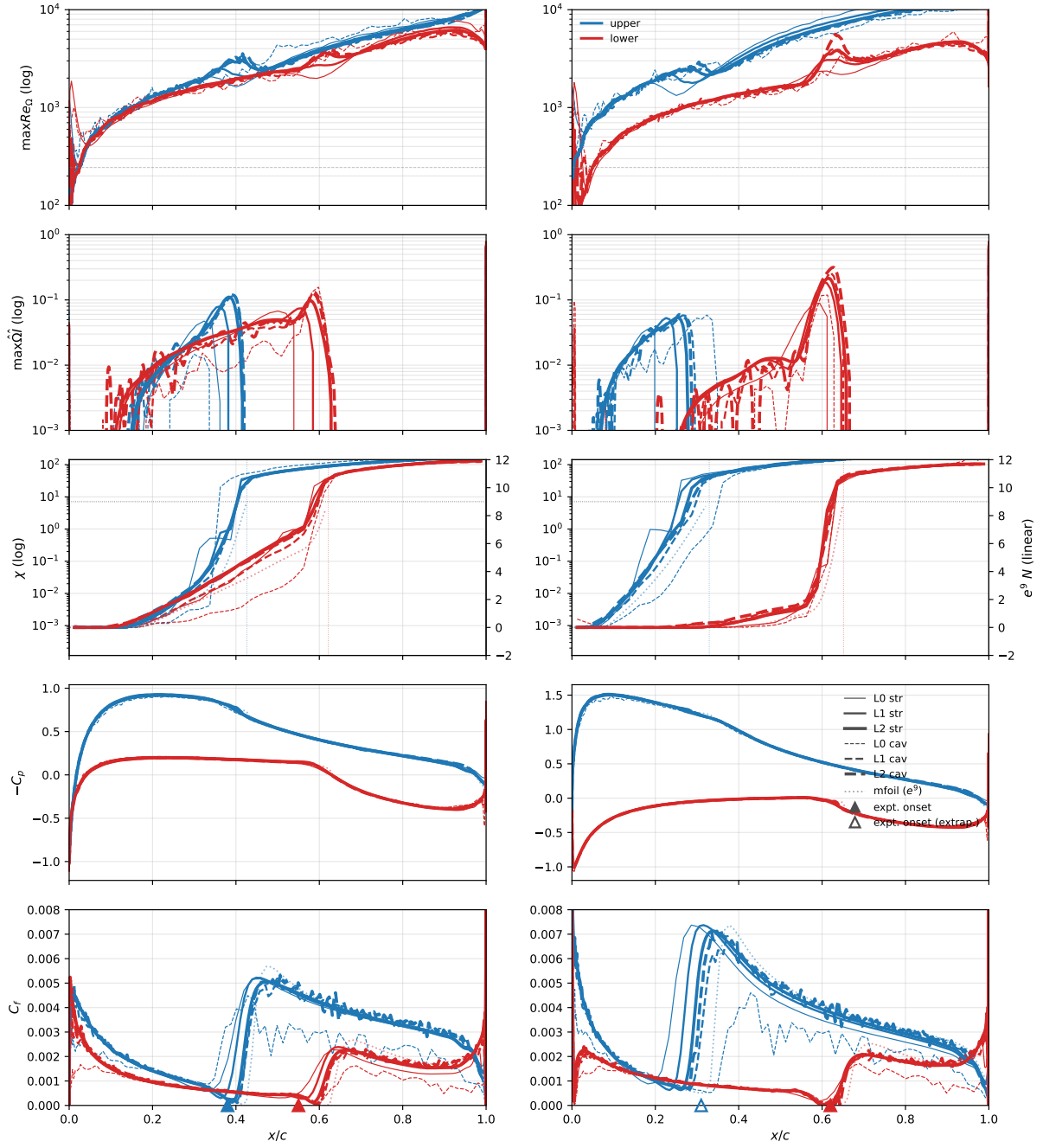


Figure 10: NLF(1)-0416, $\alpha = 0^\circ, 4^\circ$, $Re = 4 \times 10^6$.

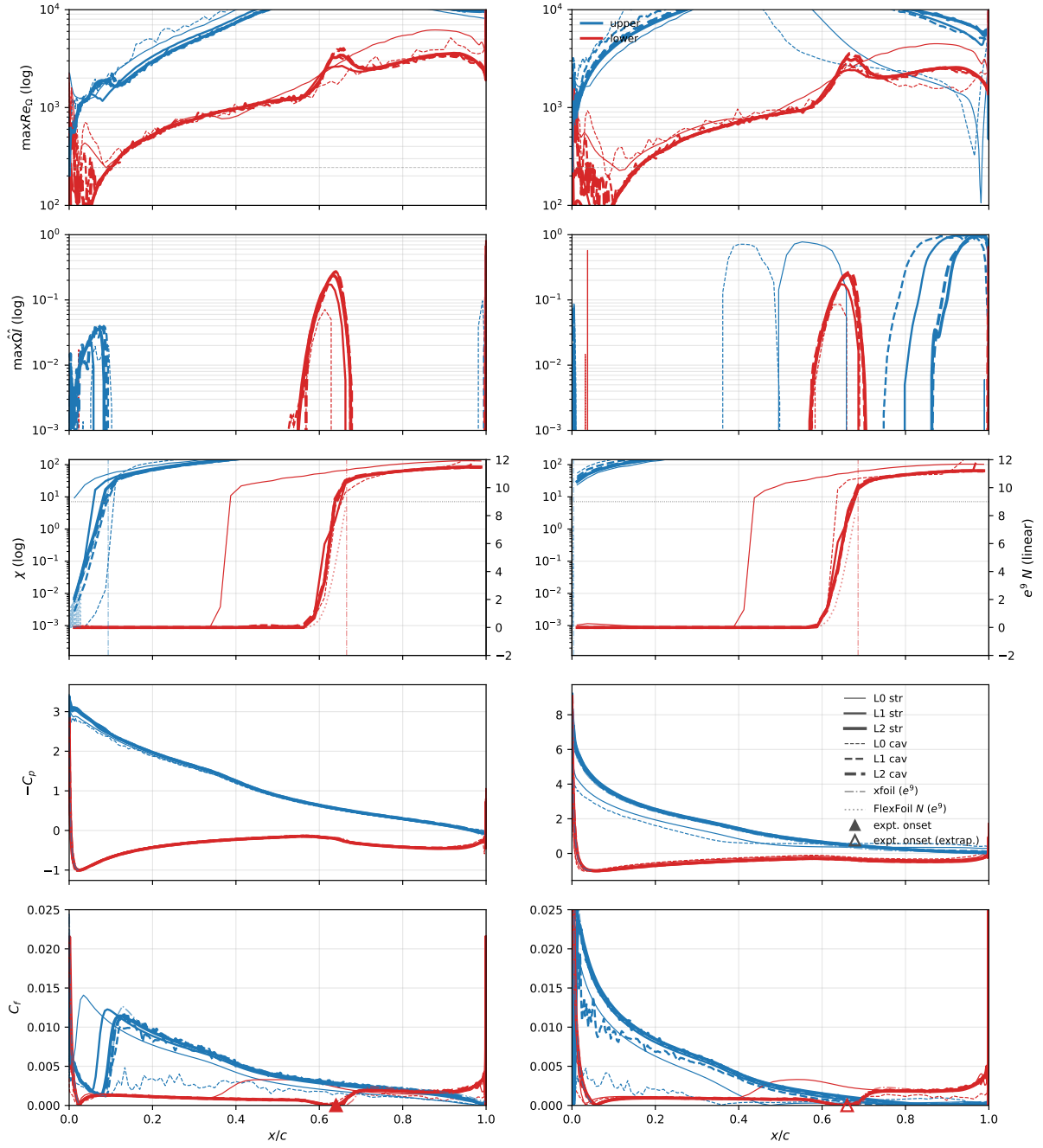


Figure 11: NLF(1)-0416, $\alpha = 9^\circ, 15^\circ$, $Re = 4 \times 10^6$.

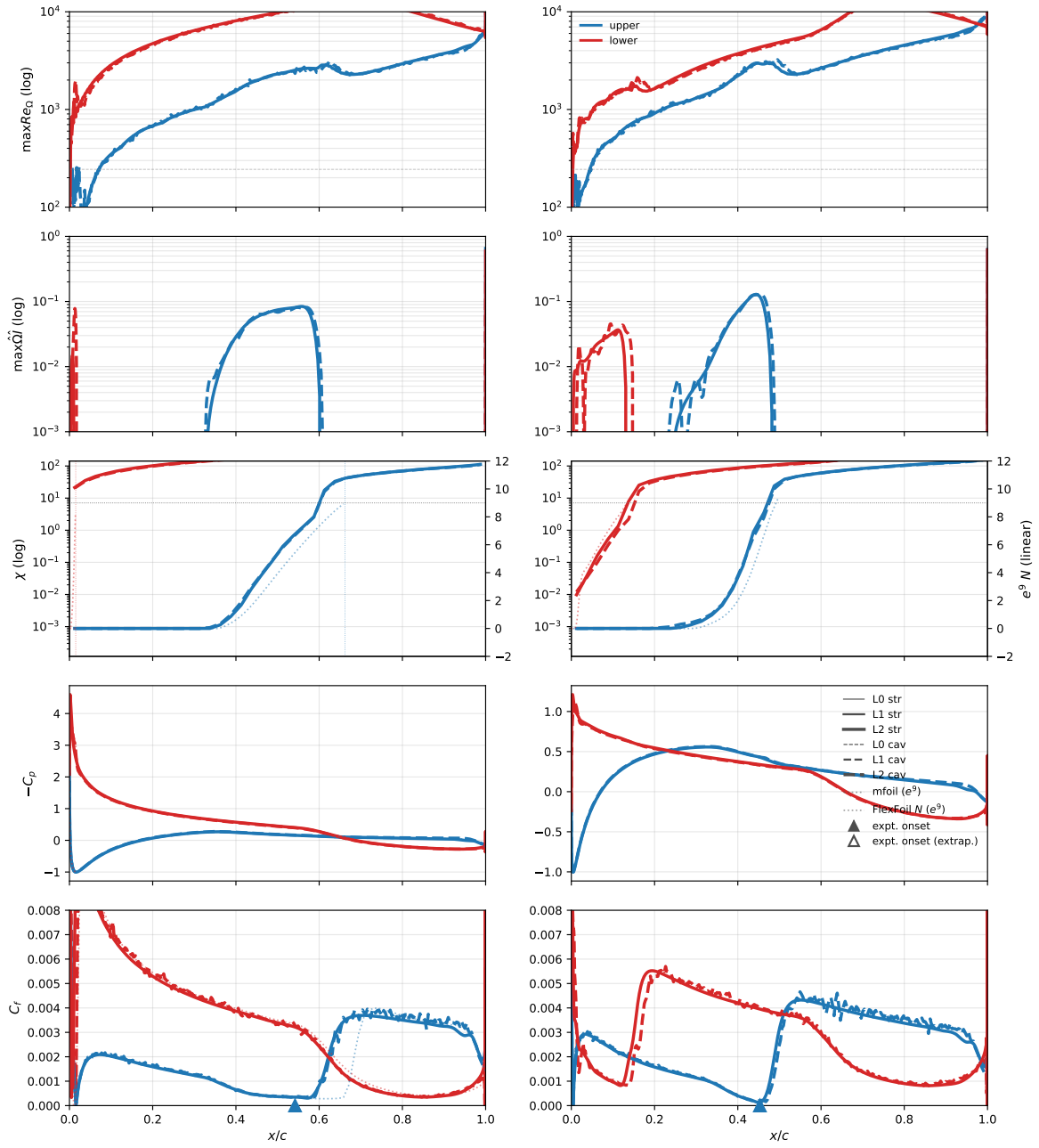


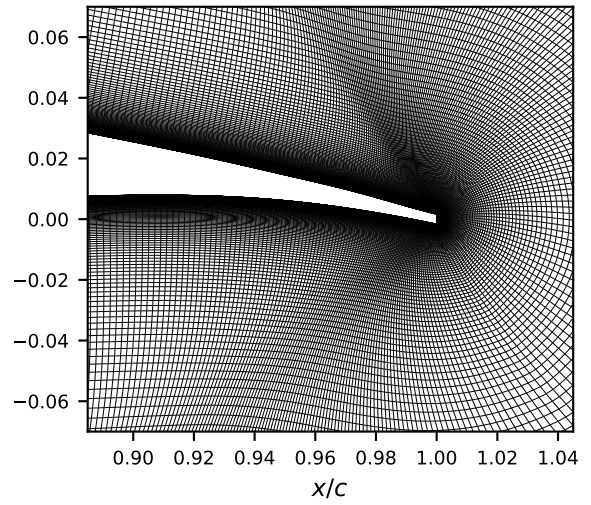
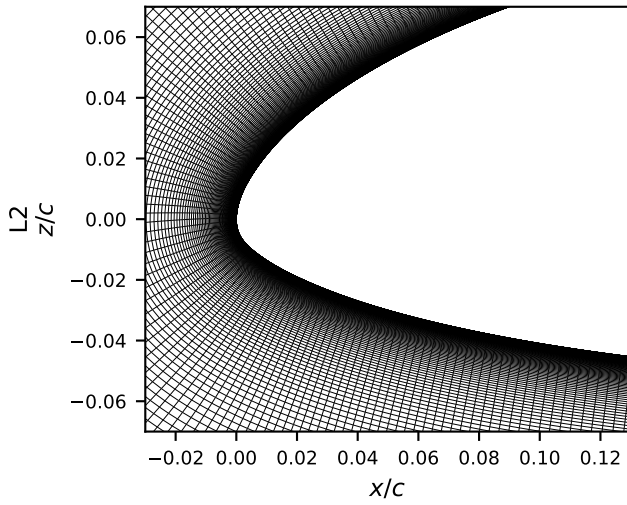
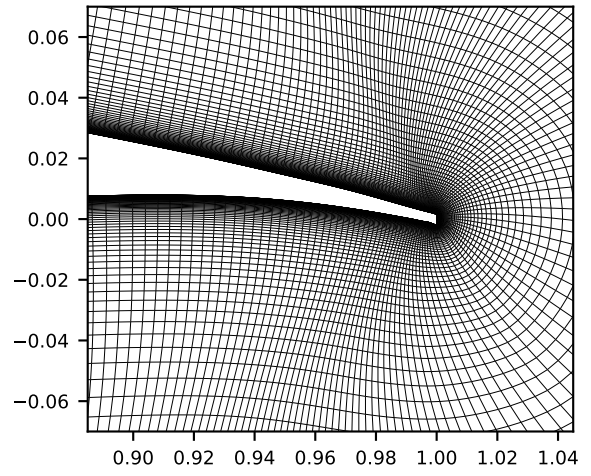
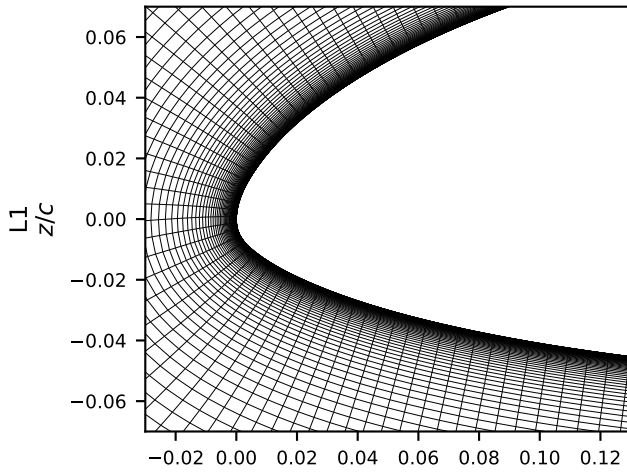
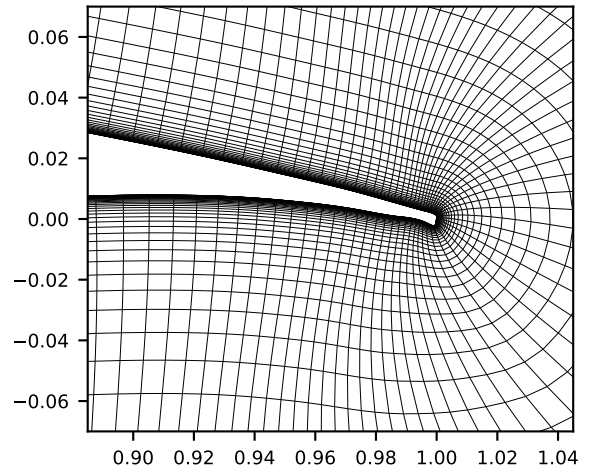
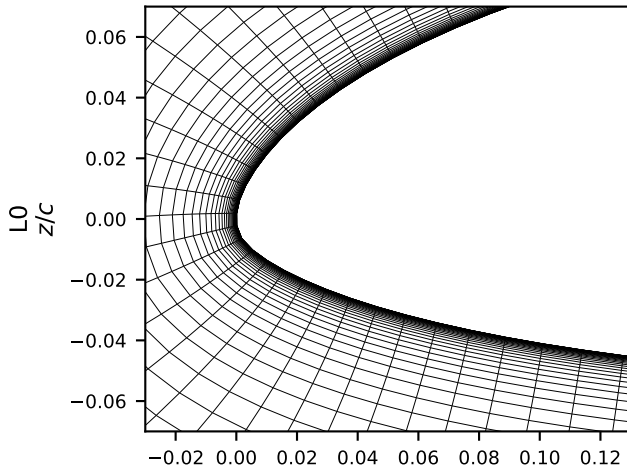
Figure 12: NLF(1)-0416, $\alpha = -8^\circ, -4^\circ$, $Re = 4 \times 10^6$.

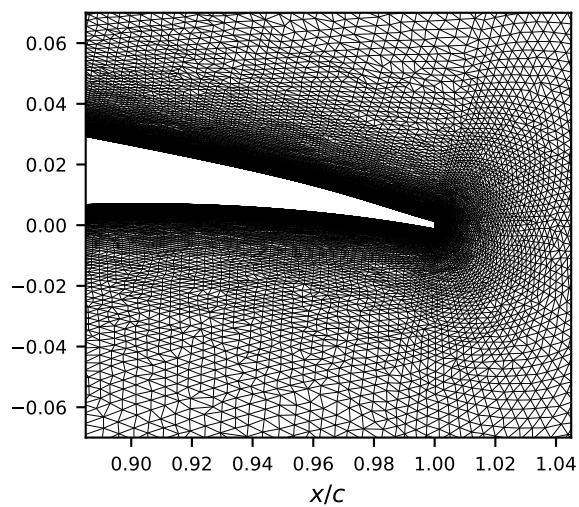
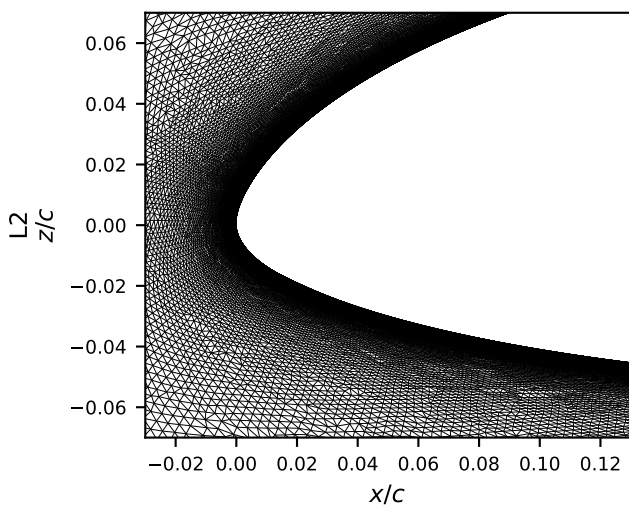
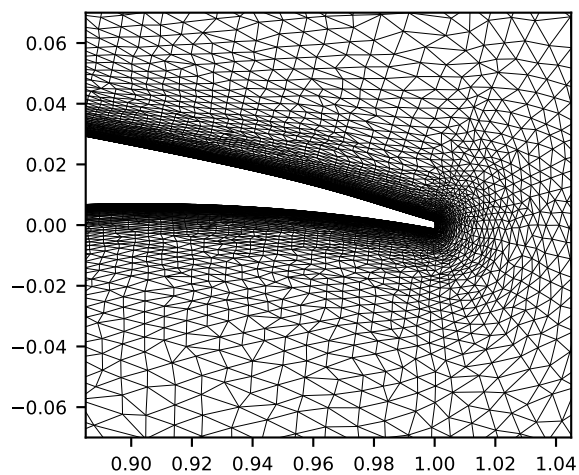
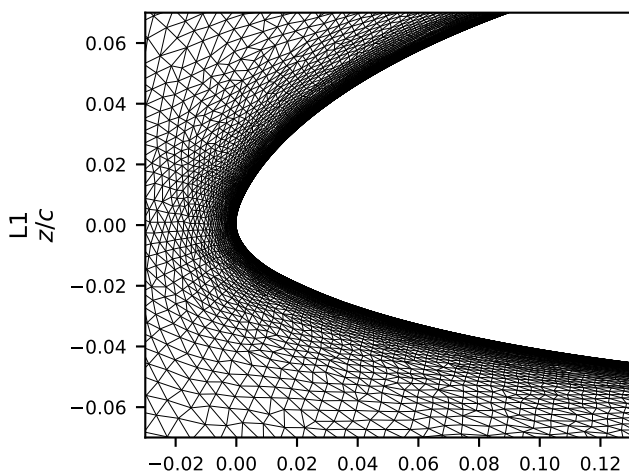
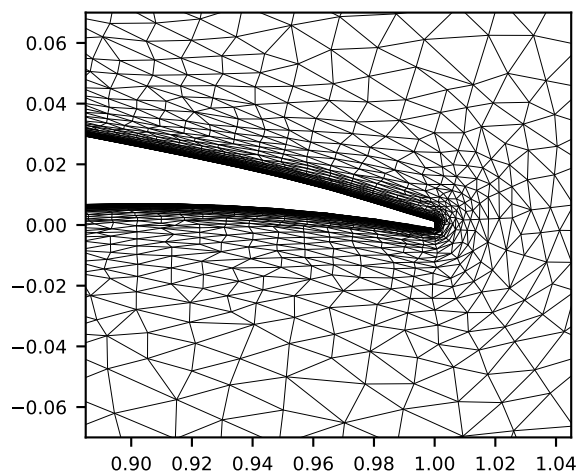
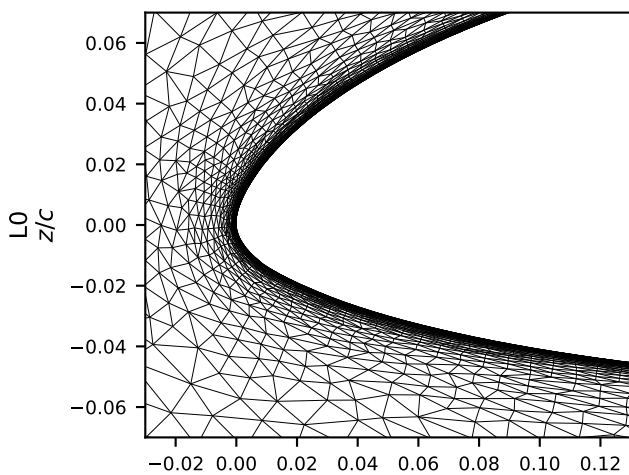
Table 3: NLF(1)-0416, $Re=4\times 10^6$: computed forces and transition locations (near-wall $\chi=1$ front stations).

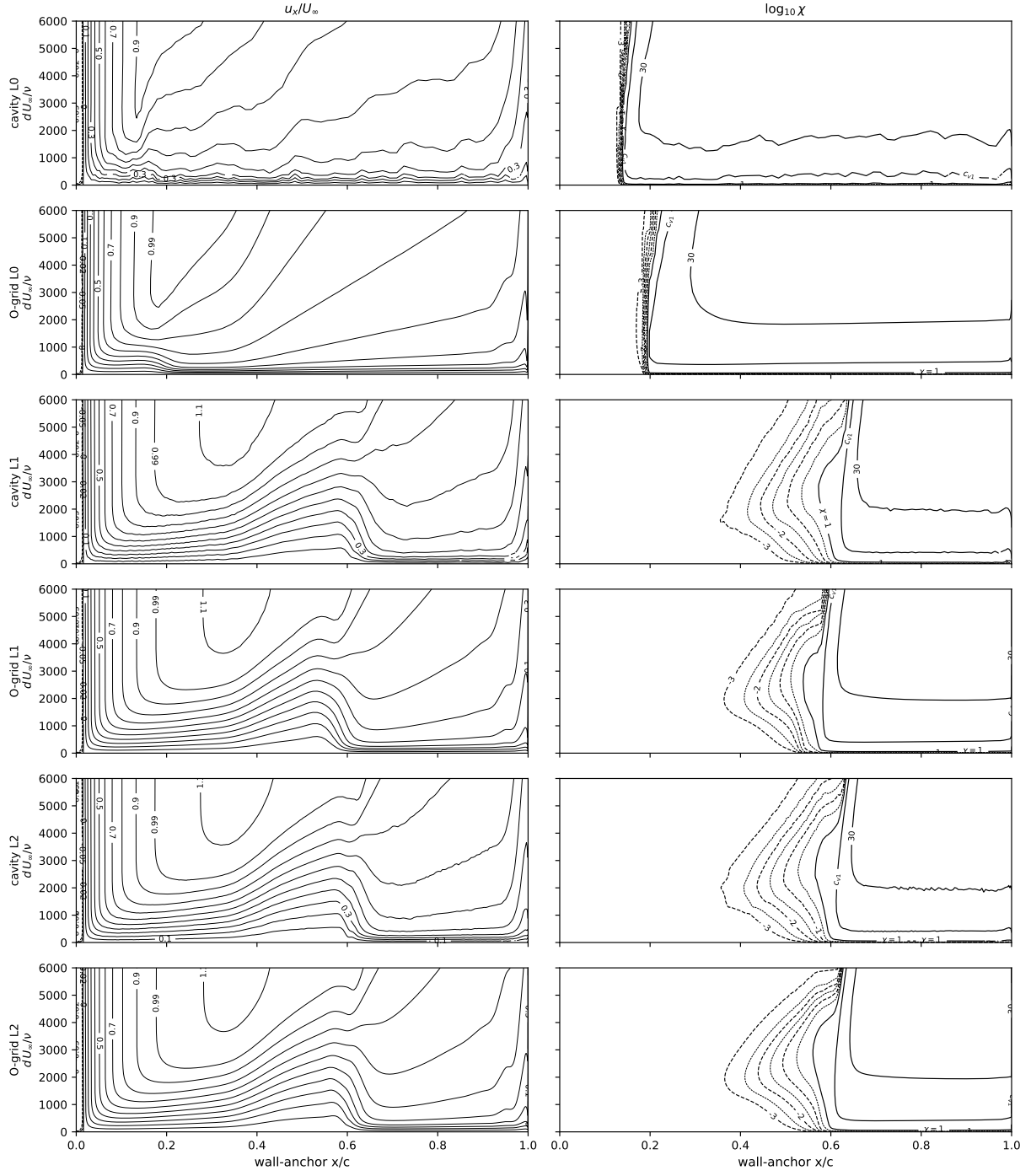
α	grid	structured O-grid				unstructured cavity			
		C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}	C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}
-8	L0	-0.4426	0.01830	0.195	0.011	-0.4689	0.01869	0.140	0.012
-8	L1	-0.4656	0.01093	0.538	0.006	-0.4397	0.01150	0.571	0.011
-8	L2	-0.4646	0.00962	0.559	0.004	-0.4426	0.00964	0.561	0.010
-4	L0	0.0469	0.01078	0.435	0.100	0.0197	0.01025	0.476	0.083
-4	L1	0.0142	0.00790	0.435	0.102	0.0357	0.00832	0.452	0.126
-4	L2	0.0208	0.00718	0.447	0.109	0.0363	0.00739	0.453	0.122
0	L0	0.5458	0.00803	0.400	0.590	0.4896	0.00808	0.355	0.592
0	L1	0.5048	0.00607	0.395	0.553	0.5209	0.00660	0.392	0.579
0	L2	0.5139	0.00564	0.387	0.566	0.5219	0.00606	0.391	0.569
4	L0	1.0001	0.01090	0.250	0.645	0.9513	0.01062	0.330	0.621
4	L1	0.9678	0.00800	0.238	0.610	0.9839	0.00805	0.269	0.614
4	L2	0.9805	0.00730	0.255	0.610	0.9848	0.00772	0.260	0.621
9	L0	1.4483	0.02252	0.015	0.385	1.4251	0.02345	0.105	0.641
9	L1	1.4886	0.01436	0.065	0.625	1.5112	0.01353	0.085	0.635
9	L2	1.5146	0.01245	0.073	0.628	1.5139	0.01286	0.078	0.638
15	L0	1.5172	0.07109	0.010	0.440	1.3976	0.11602	0.005	0.623
15	L1	1.9317	0.03159	0.004	0.650	1.9232	0.03454	0.001	0.659
15	L2	2.0057	0.02628	0.005	0.659	1.9924	0.02772	0.000	0.662

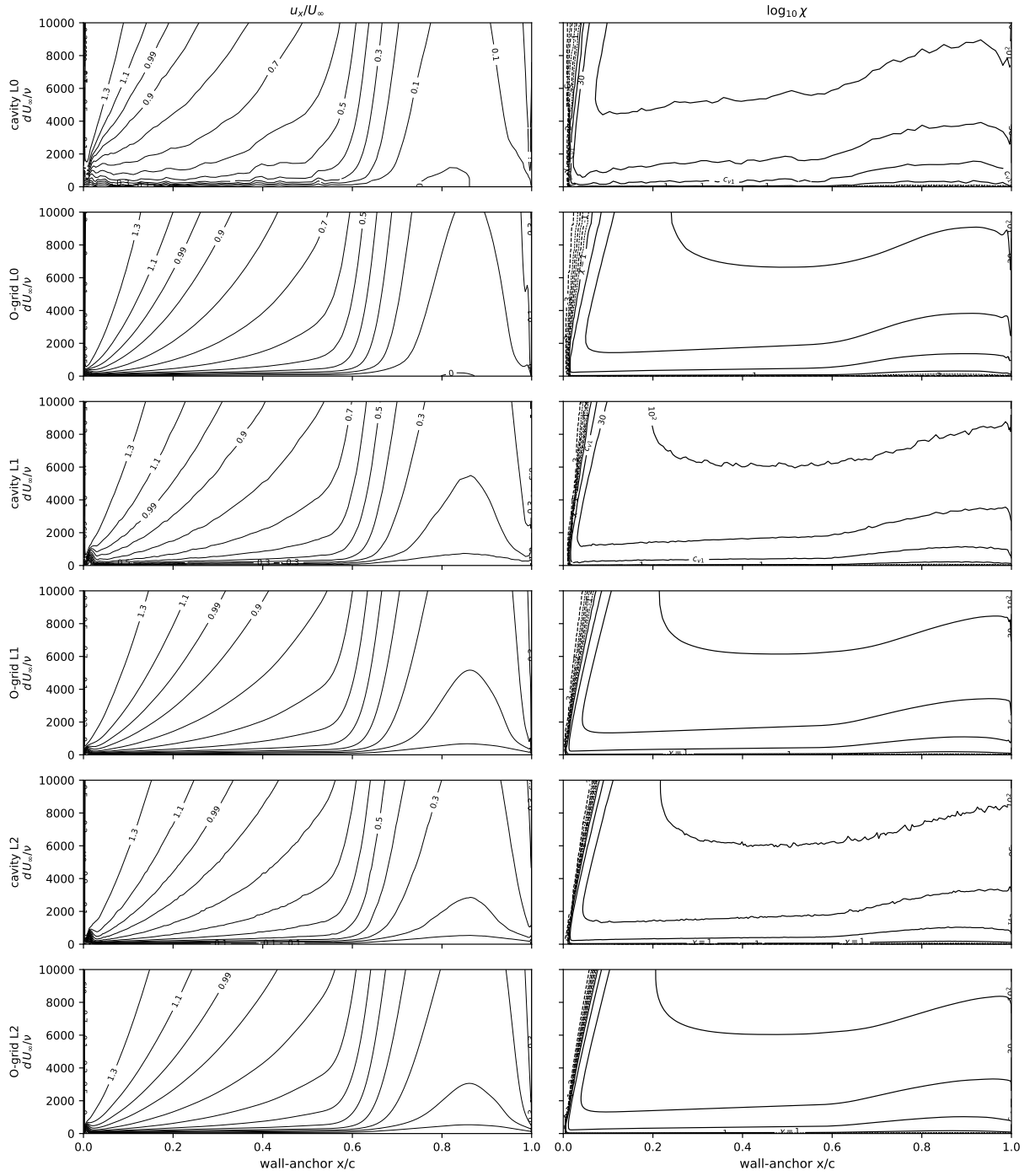
Table 4: NLF(1)-0416: the independent OpenFOAM implementation on the structured L0, L1 and L2 grids, fronts at the same near-wall $\chi=1$ convention (the port omits the f_{v1} bypass).

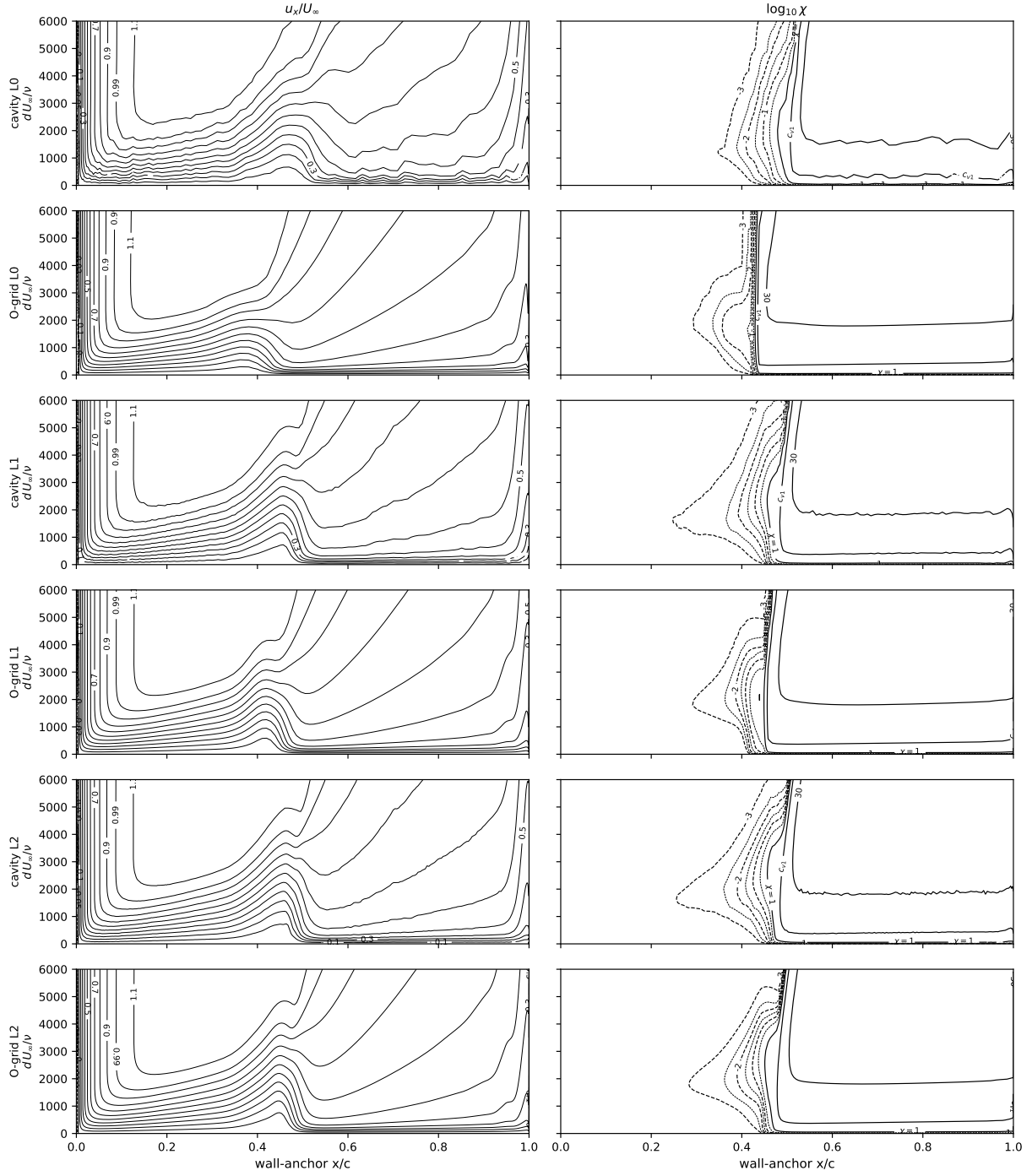
α	grid	C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}
-8	L0	-0.4460	0.01241	0.556	0.007
-8	L1	-0.4785	0.00781	0.576	0.000
-8	L2	-0.4606	0.00826	0.568	0.000
-4	L0	0.0388	0.00748	0.424	0.018
-4	L1	0.0005	0.00634	0.464	0.117
-4	L2	0.0239	0.00693	0.459	0.111
0	L0	0.5473	0.00669	0.387	0.564
0	L1	0.5034	0.00603	0.393	0.572
0	L2	0.5153	0.00573	0.395	0.568
4	L0	1.0114	0.00955	0.271	0.618
4	L1	0.9670	0.00778	0.266	0.616
4	L2	0.9793	0.00749	0.261	0.625
9	L0	1.5277	0.01496	0.063	0.618
9	L1	1.5066	0.01273	0.076	0.633
9	L2	1.5151	0.01270	0.075	0.644
15	L0	1.7210	0.05409	0.000	0.634
15	L1	1.9780	0.02754	0.000	0.665
15	L2	2.0221	0.02564	0.000	0.667

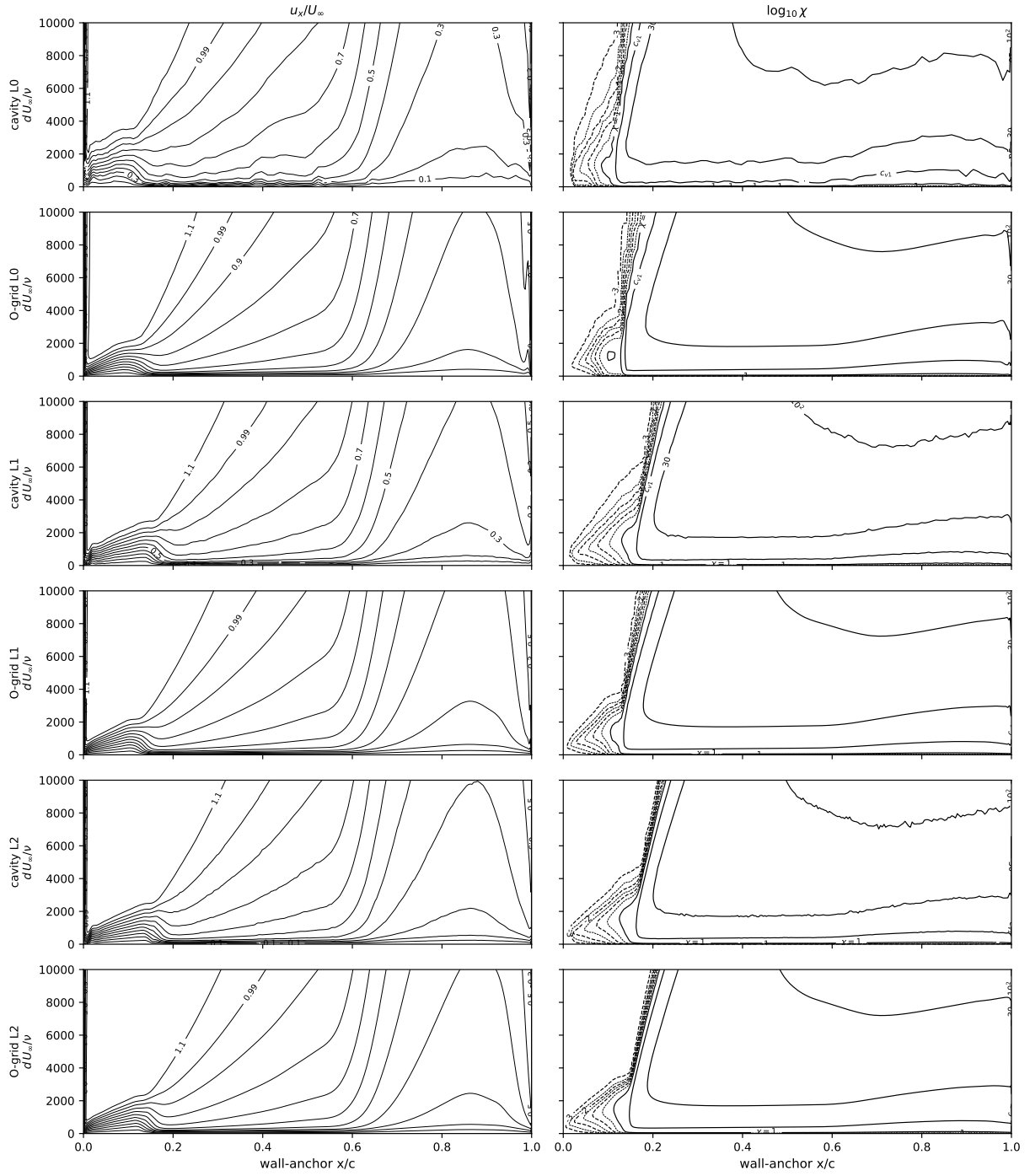


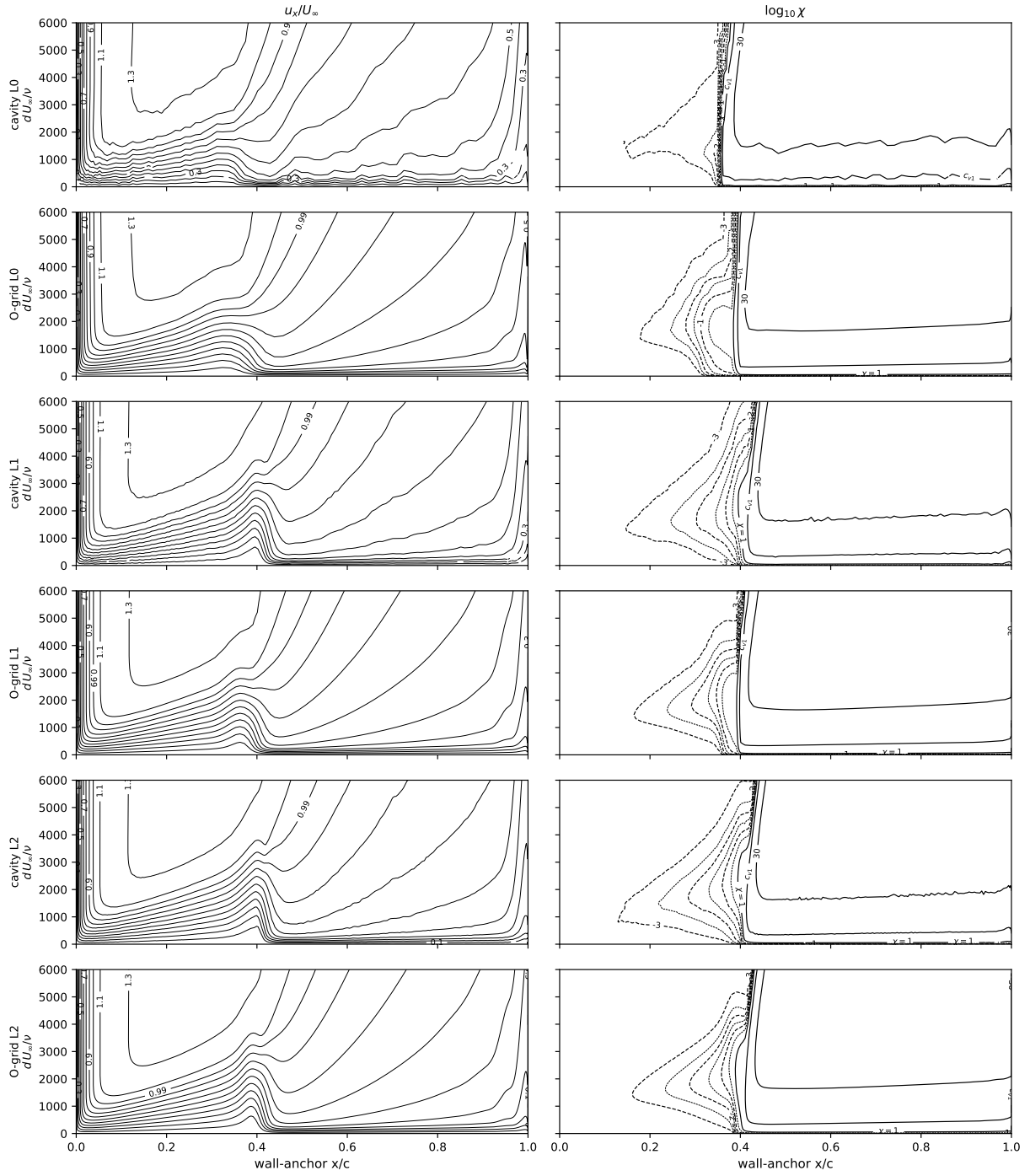


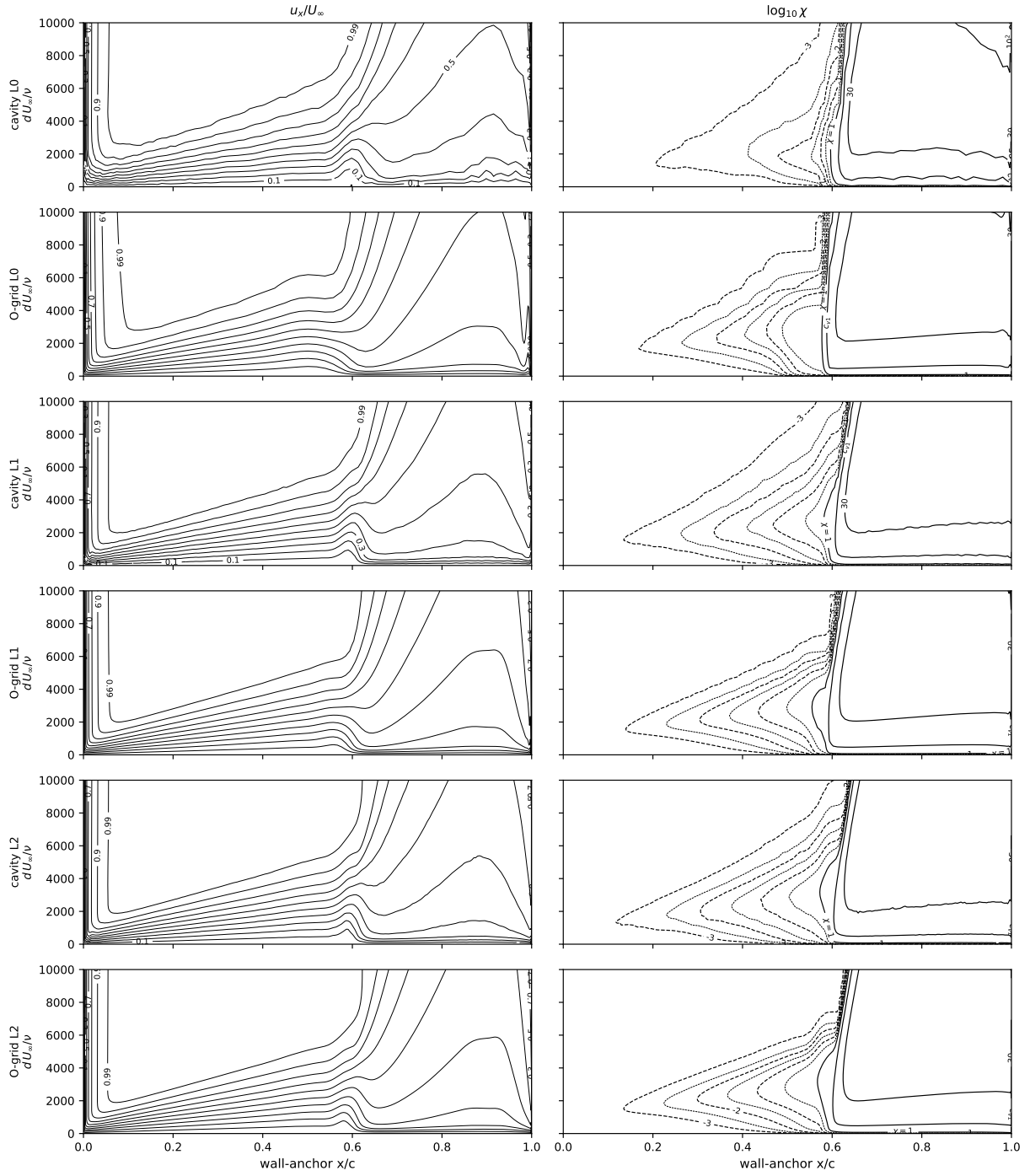


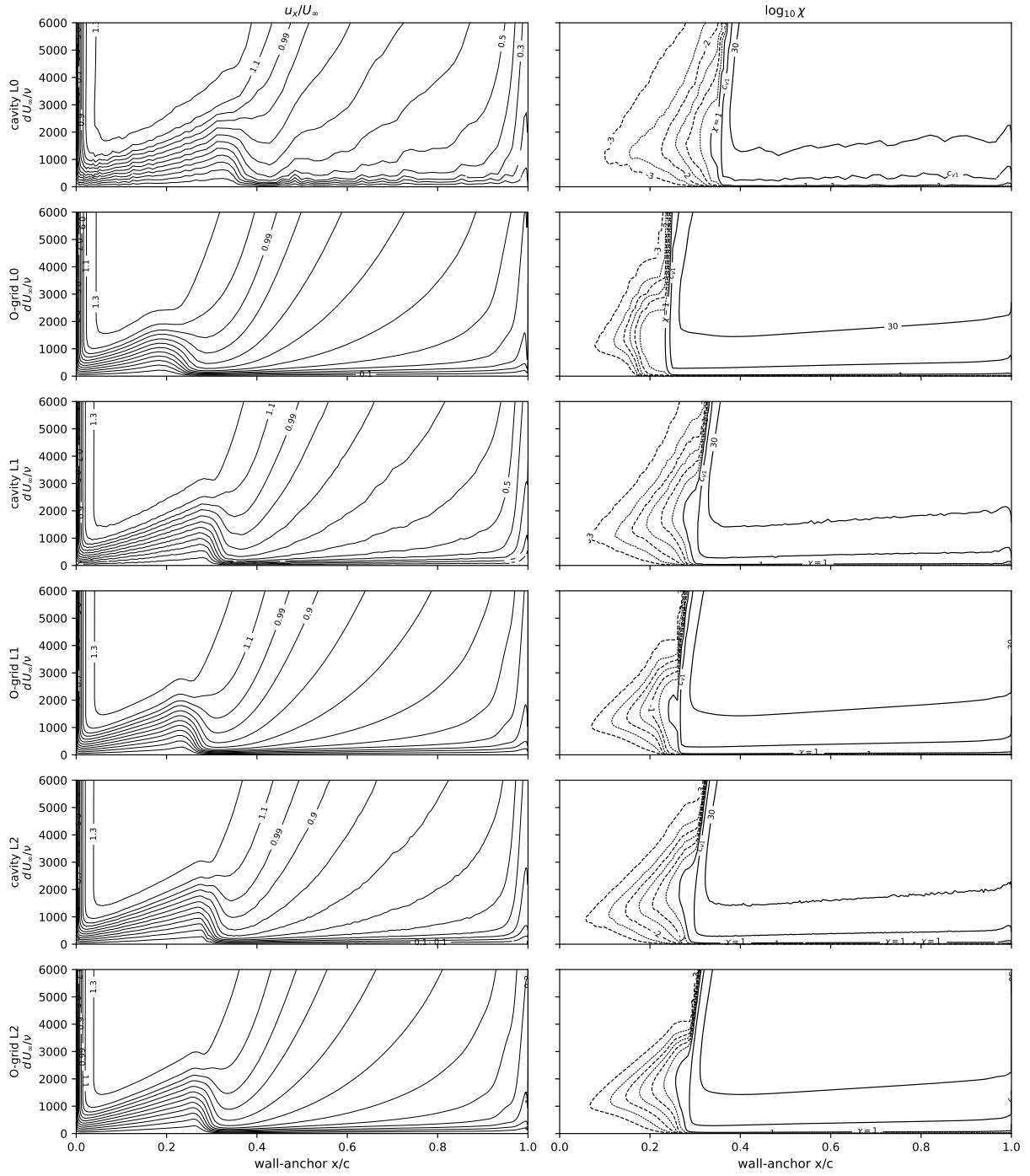


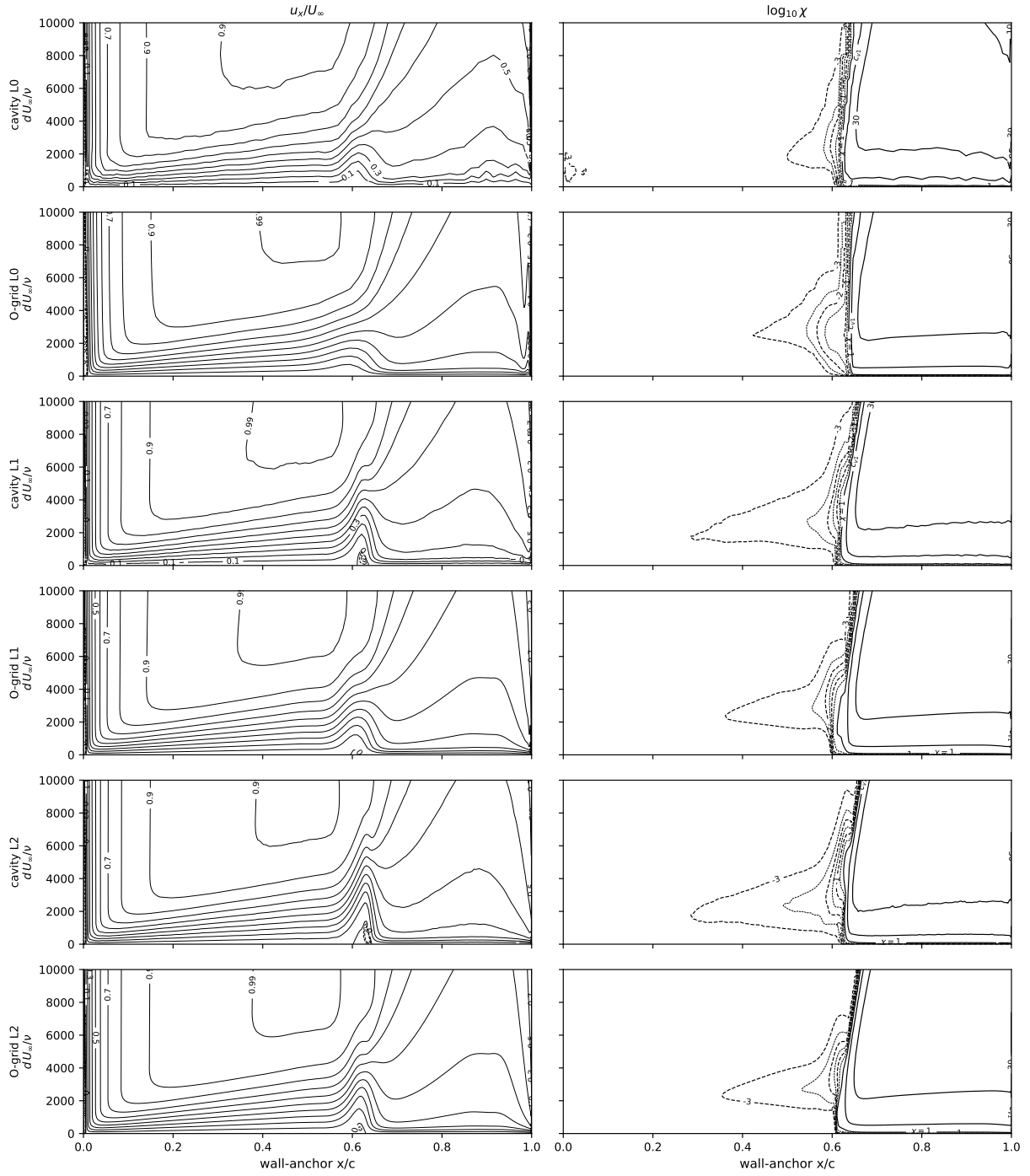


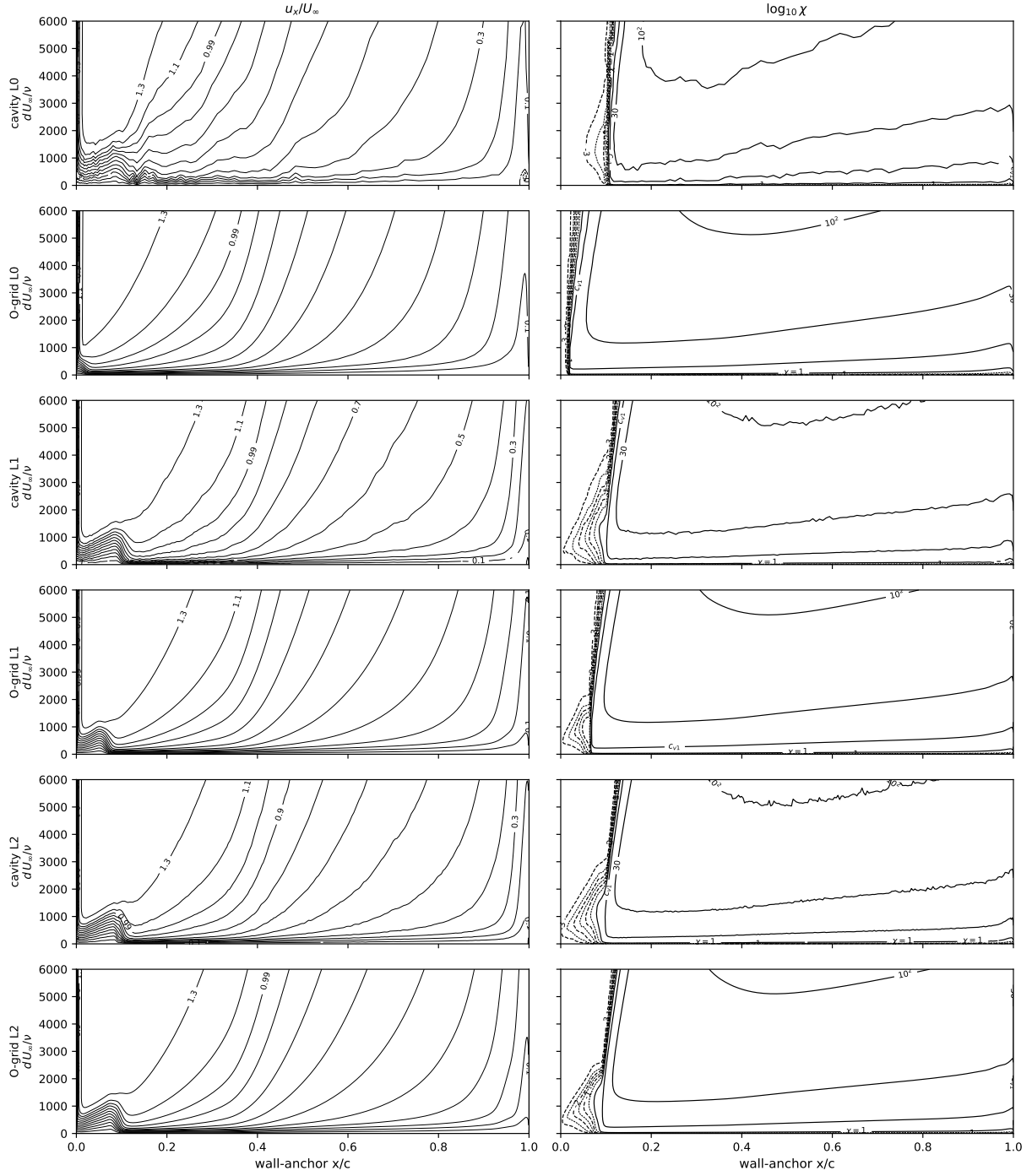


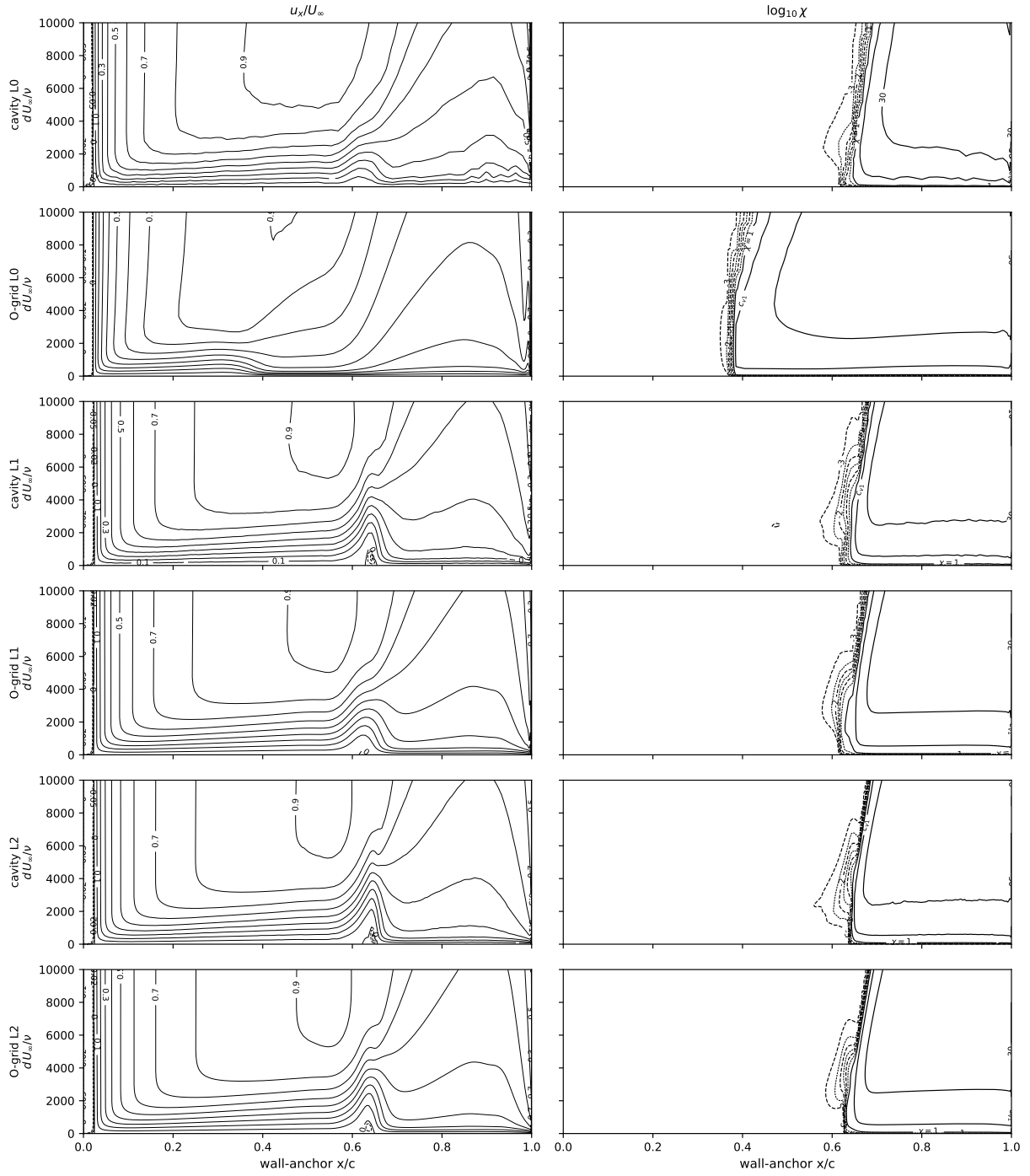


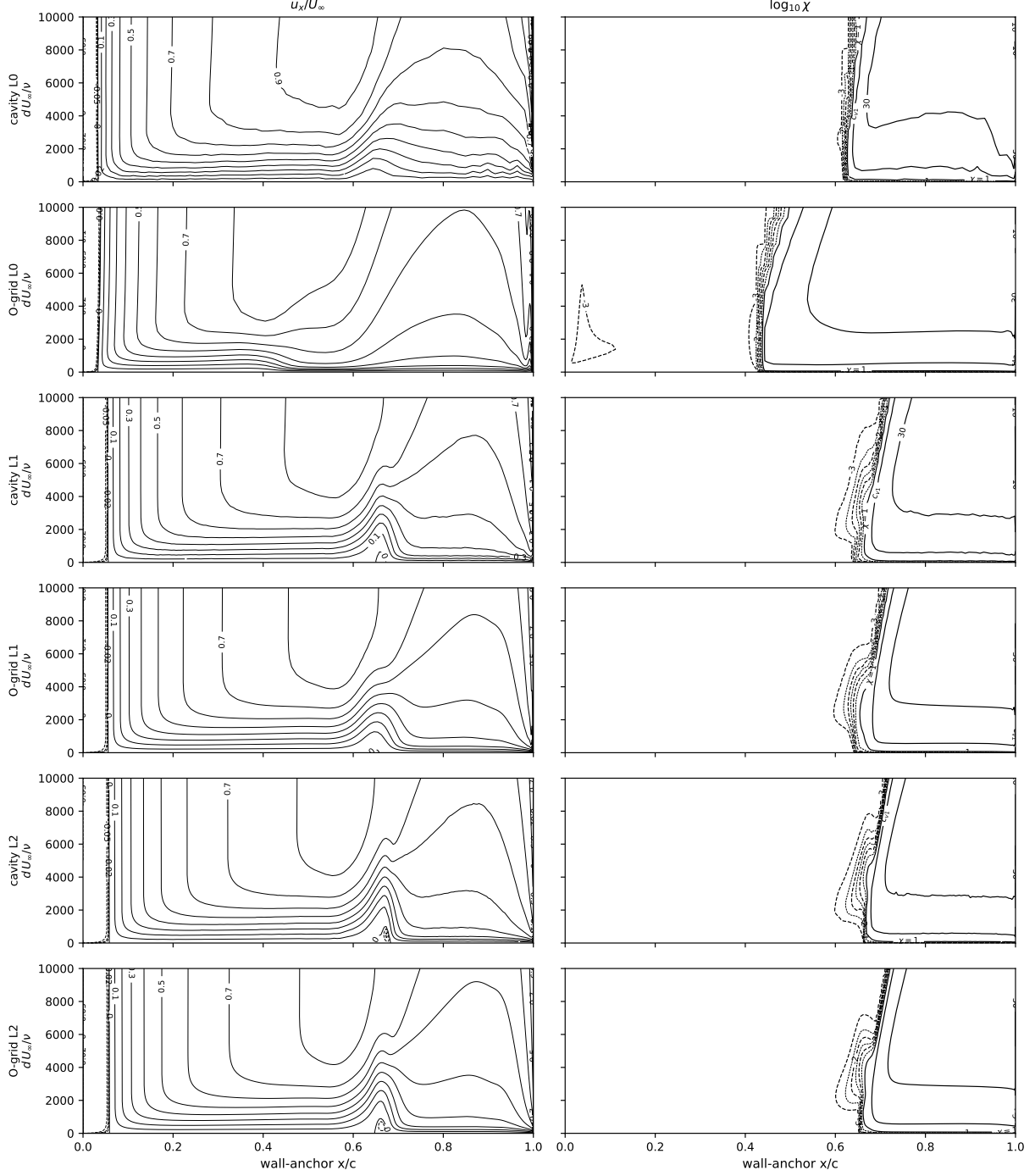












2.3 Eppler 387 at $Re = 2 \times 10^5$

$\alpha \in \{0^\circ, 2^\circ, 5^\circ, 7^\circ\}$ on both families at L0–L2 plus six extension incidences ($-2^\circ, 1^\circ, 3^\circ, 4^\circ, 6^\circ, 8.5^\circ$) on the finest pair; same seed. Figure 13: drag polar against the LTPT measurement (McGhee et al., TM-4062), the e^9 reference, fully turbulent SA, and the digitized $\gamma-Re_\theta$ and SA-BC curves (-6.4% to $+3.7\%$ over $\alpha = 0^\circ$ – 7°). Figure 14: bubble stations against the oil flow, Selig et al., Cole–Mueller, and the e^9 references—reattachment 0.02 – $0.04c$ late at $\alpha \leq 5^\circ$, at the edge of bubble collapse at 7° . Figures 15–16: surface suites. Table 5: forces. Table 6: bubble stations. Table 7: at L2 lift within 0.5% , separation

0.030–0.037 c ahead of and reattachment 0.033–0.045 c behind the SA-AI stations.

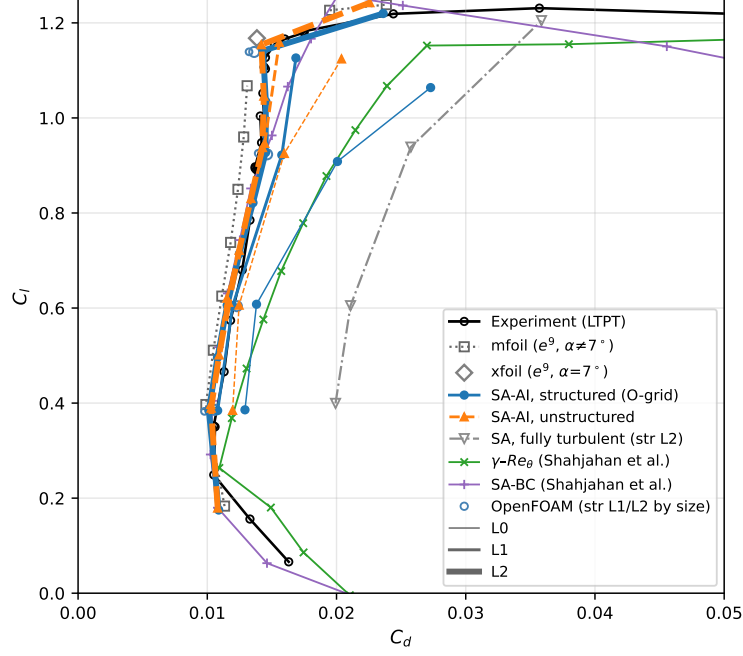


Figure 13: Eppler 387 drag polar, $Re = 2 \times 10^5$. Fully turbulent SA ($\chi_\infty = 3$) 77–97% above the SA-AI drag in the bucket, +151% at $\alpha = 7^\circ$; mesh families agree to $\Delta C_d \leq 2.4 \times 10^{-4}$ over $\alpha = -2^\circ$ – 7° and 1.0×10^{-3} at 8.5° ; OpenFOAM port omits the bypass, worth -6 to -10 counts at $\alpha = 0$ – 5° .

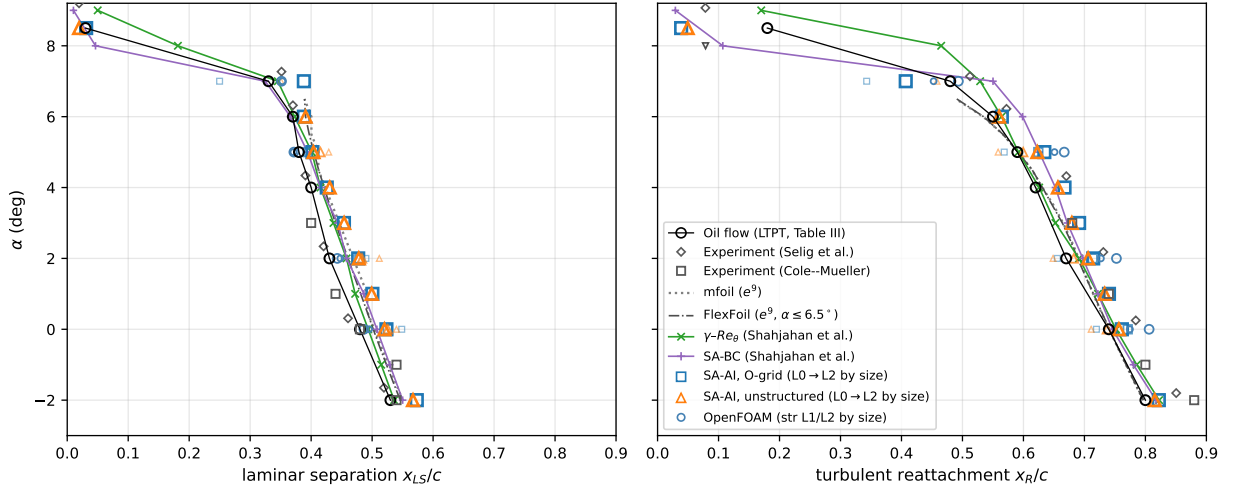


Figure 14: Eppler 387 bubble stations, $Re = 2 \times 10^5$; signed- C_f zero crossings against oil flow (TM-4062 Table III), Selig, Cole–Mueller, e^9 . At $\alpha = 7^\circ$ XFOIL $C_l = 1.17$, mfoil $C_l = 0.928$.

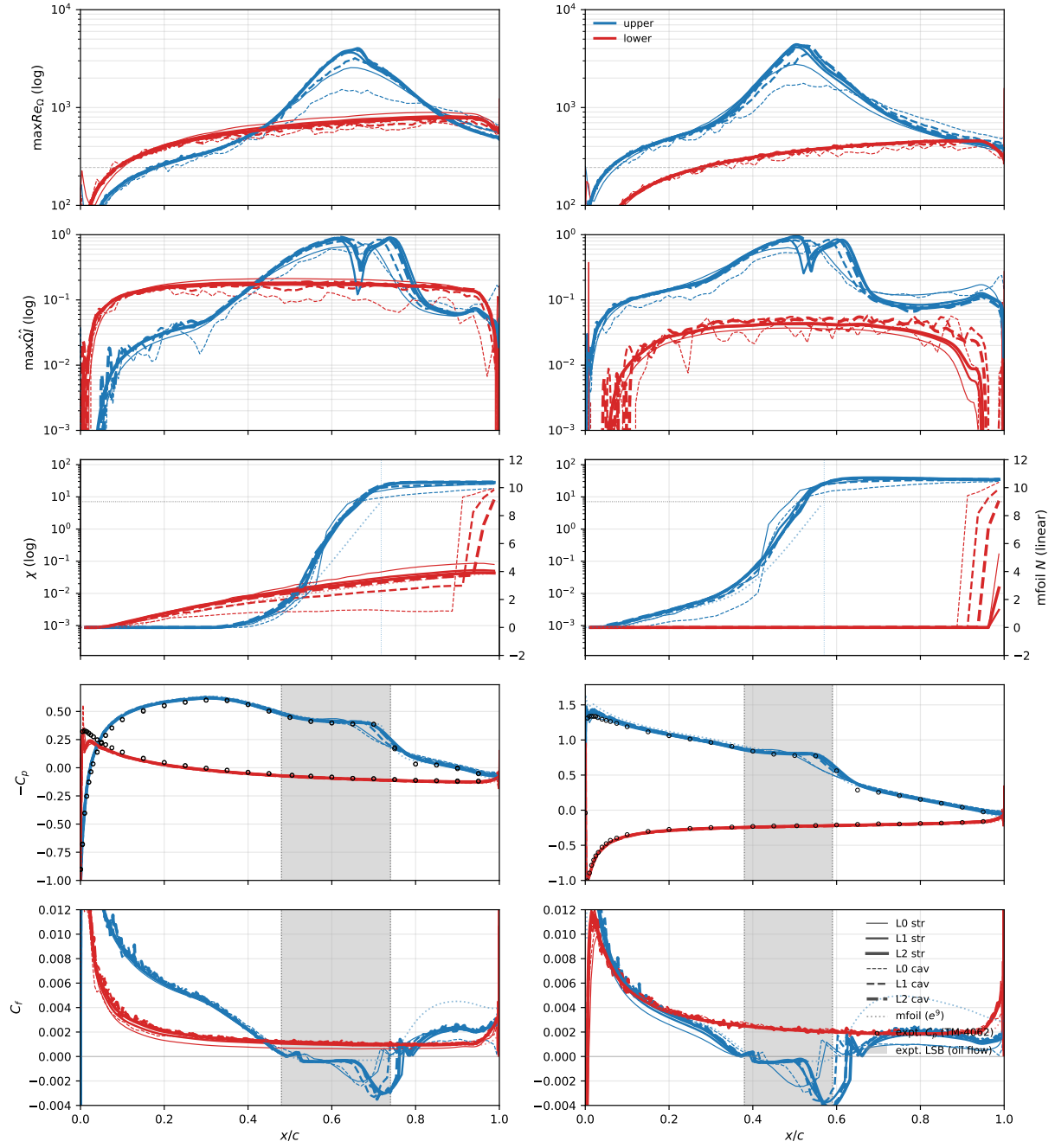


Figure 15: Eppler 387, $\alpha = 0^\circ, 5^\circ$, $Re = 2 \times 10^5$.

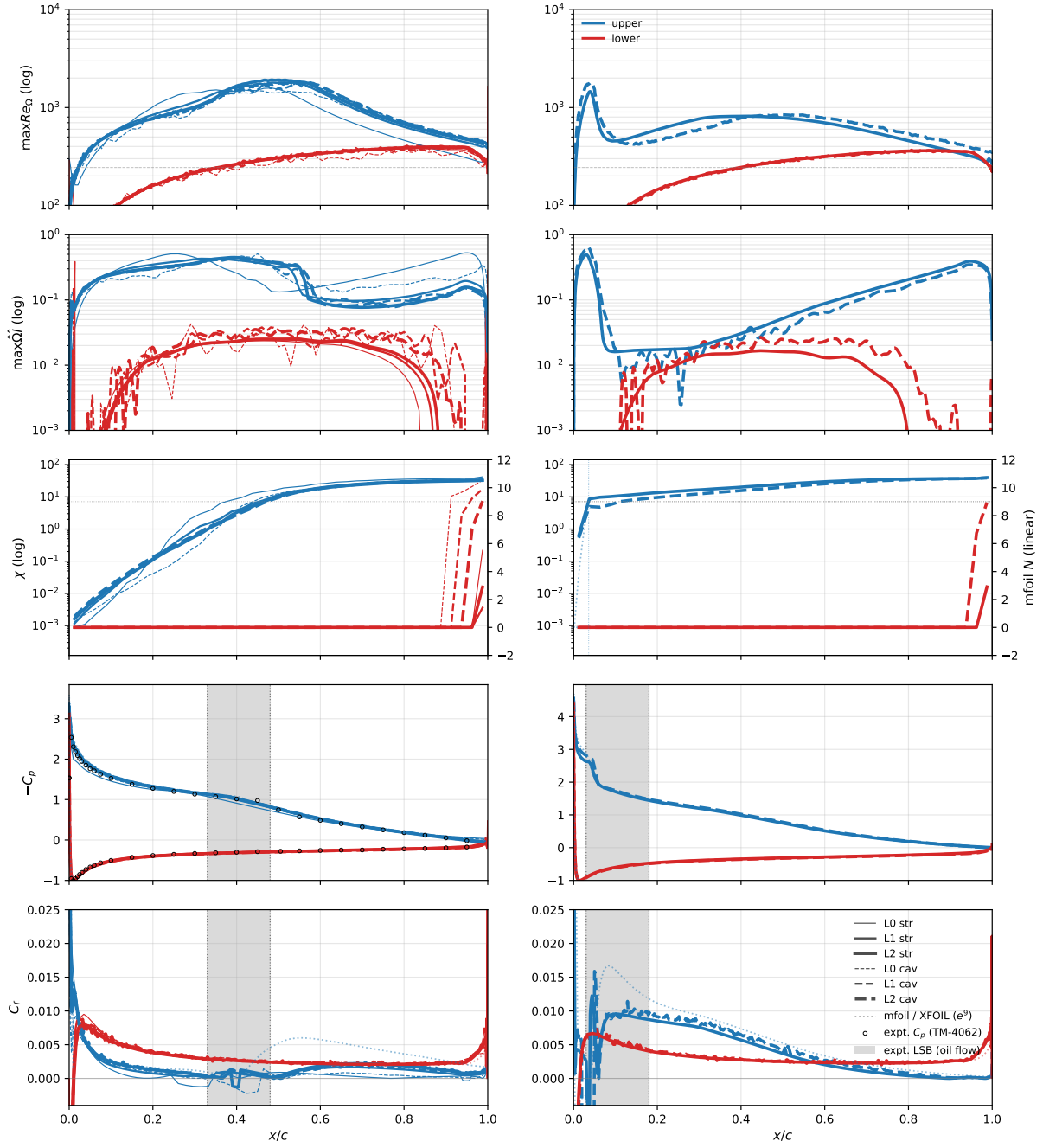


Figure 16: Eppler 387, $\alpha = 7^\circ, 8.5^\circ$, $Re = 2 \times 10^5$.

Table 5: Eppler 387, $Re = 2 \times 10^5$: computed forces (the -2° – 8.5° extension incidences exist on the finest pair only).

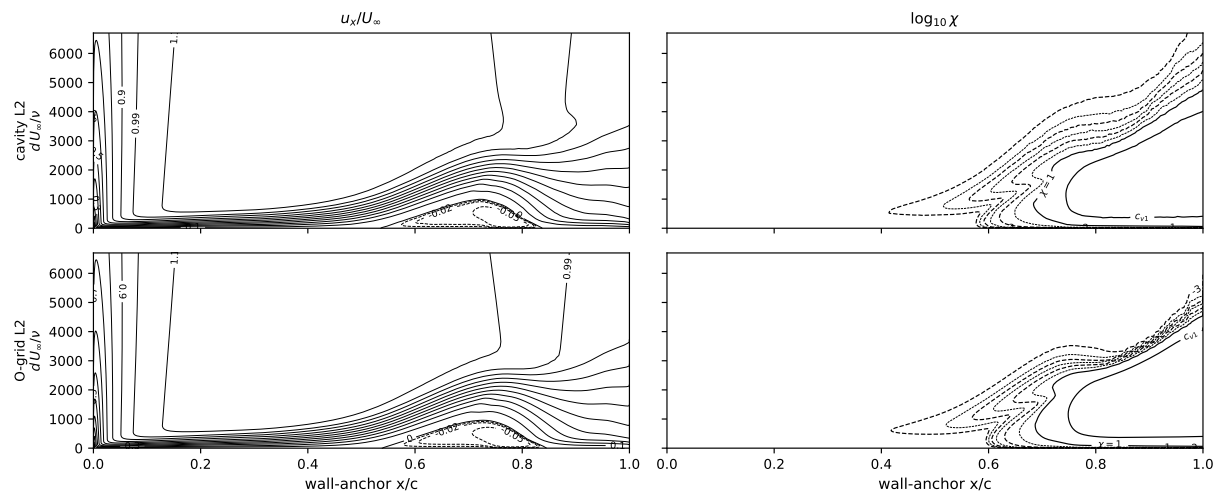
α	grid	structured C_L	O-grid C_D	unstructured C_L	cavity C_D
−2	L2	0.1752	0.01089	0.1791	0.01080
0	L0	0.3859	0.01292	0.3847	0.01196
0	L1	0.3843	0.01080	0.3969	0.01026
0	L2	0.3844	0.01010	0.3894	0.01030
1	L2	0.4957	0.01078	0.5012	0.01092
2	L0	0.6079	0.01381	0.6061	0.01246
2	L1	0.6048	0.01196	0.6207	0.01152
2	L2	0.6059	0.01154	0.6124	0.01167
3	L2	0.7146	0.01247	0.7218	0.01245
4	L2	0.8218	0.01354	0.8304	0.01341
5	L0	0.9083	0.02007	0.9253	0.01596
5	L1	0.9218	0.01575	0.9465	0.01444
5	L2	0.9274	0.01451	0.9375	0.01428
6	L2	1.0346	0.01453	1.0456	0.01439
7	L0	1.0639	0.02728	1.1241	0.02038
7	L1	1.1262	0.01686	1.1593	0.01557
7	L2	1.1415	0.01431	1.1535	0.01419
8.5	L2	1.2190	0.02310	1.2611	0.02150

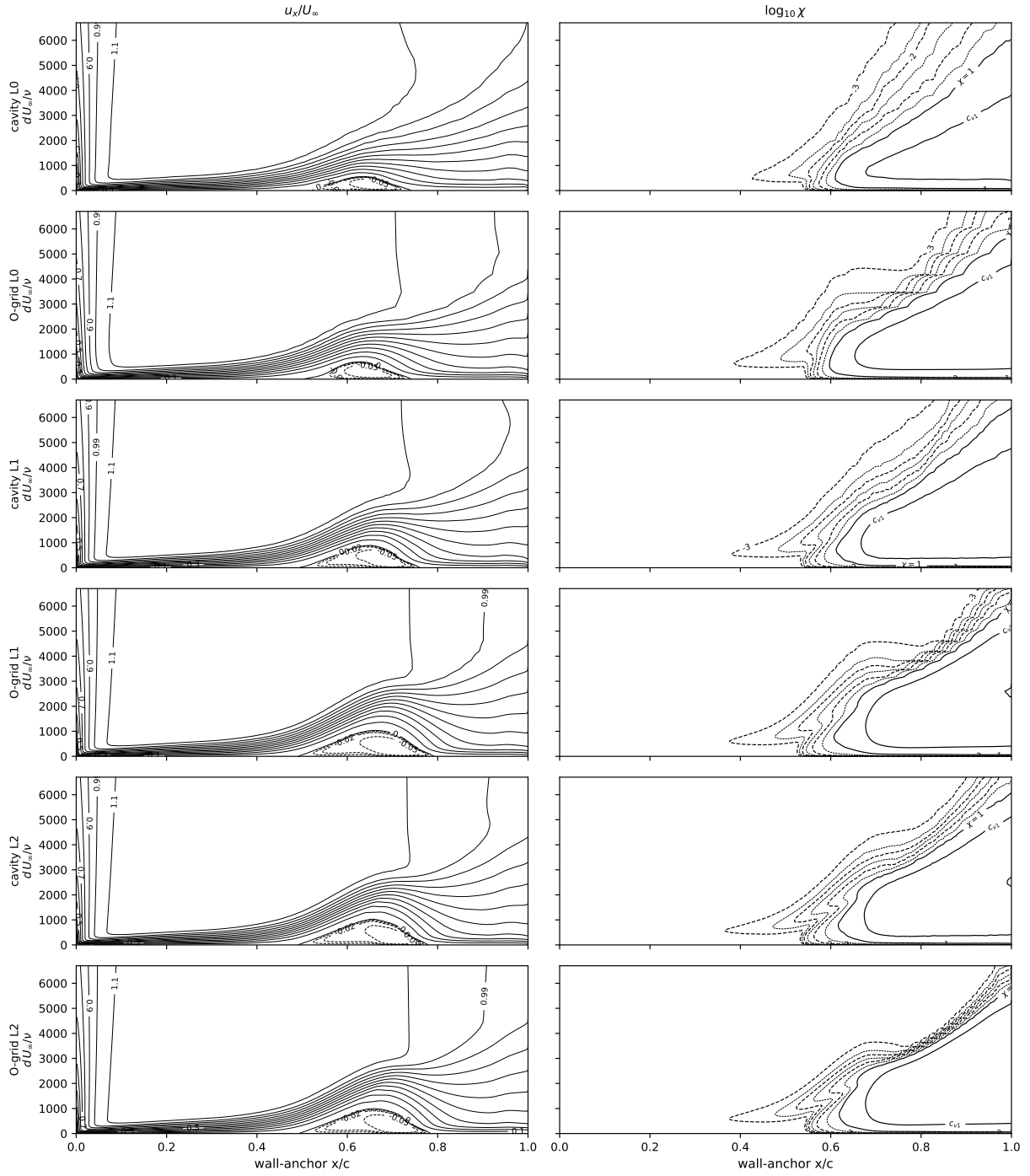
Table 6: Eppler 387, $Re = 2 \times 10^5$: computed upper-surface laminar-separation and turbulent-reattachment stations (signed- C_f zero crossings; missing entries transition ahead of separation and have no bubble). The measured oil-flow reattachments (TM-4062 Table III) at $\alpha = 0^\circ/2^\circ/5^\circ$ are 0.74/0.67/0.59 c .

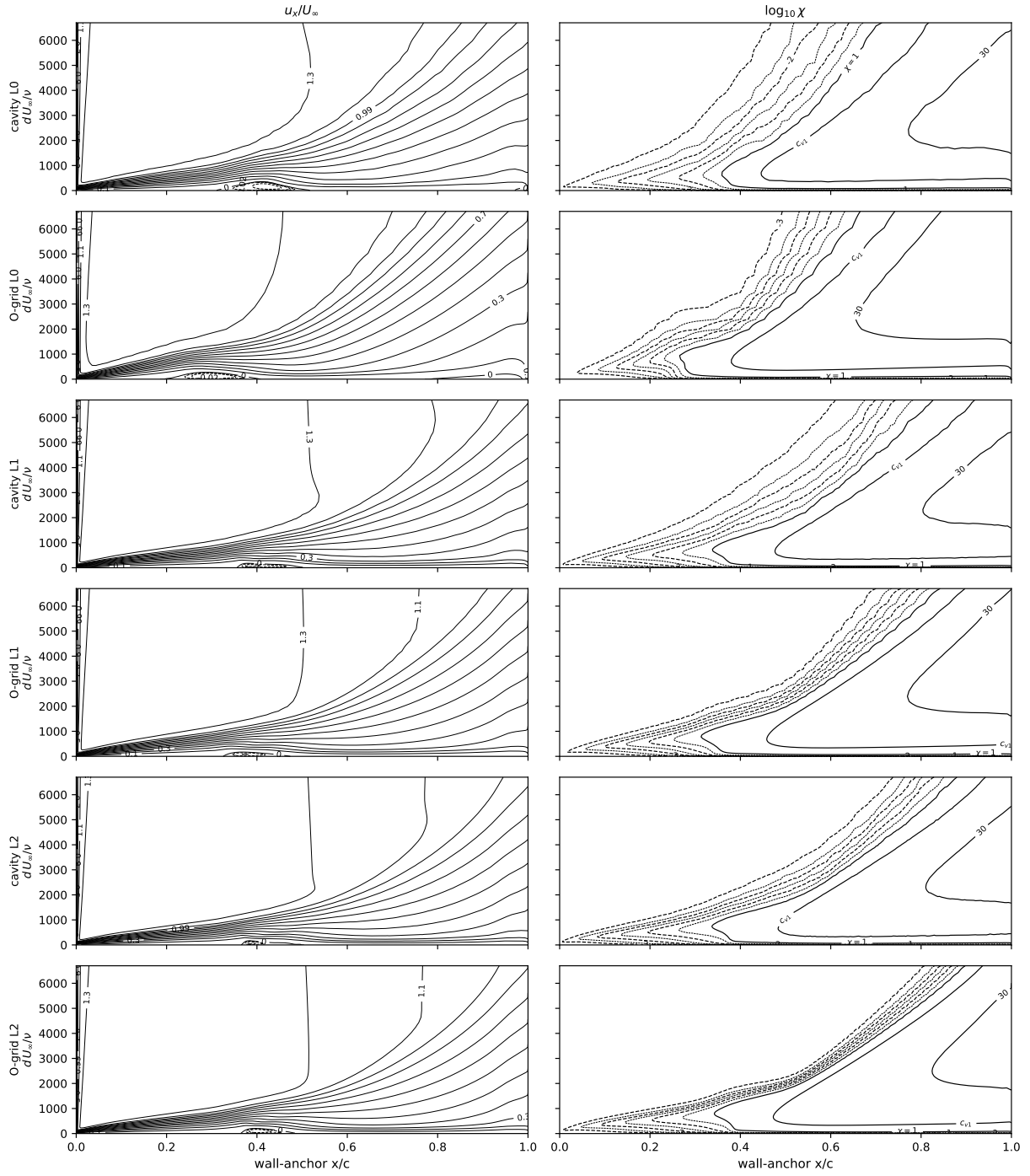
α	grid	structured x_{LS}/c	O-grid x_R/c	unstructured x_{LS}/c	cavity x_R/c
−2	L2	0.573	0.822	0.567	0.815
0	L0	0.549	0.720	0.540	0.711
0	L1	0.526	0.771	0.526	0.737
0	L2	0.523	0.761	0.520	0.757
1	L2	0.500	0.740	0.499	0.733
2	L0	0.490	0.656	0.512	0.649
2	L1	0.479	0.713	0.483	0.682
2	L2	0.477	0.714	0.478	0.706
3	L2	0.454	0.691	0.454	0.679
4	L2	0.426	0.667	0.430	0.657
5	L0	0.393	0.568	0.429	0.558
5	L1	0.394	0.624	0.415	0.600
5	L2	0.402	0.634	0.404	0.623
6	L2	0.388	0.565	0.391	0.560
7	L0	0.250	0.343	0.351	0.458
7	L1	—	—	—	—
7	L2	0.388	0.407	—	—
8.5	L2	0.031	0.039	0.019	0.049

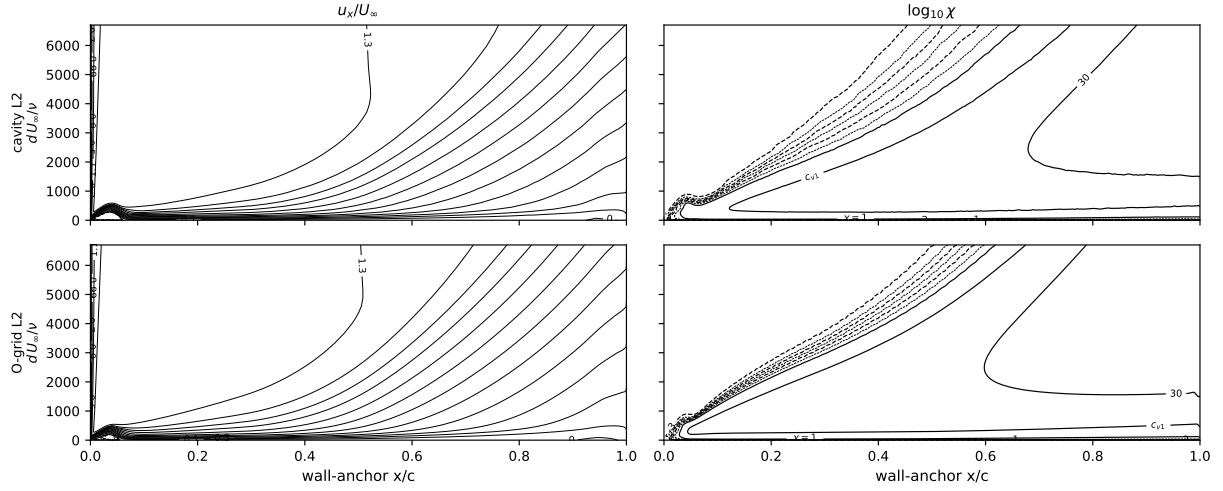
Table 7: Eppler 387: the independent OpenFOAM implementation on the structured L0, L1 and L2 grids (fronts at the near-wall $\chi = 1$ convention, stations at the signed- C_f zero crossings; the port omits the f_{v1} bypass).

α	grid	C_L	C_D	x_{tr}^{up}	x_{LS}	x_R
0	L0	0.3887	0.01167	0.638	0.493	0.784
0	L1	0.3826	0.00978	0.605	0.495	0.775
0	L2	0.3857	0.01058	0.621	0.486	0.806
2	L0	0.6133	0.01406	0.568	0.447	0.739
2	L1	0.6037	0.01174	0.543	0.448	0.727
2	L2	0.6039	0.01229	0.564	0.443	0.752
5	L0	0.9292	0.01665	0.457	0.344	0.654
5	L1	0.9256	0.01394	0.468	0.371	0.651
5	L2	0.9234	0.01463	0.487	0.372	0.667
7	L0	1.1173	0.01817	0.139	–	0.201
7	L1	1.1391	0.01323	0.329	0.353	0.453
7	L2	1.1388	0.01363	0.379	0.351	0.493









2.4 Eppler 387 Reynolds sweep at $\alpha = 5^\circ$

$Re \in \{6 \times 10^4, 10^5, 3 \times 10^5, 4.6 \times 10^5\}$ on the same grids (24 further solutions). Figure 17: the refinement fan closes onto the measurement at $2\text{--}4.6 \times 10^5$ and opens at 10^5 , the model's bursting boundary, one Reynolds step above the measurement's (6×10^4). Figures 18–19: surface suites. Table 8: L2 coefficients.

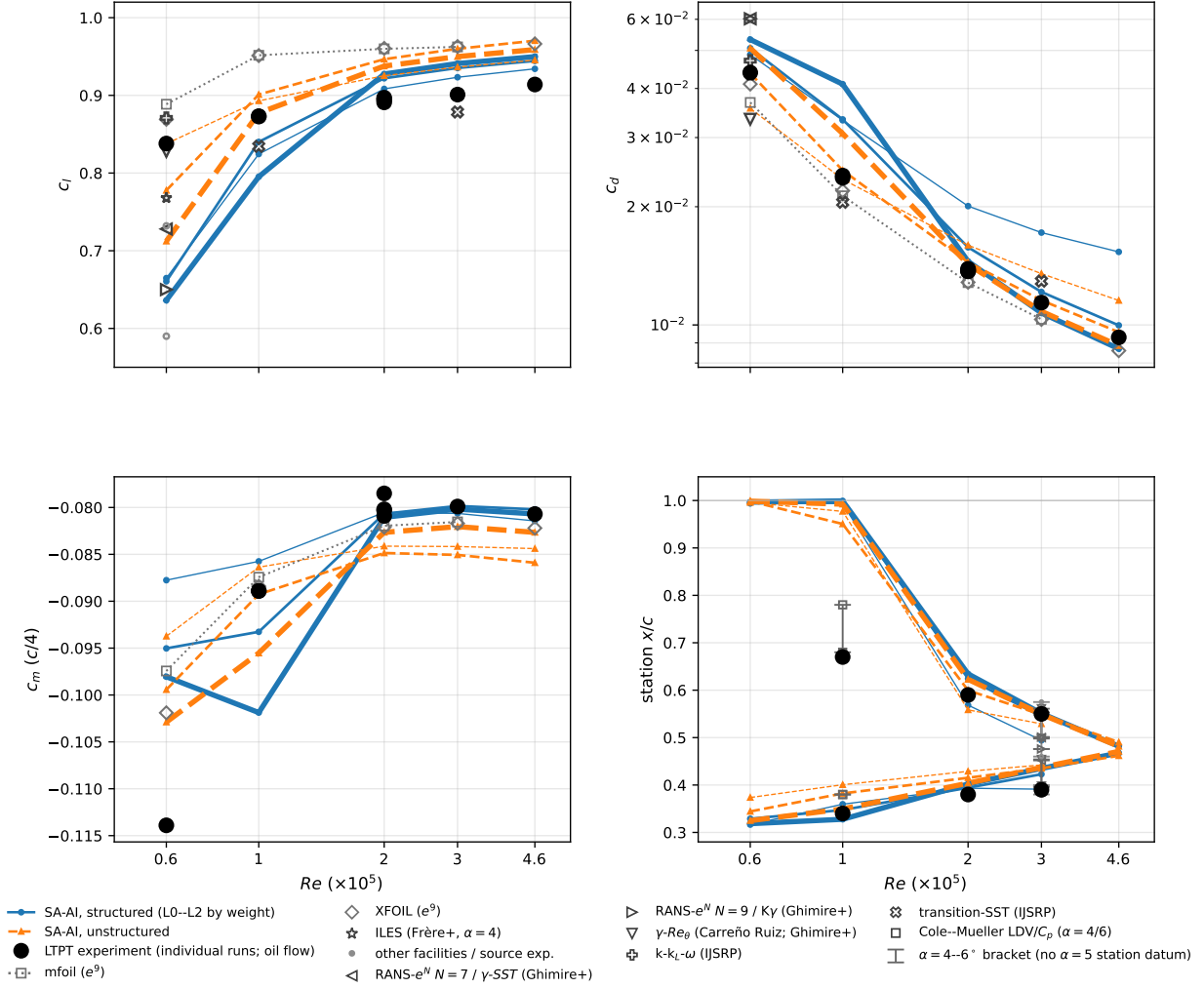


Figure 17: Reynolds-sweep forces and stations, $\alpha = 5^\circ$, $Re = 6 \times 10^4 - 4.6 \times 10^5$. LTPT runs 1/2/4/1/1 at $0.6/1/2/3/4.6 \times 10^5$.

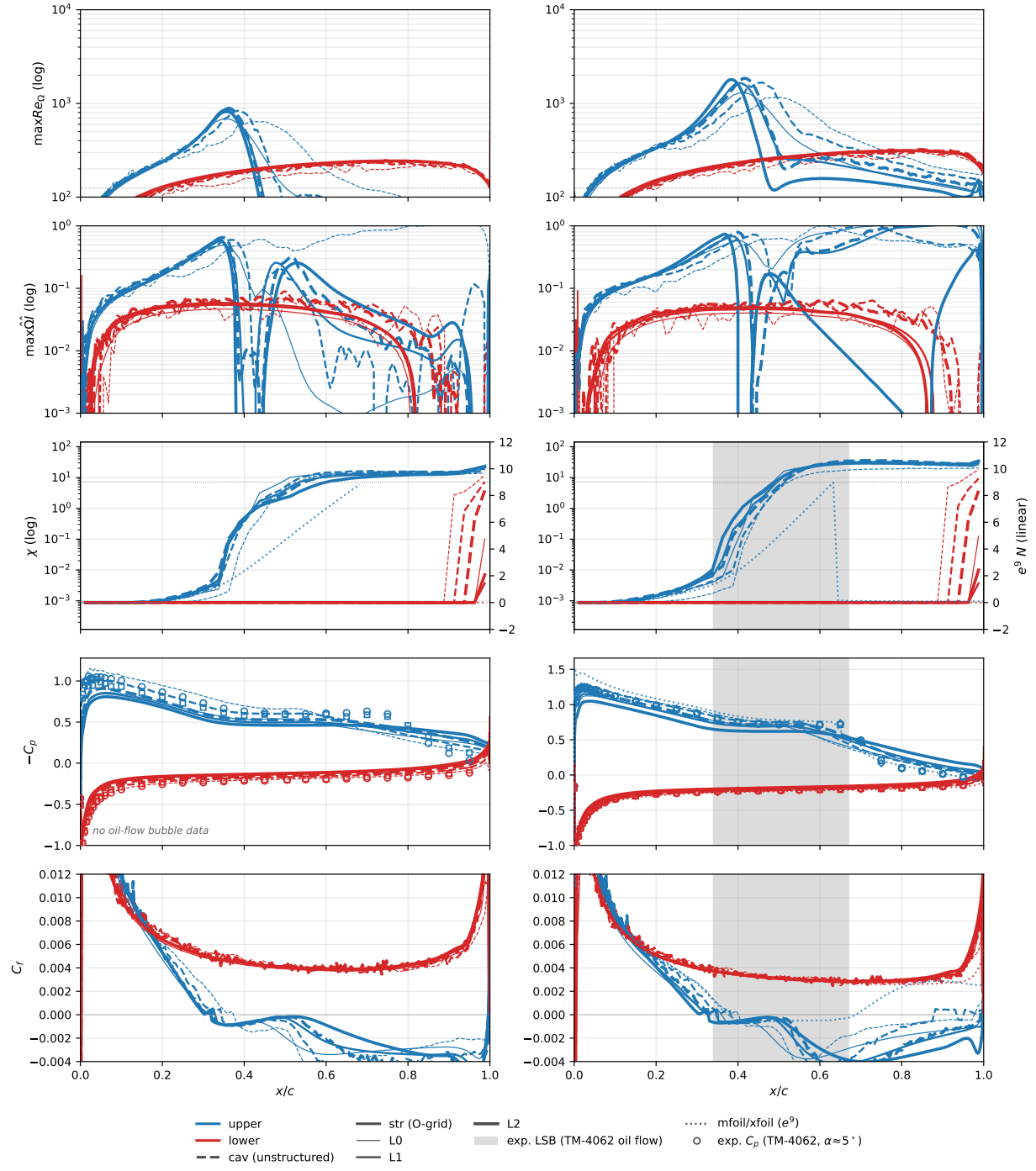


Figure 18: Reynolds sweep, $Re = 6 \times 10^4, 10^5$.

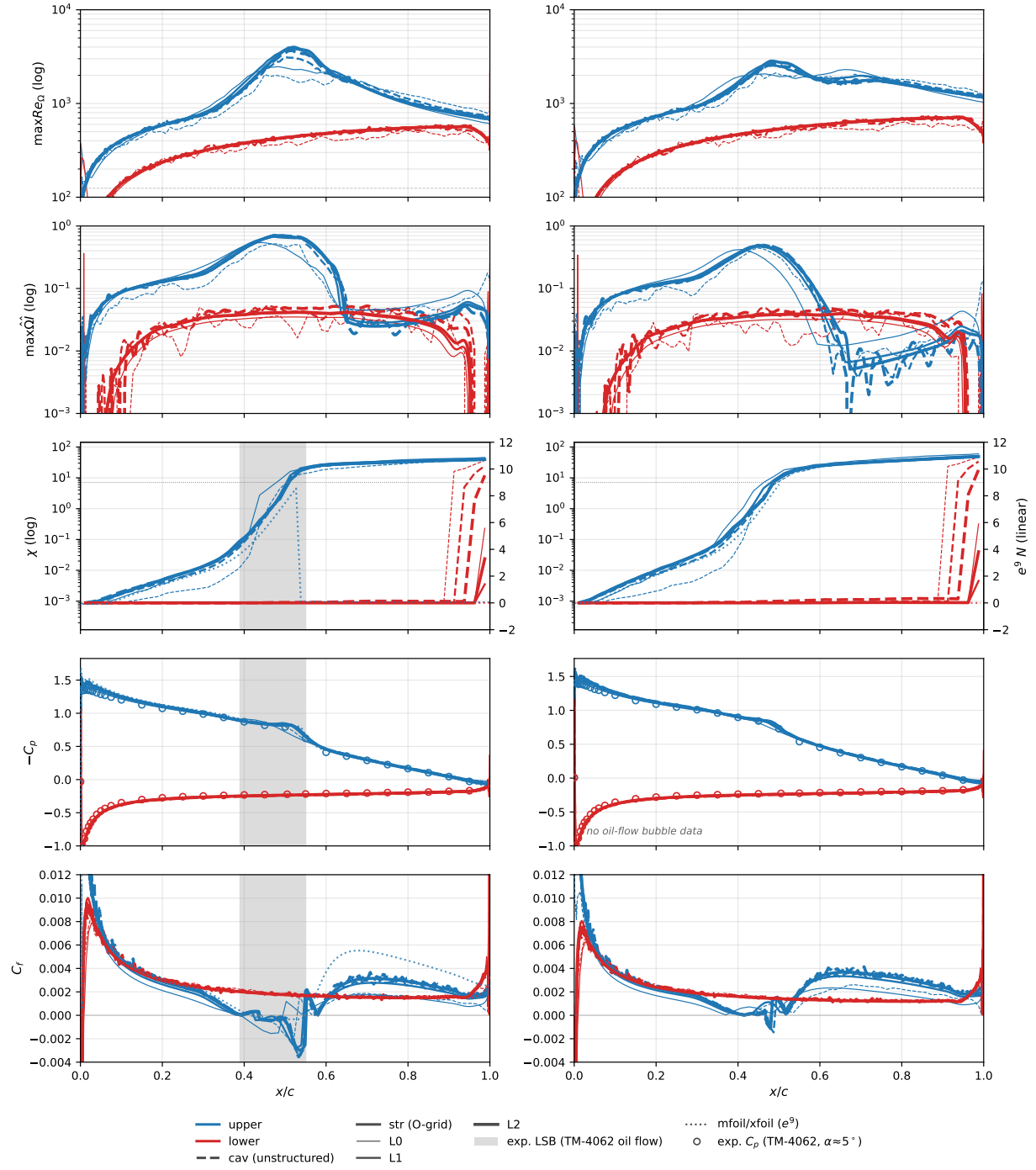
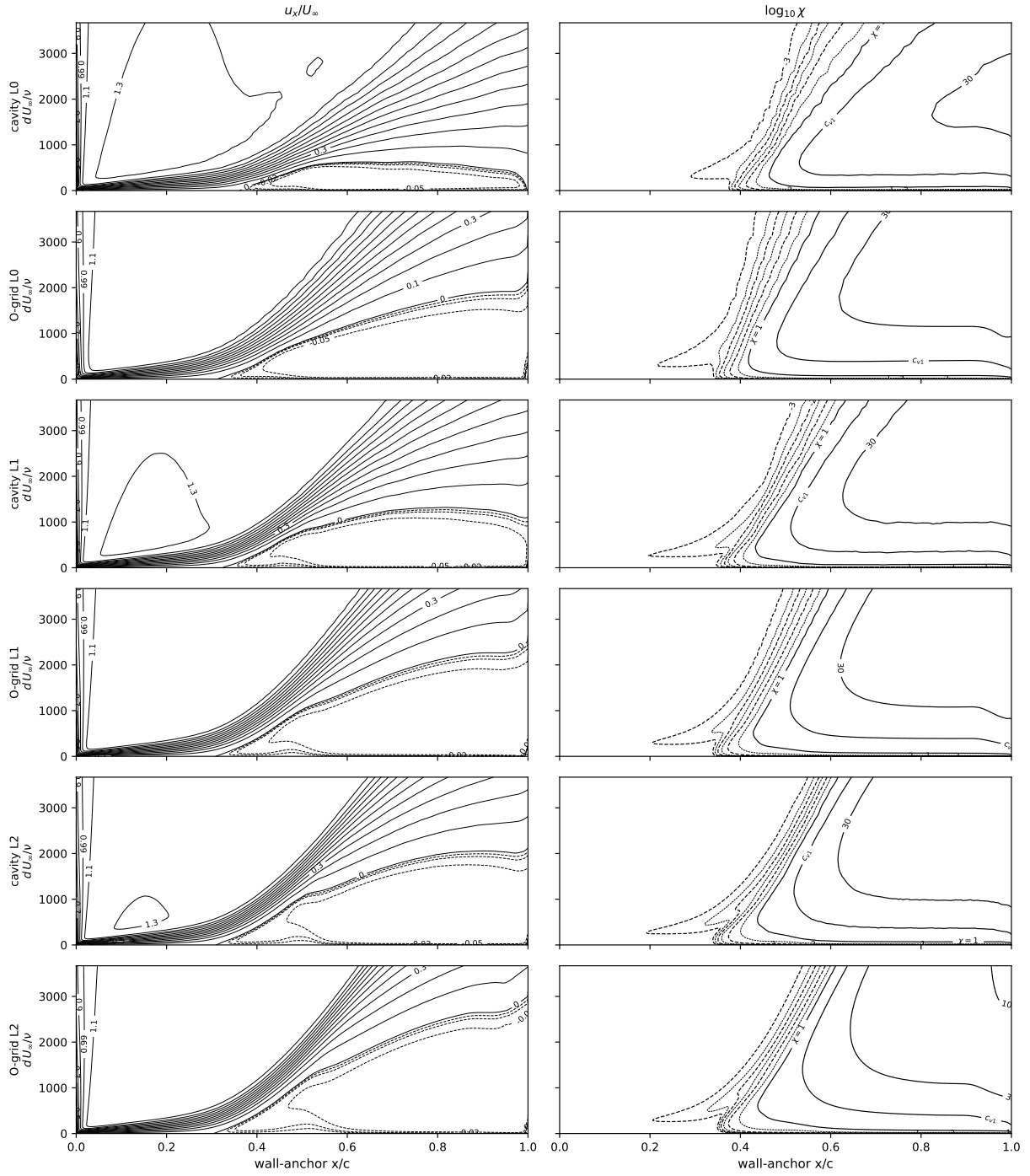


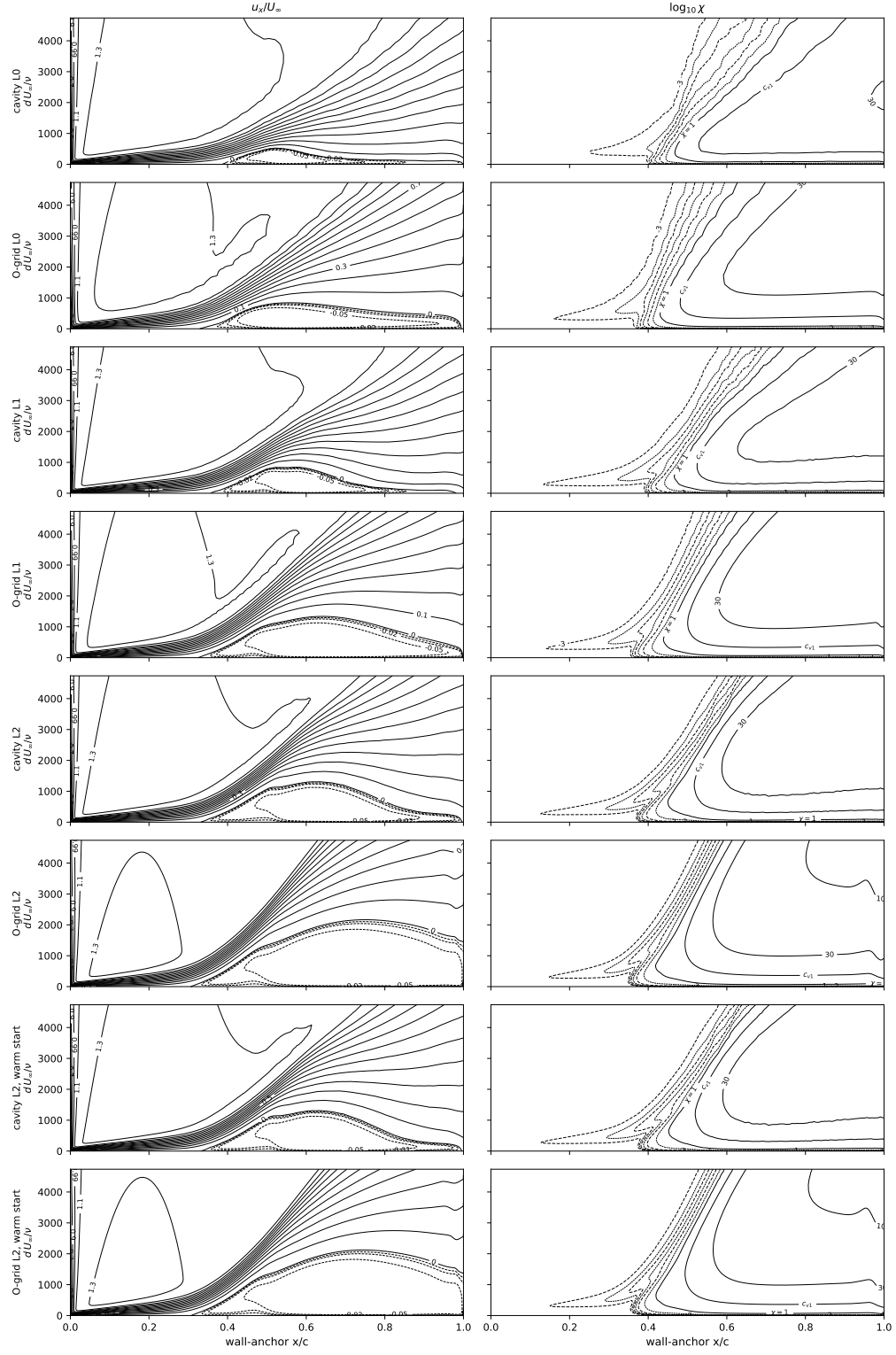
Figure 19: Reynolds sweep, $Re = 3, 4.6 \times 10^5$.

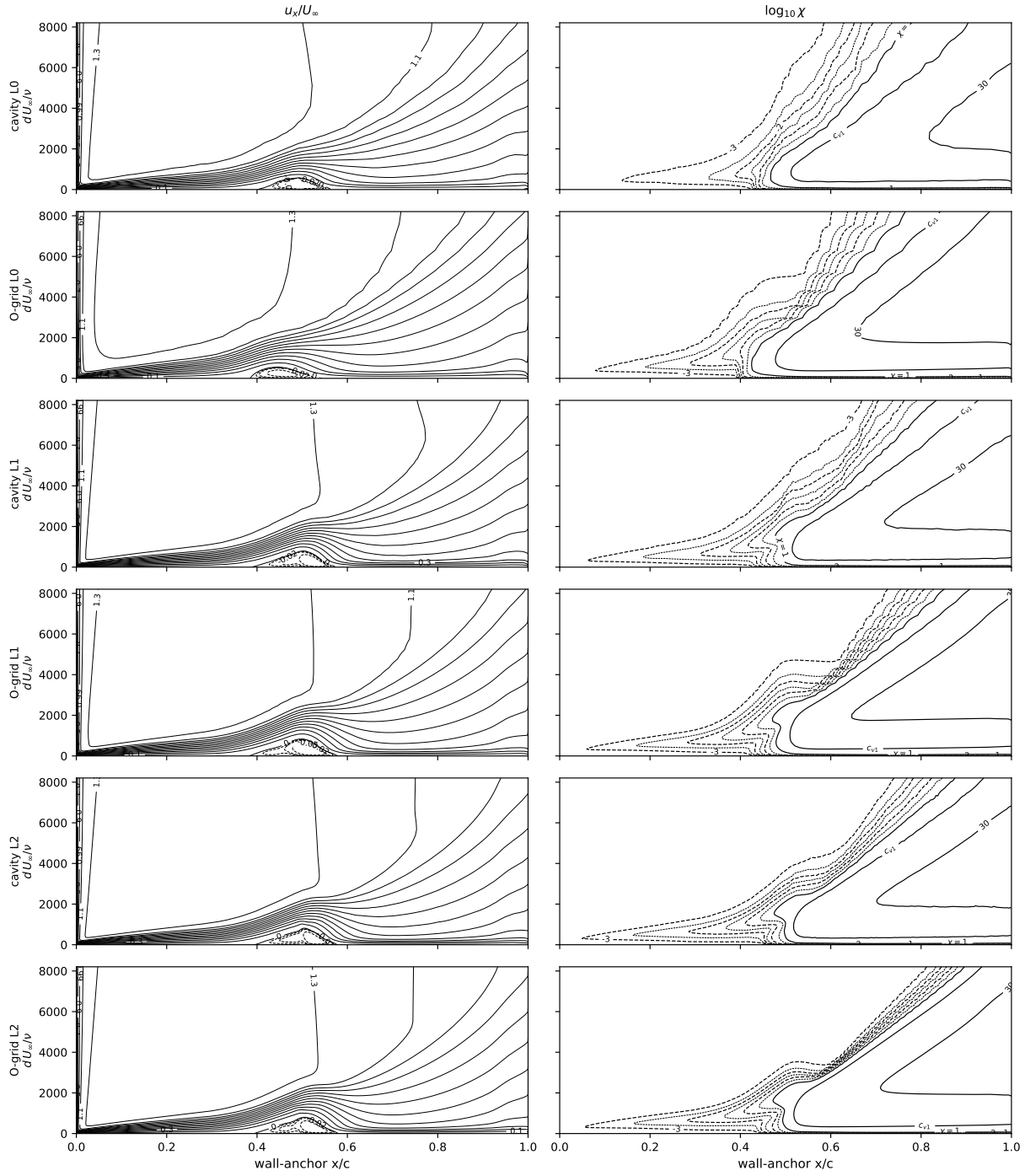
Table 8: Eppler 387 at $\alpha = 5^\circ$ across the Reynolds sweep: L2 section coefficients against the e^9 panel reference and the LTPT measurement (read at the tabulated angle nearest 5°). Computed moments transferred to the quarter chord.

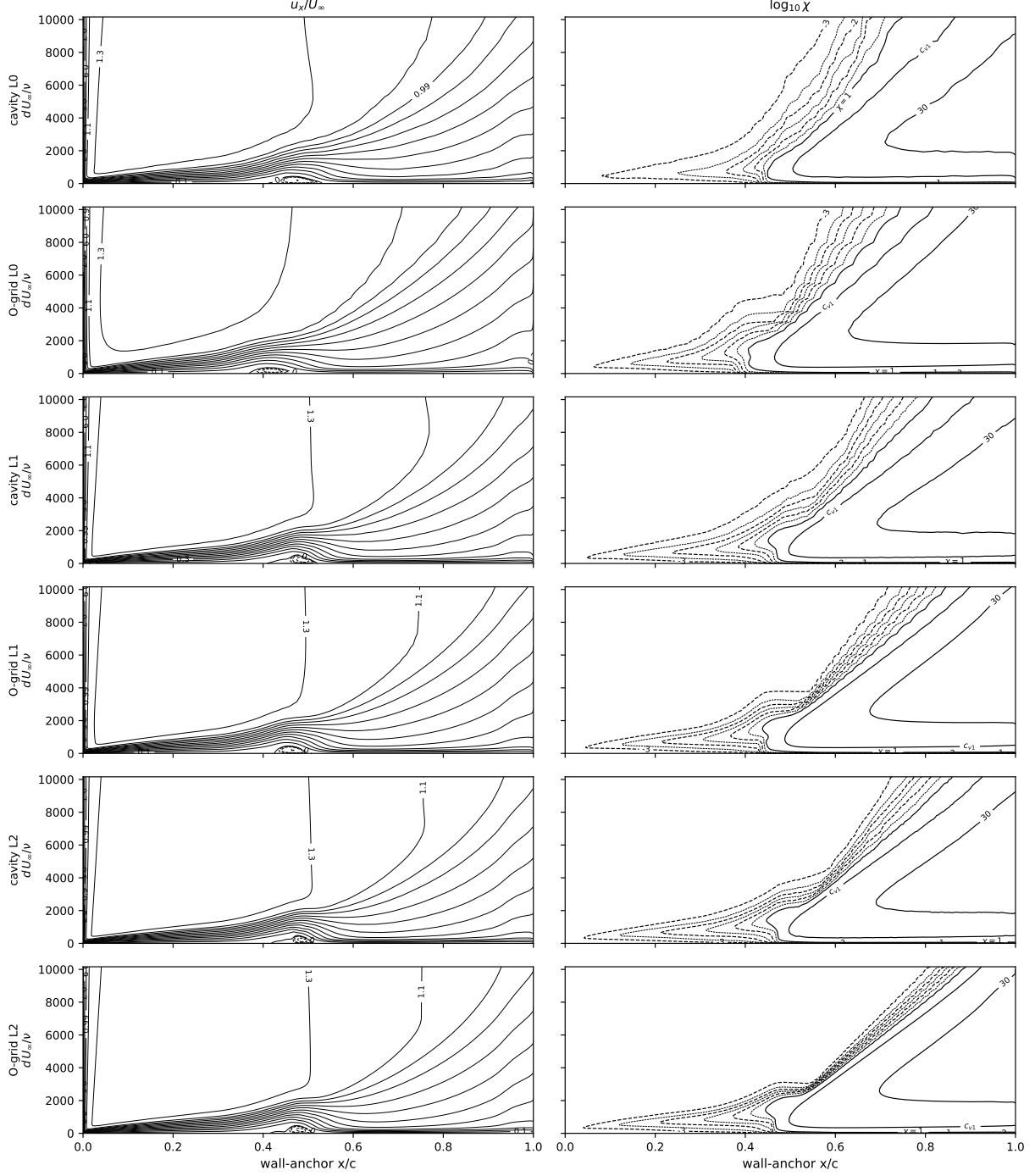
Re	SA-AI, structured L2			SA-AI, unstructured L2			e^9 panel			Exp (LTPT)		
	c_l	c_d	c_m	c_l	c_d	c_m	c_l	c_d	c_m	c_l	c_d	c_m
6×10^4	0.636	0.0533	-0.098	0.712	0.0506	-0.103	0.869 [‡]	0.0411 [‡]	-0.102 [‡]	0.838	0.0439	-0.114
1×10^5	0.795	0.0410	-0.102	0.877	0.0307	-0.096	0.952	0.0213	-0.087	0.873	0.0237	-0.089
2×10^5	0.927	0.0145	-0.081	0.938	0.0143	-0.083	0.960	0.0128	-0.082	0.891	0.0138	-0.081
3×10^5	0.941	0.0107	-0.080	0.950	0.0108	-0.082	0.962	0.0103	-0.082	0.901	0.0114	-0.080
4.6×10^5	0.950	0.0087	-0.081	0.959	0.0089	-0.083	0.966	0.0086	-0.082	0.914	0.0093	-0.081

[‡]XFOIL, whose forces sit closest to the measurement, is the reference at this bistable condition; mfoil gives $c_l = 0.889$, $c_d = 0.0368$, $c_m = -0.097$, and FlexFoil sides with XFOIL ($c_l = 0.868$, $c_d = 0.0406$, $c_m = -0.102$).









2.5 Cylinder drag crisis, $Re_D = 1\text{--}10^{10}$

A steady-branch traverse over ten decades at a single seed ($\chi_\infty = 1.1 \times 10^{-2}$; Mack's map at $Tu = 0.2\%$), swept as one upward continuation ladder on matched quasi-two-dimensional O-grids ($y^+ \lesssim 1$; a large-domain laminar grid for $Re_D \leq 10^3$, a 1200×148 working grid, and 1600×170 fine grids for the supercritical and ultra range), twenty-five points $Re_D = 1\text{--}10^{10}$. Every case lands on the symmetric branch, $|C_L| < 2 \times 10^{-6}$.

Figure 20: the crisis falls between $Re_D = 3 \times 10^5$ ($C_d = 0.773$) and 1.78×10^6 (0.228); the supercritical floor

settles near $C_d = 0.21\text{--}0.23$ ($1.78 \times 10^6\text{--}10^7$) and stays flat to 10^{10} . Figure 21: transition and separation angles. Figure 22: fields across the traverse. Table 9: the up-ladder C_d . Below shedding onset ($Re_D \approx 46$) the steady solution is physical: $C_d = 10.67$ at $Re_D = 1$, 2.797 at 10 (-1.7% vs Dennis & Chang).

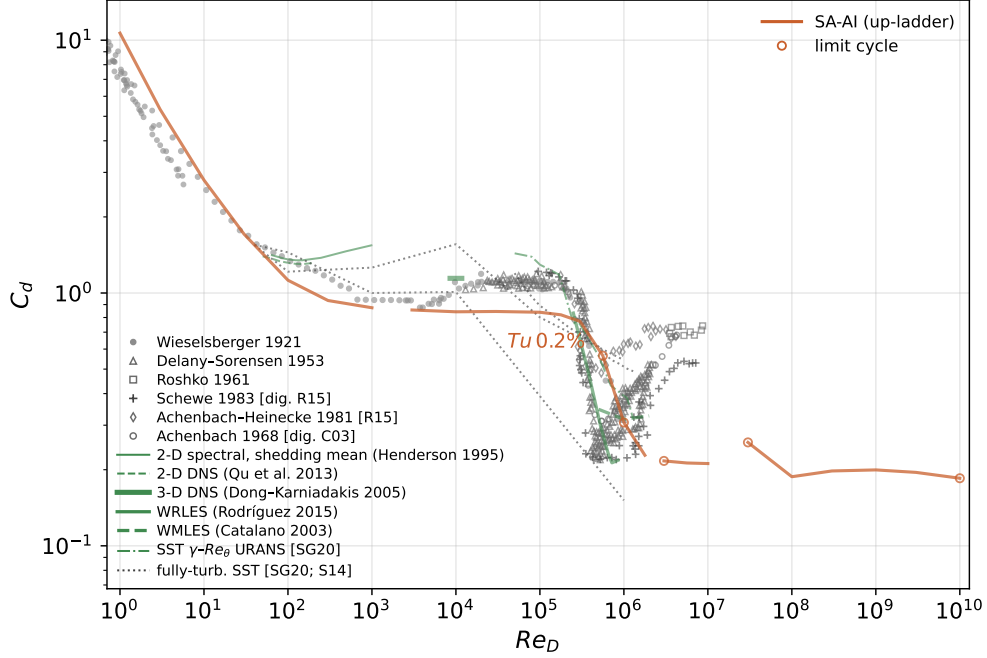


Figure 20: Cylinder drag crisis, $Re_D = 1\text{--}10^{10}$; single seed $Tu = 0.2\%$, up-ladder (clean systematic sweep, matched Re_D grid). Symbols experiments, lines computations; open rings mark the cases the steady monitor did not accept.

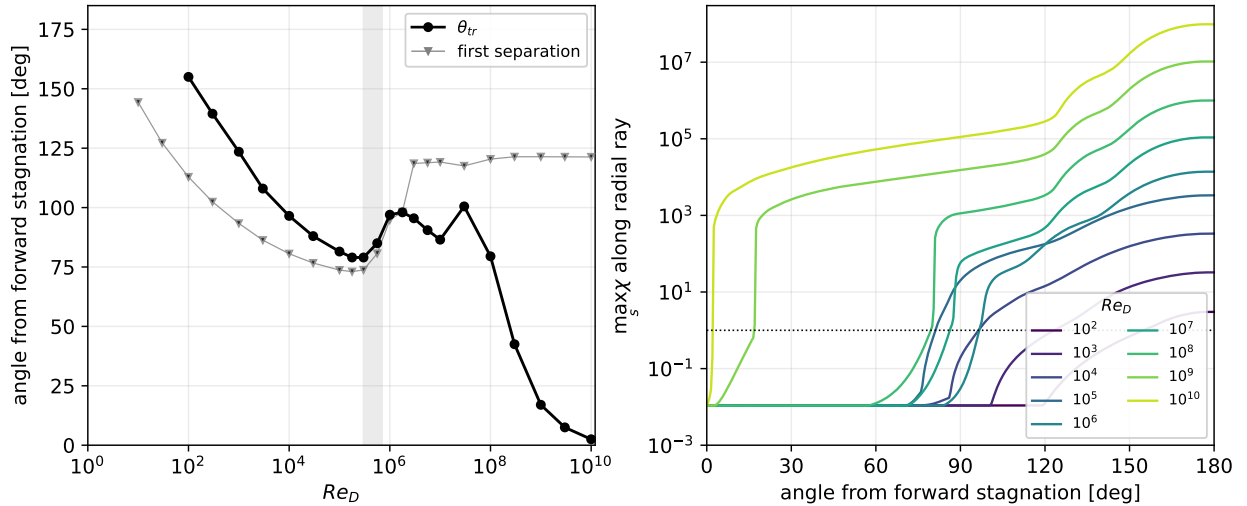


Figure 21: $Tu = 0.2\%$, $Re_D = 1\text{--}10^{10}$, up-ladder. (a) transition angle θ_{tr} ($\chi = 1$) and first wall separation vs Re_D (below the crisis θ_{tr} is the wake $\chi = 1$ front); the front marches from $\sim 100^\circ$ to $\sim 2^\circ$ by 10^{10} . (b) $\max_s \chi(\theta)$ per decade $10^2\text{--}10^{10}$.

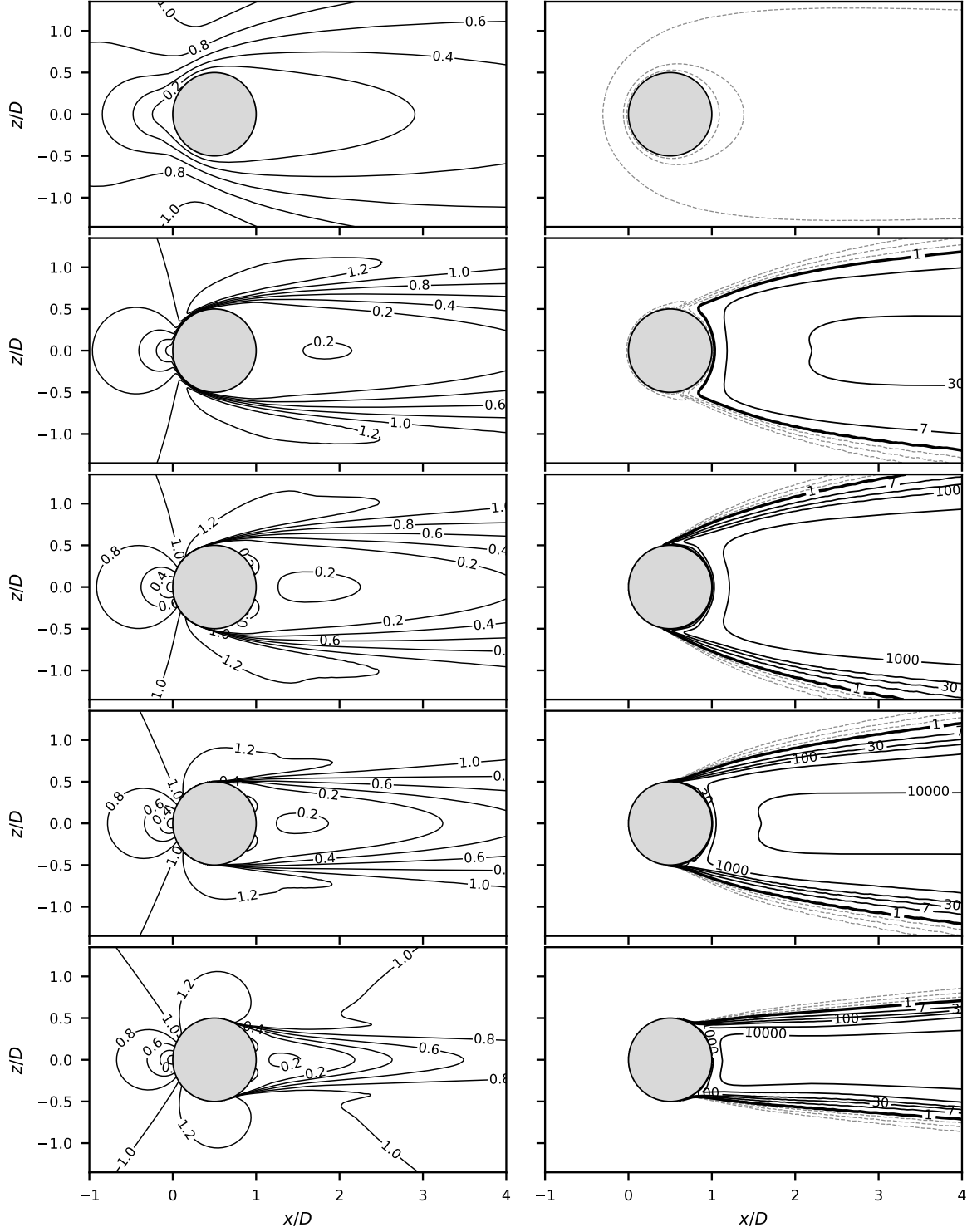


Figure 22: Drag-crisis fields, $Re_D = 10, 10^3, 10^5, 5 \times 10^5, 2 \times 10^6$ (rows, up-ladder $Tu = 0.2\%$); columns $|\mathbf{u}|/U_\infty$ and χ (dashed $\chi < 1$, solid $\chi = 1$, c_{v1} , 30, 10^2 – 10^4). $C_d = 2.80, 0.87, 0.84, 0.62, 0.22$.

Table 9: Cylinder steady drag-crisis traverse, systematic $Tu = 0.2\%$ up-ladder (C_d median over the tail window, [peak-to-peak] bracketed where the steady monitor did not accept; θ_{tr} the $\chi = 1$ transition angle, θ_{sep} the first separation; $|C_L| < 2 \times 10^{-6}$ throughout).

Re_D	grid	C_d	$\theta_{tr} (^\circ)$	$\theta_{sep} (^\circ)$
1	lowre	10.665	—	—
3	lowre	5.330	—	—
10	lowre	2.797	—	144.3
30	lowre	1.710	—	127.2
100	lowre	1.123	155.0	112.9
300	lowre	0.934	139.5	102.4
1000	lowre	0.874	123.5	93.4
3000	pilot	0.857	108.0	86.3
10^4	pilot	0.843	96.5	80.6
3×10^4	pilot	0.845	88.0	76.7
10^5	pilot	0.839	81.5	73.7
1.78×10^5	pilot	0.820	79.0	73.0
3×10^5	pilot	0.773	79.0	73.8
5.62×10^5	pilot	0.565 [0.075]	85.0	80.7
10^6	pilot	0.307 [0.044]	97.0	94.8
1.78×10^6	pilot	0.228	98.0	98.3
3×10^6	highre	0.217 [0.010]	95.5	118.6
5.62×10^6	highre	0.213	90.5	118.9
10^7	highre	0.212	86.5	119.2
3×10^7	ultra	0.256 [0.171]	100.5	117.6
10^8	ultra	0.188	79.5	120.4
3×10^8	ultra	0.198	42.5	121.4
10^9	ultra	0.200	17.0	121.4
3×10^9	ultra	0.195	7.5	121.3
10^{10}	ultra	0.185 [0.009]	2.5	121.3

2.6 Daedalus wing (three-dimensional deployment)

The Daedalus human-powered-aircraft wing: half-span 17.07 m, root chord 0.917 m, aspect ratio 37.8, DAE-11/21/31 sections; $Re_{root} = 5 \times 10^5$, $\alpha \in \{4^\circ, 5^\circ, 6^\circ\}$ ($C_L \approx 1.02\text{--}1.23$), flight-quiet seed $\chi_\infty = 8.76 \times 10^{-6}$ ($N_{crit} \approx 13.6$); two mesh families (spanwise-stacked structured O-grid, 0.72–36.7M nodes; unstructured prism/tet/hex, 1.70–54.9M nodes) at three levels, eighteen solutions; the reference is AVL with strip-wise XFOIL section polars at matched local c_l and N_{crit} , trimmed to the RANS lift. Figure 23: the families agree at L2 to 0.35–0.63% in C_L and 0.7–1.7 counts in C_D , and the finest-grid drag runs 15.6/10.3/5.6 counts (7.3/4.4/2.2%) below the matched-lift reference at $\alpha = 4^\circ/5^\circ/6^\circ$. Figure 24: surface maps at $\alpha = 4^\circ$ against the e^N strips—at $\eta = 0.31$, separation within $0.017c$ and reattachment within $0.009c$ at every incidence, the $\chi = 1$ front sits $\sim 0.09c$ ahead of the strips’ N_{crit} contour, and the tip vortex trips transition to the leading edge for $\eta \gtrsim 0.992$. Table 10: totals by level ($\alpha = 5^\circ, 6^\circ$ surface maps and section sheets below).

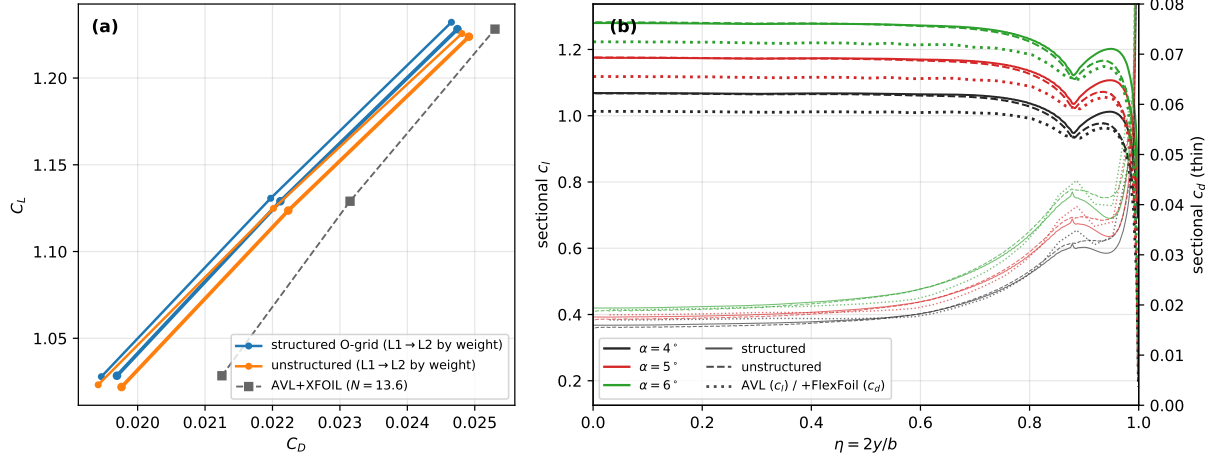


Figure 23: Daedalus polar and sections. Families agree at L1 to 0.46–0.53% in C_L and 0.4–1.6 counts in C_D , at L0 to 0.74–0.97% and 18–21 counts; taper break at $\eta = 0.88$.

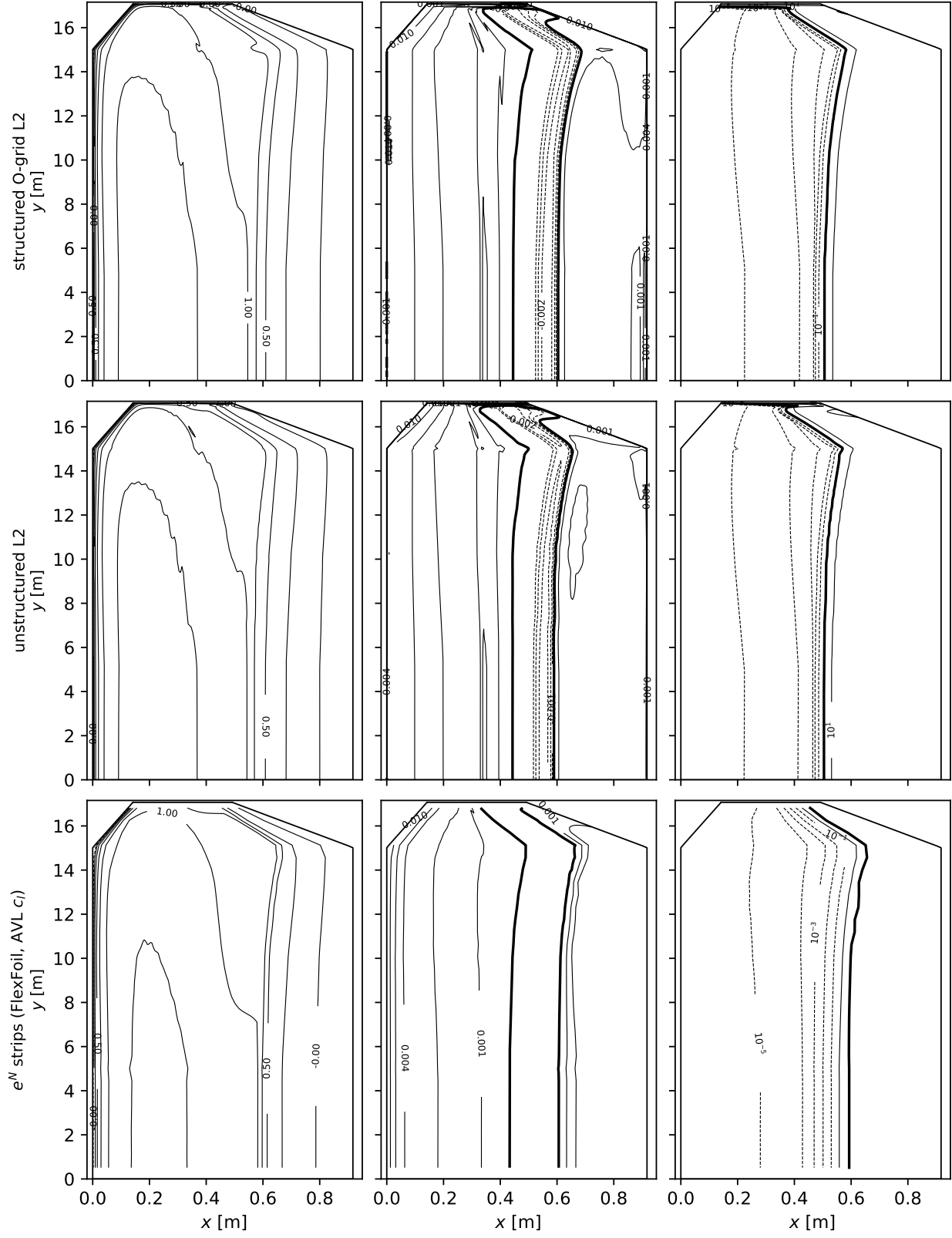
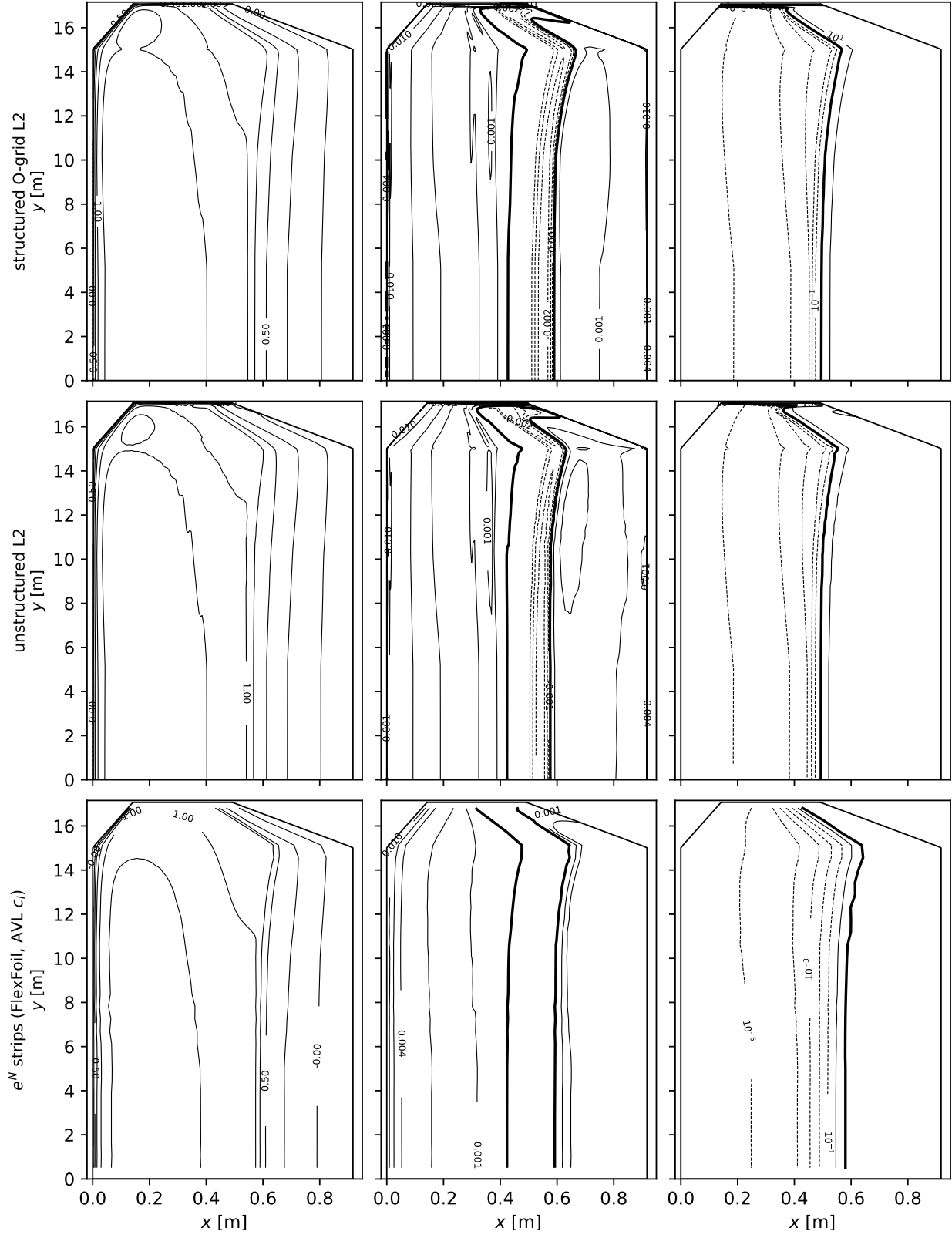
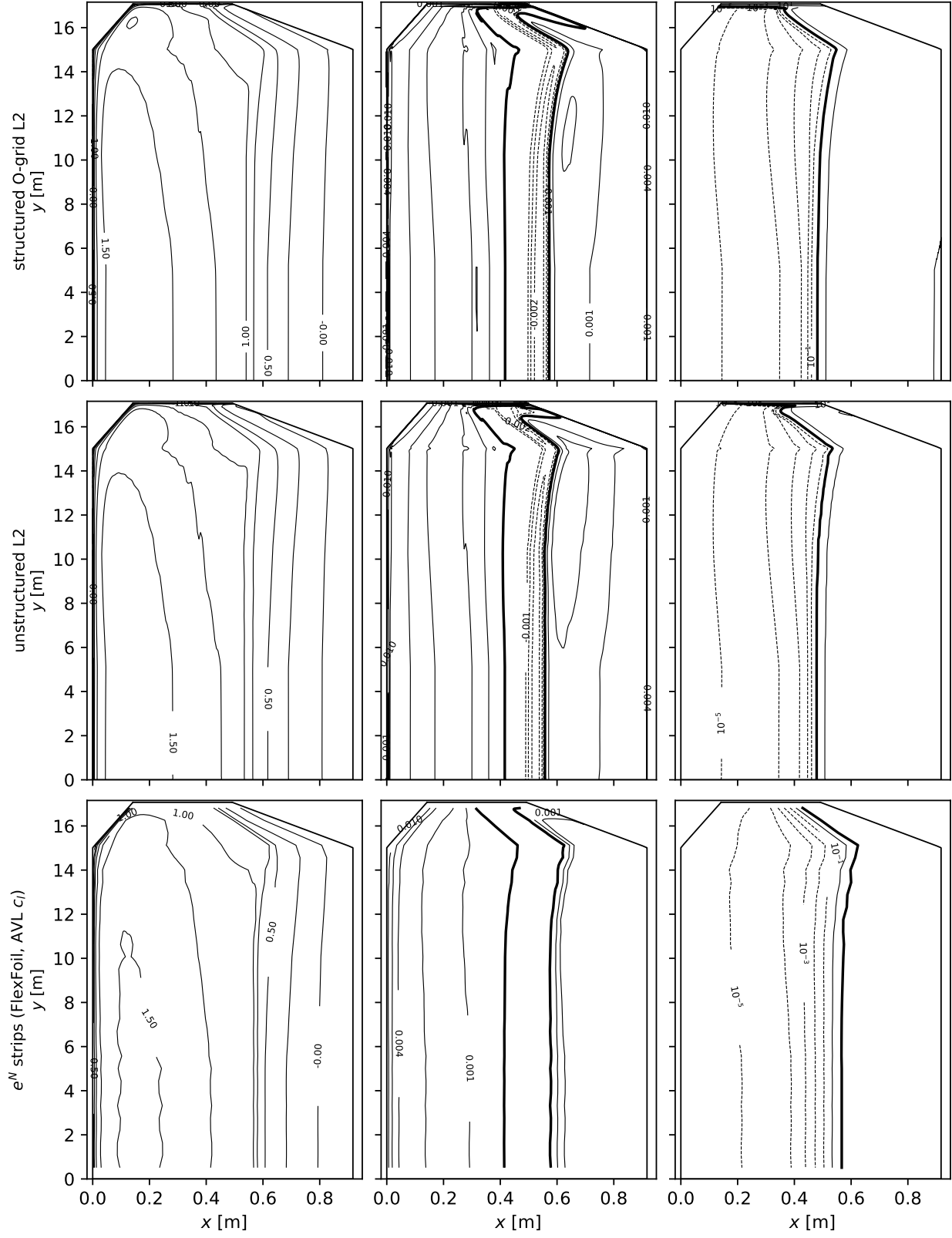


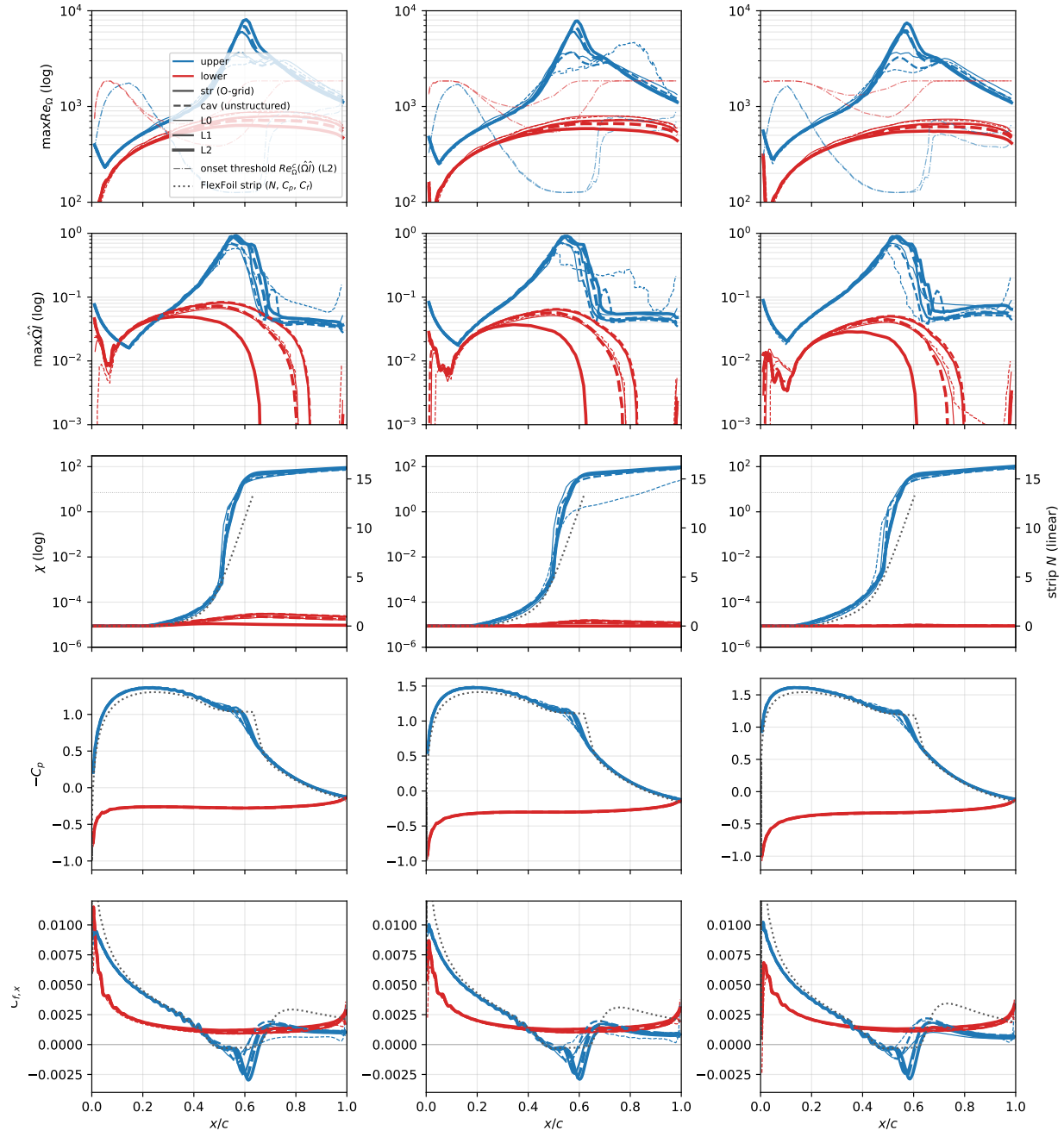
Figure 24: Daedalus surface maps, $\alpha = 4^\circ$, $Re_{\text{root}} = 5 \times 10^5$.

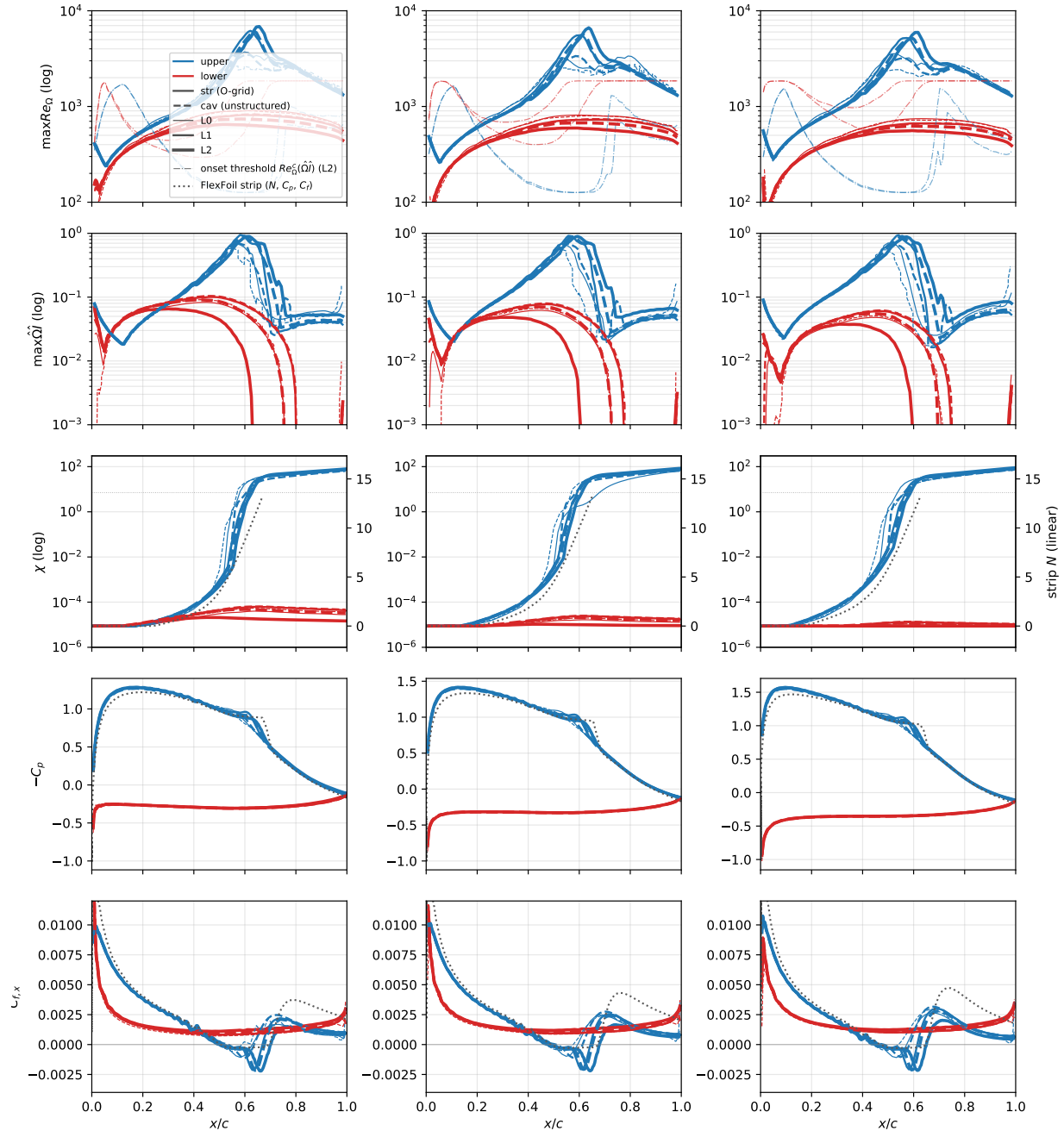
Table 10: Daedalus wing totals by grid level: SA-AI on both mesh families against the AVL+XFOIL reference trimmed to the RANS lift at each incidence.

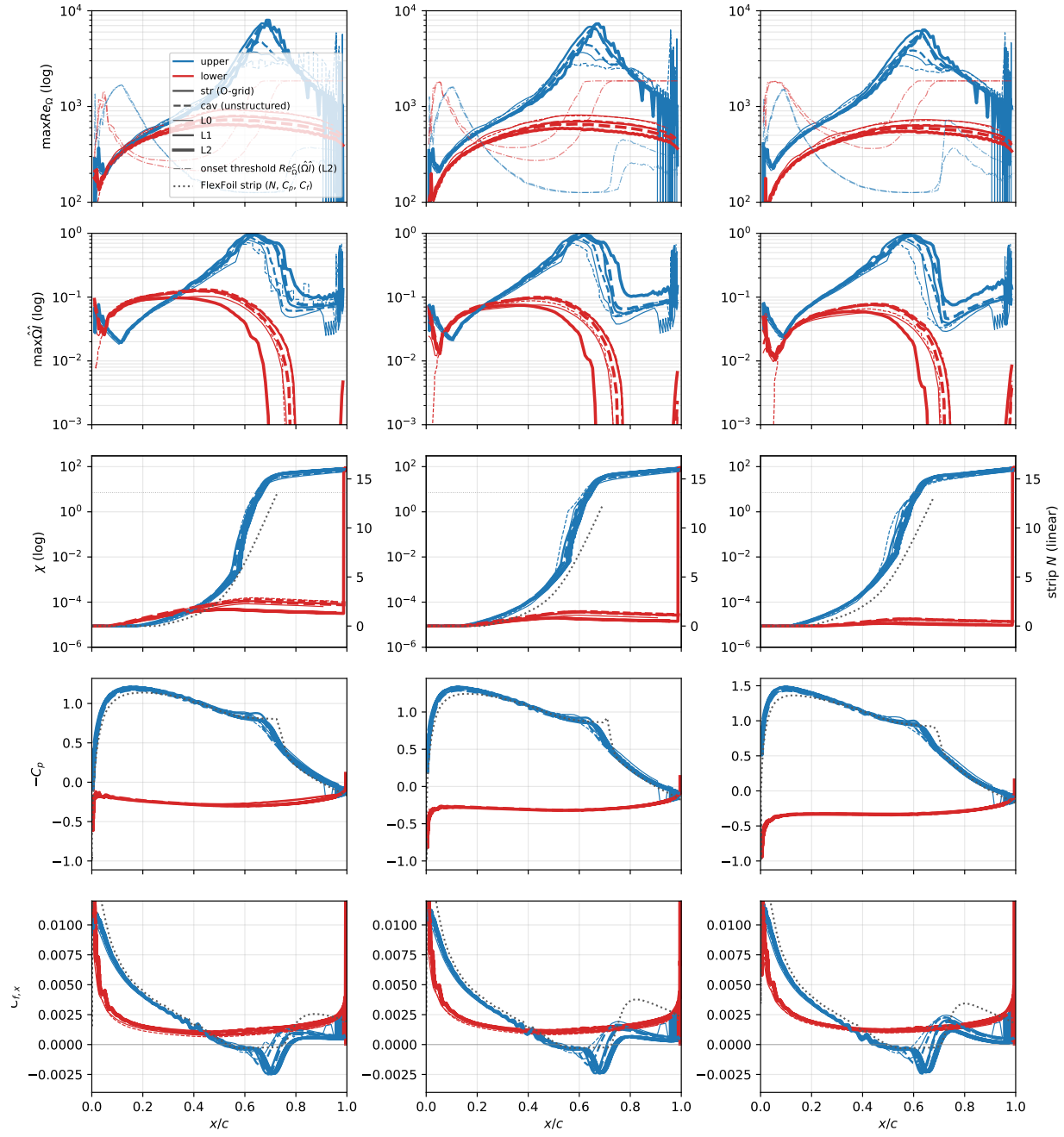
α	level	structured O-grid		unstructured	
		C_L	C_D	C_L	C_D
4°	L0	1.0244	0.01958	1.0156	0.02134
	L1	1.0279	0.01946	1.0232	0.01941
	L2	1.0284	0.01969	1.0219	0.01976
	AVL+XFOIL	1.0284	0.02125		
5°	L0	1.1254	0.02219	1.1145	0.02427
	L1	1.1307	0.02197	1.1248	0.02201
	L2	1.1290	0.02212	1.1236	0.02223
	AVL+XFOIL	1.1290	0.02315		
6°	L0	1.2262	0.02505	1.2171	0.02713
	L1	1.2321	0.02465	1.2256	0.02481
	L2	1.2282	0.02474	1.2238	0.02491
	AVL+XFOIL	1.2282	0.02530		

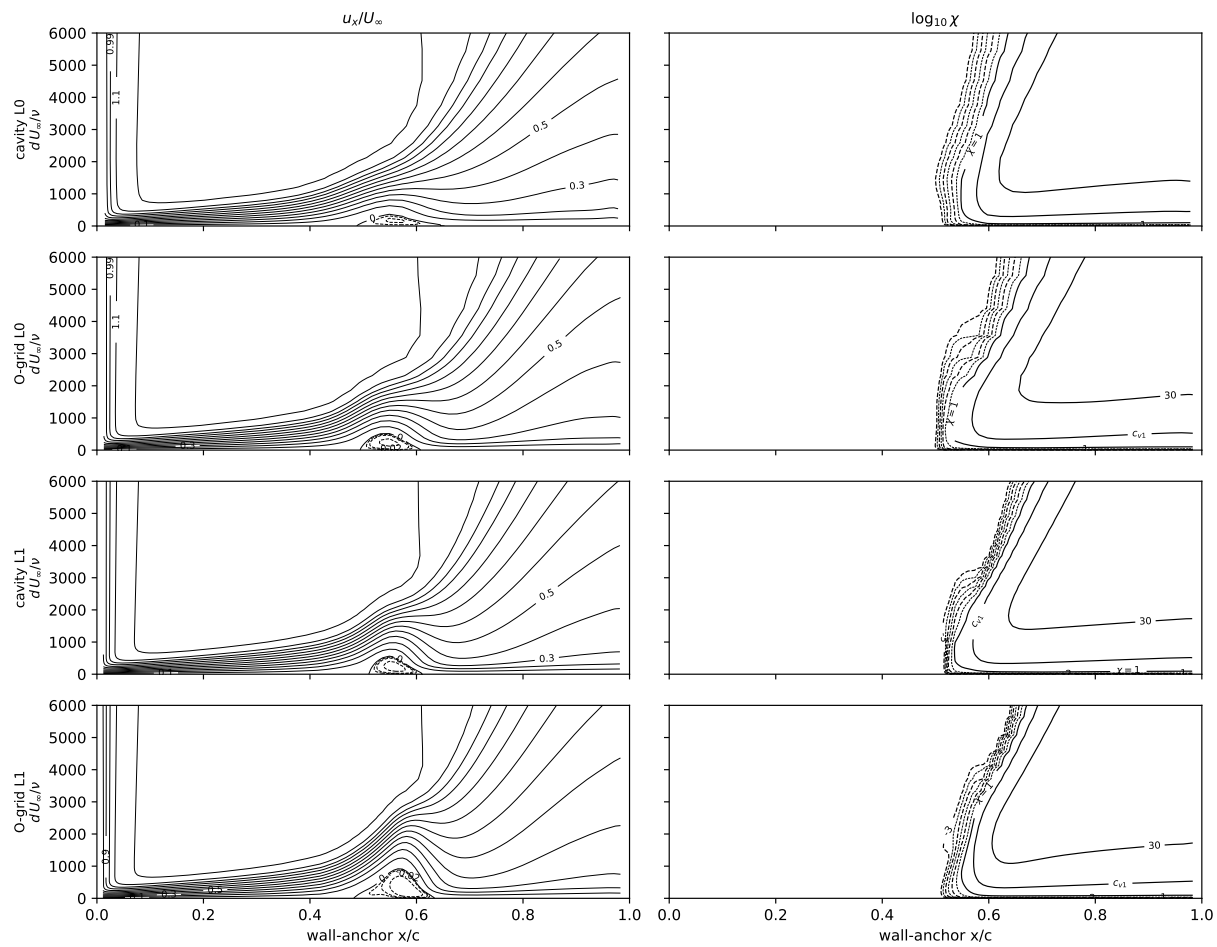


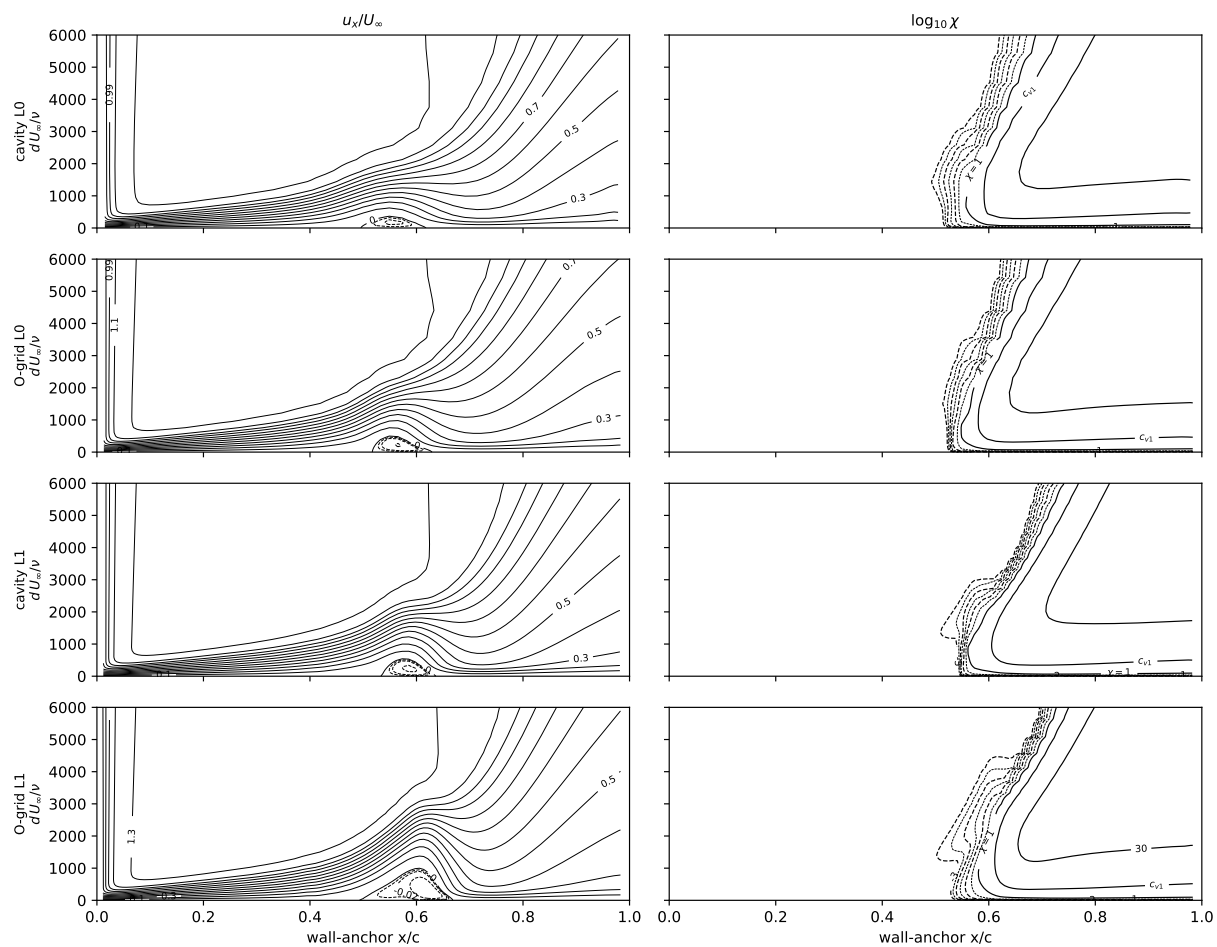


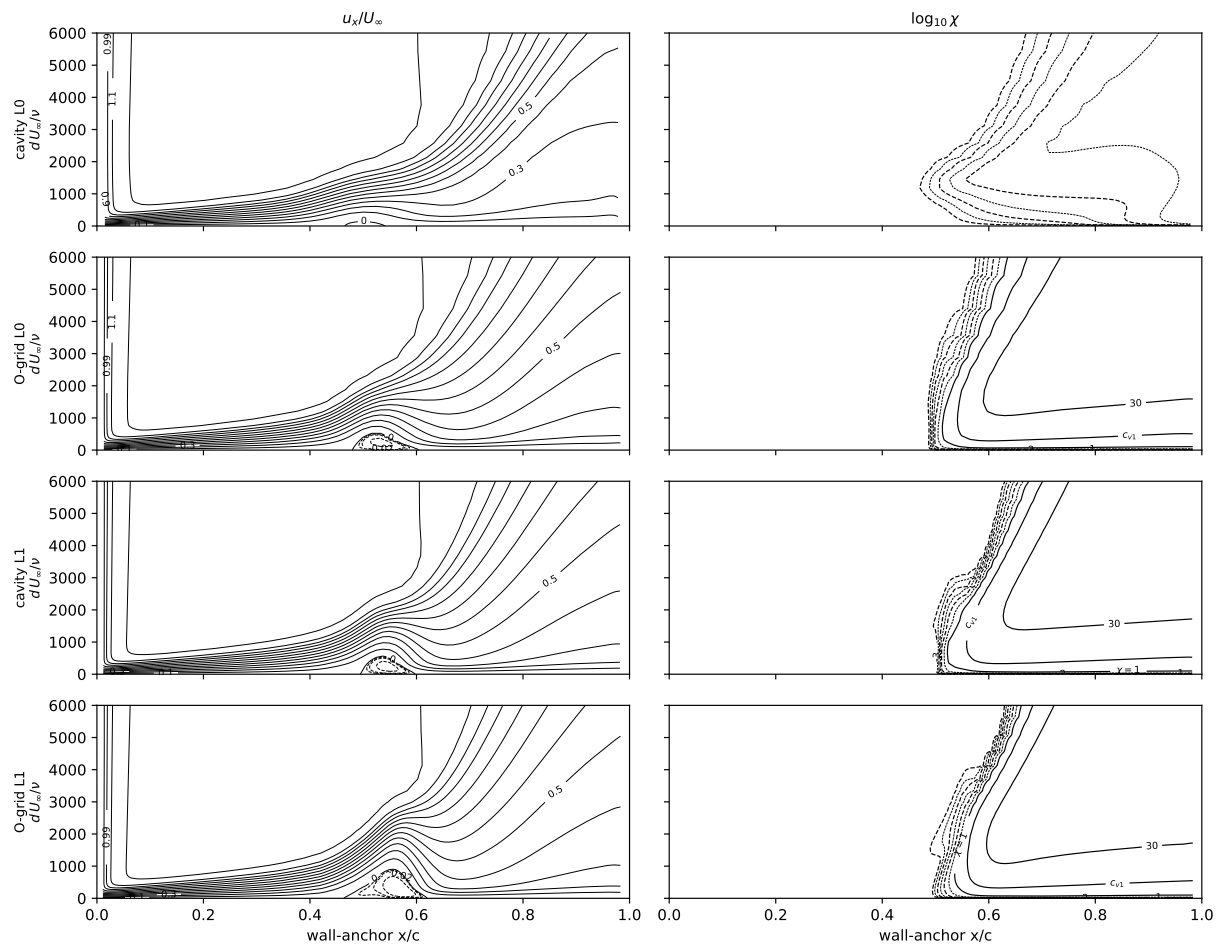


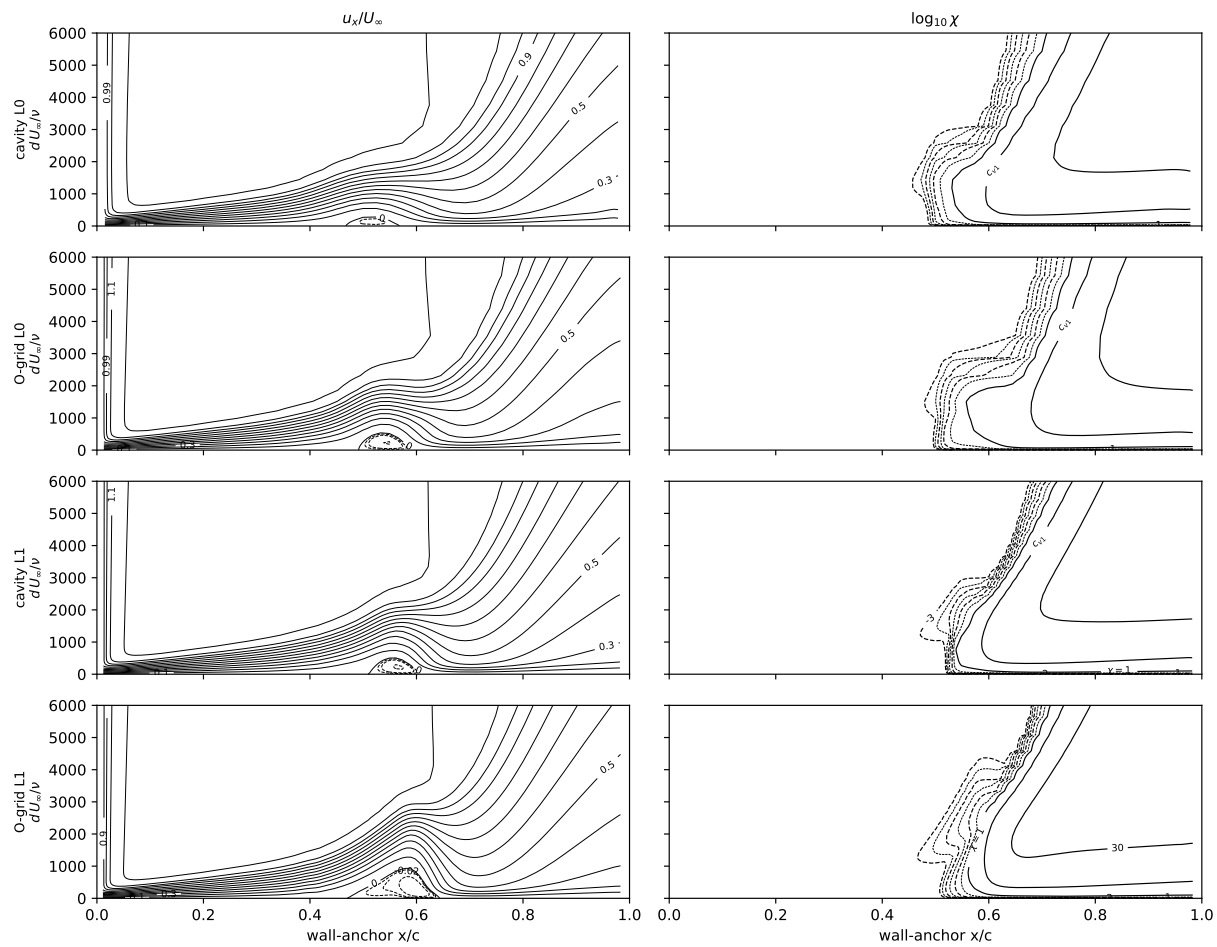


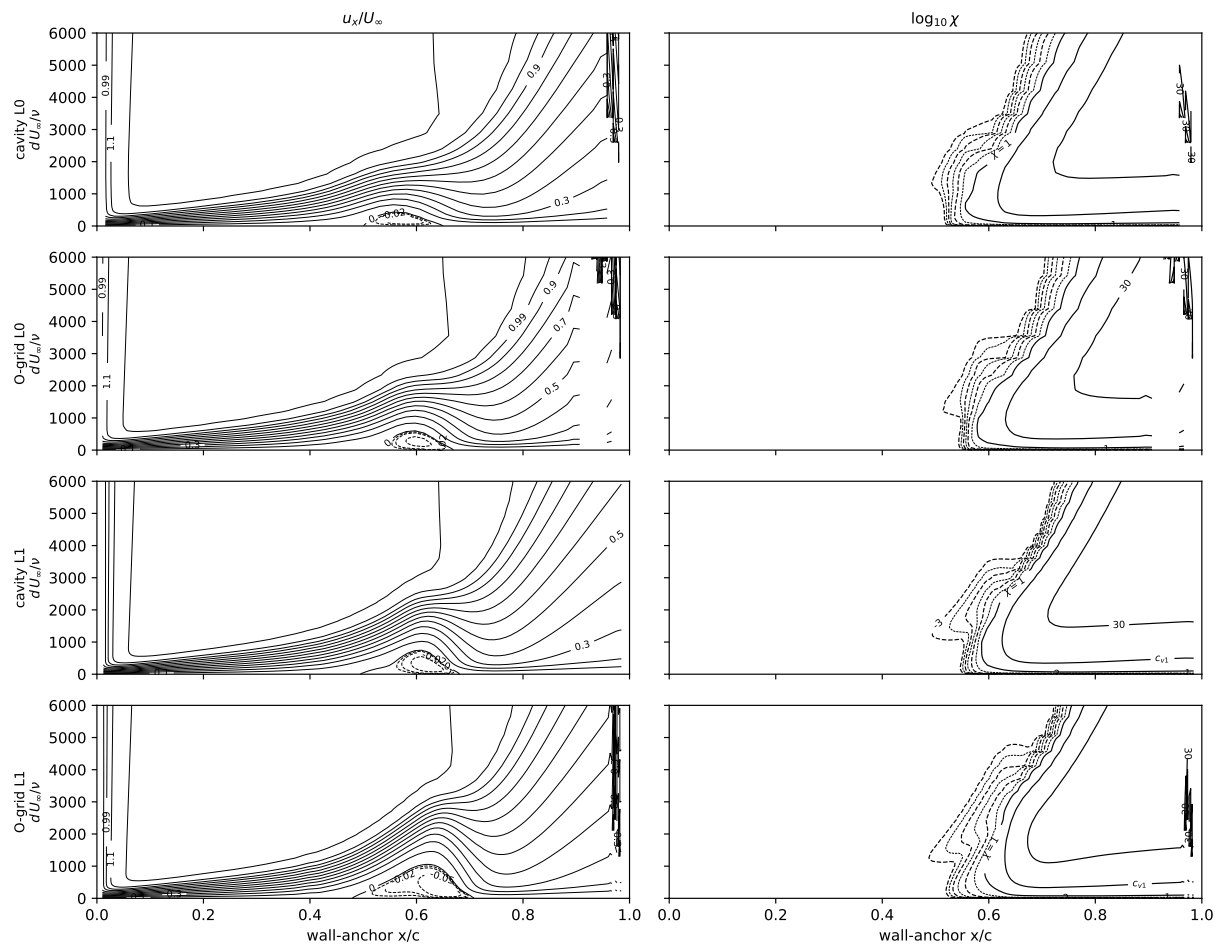


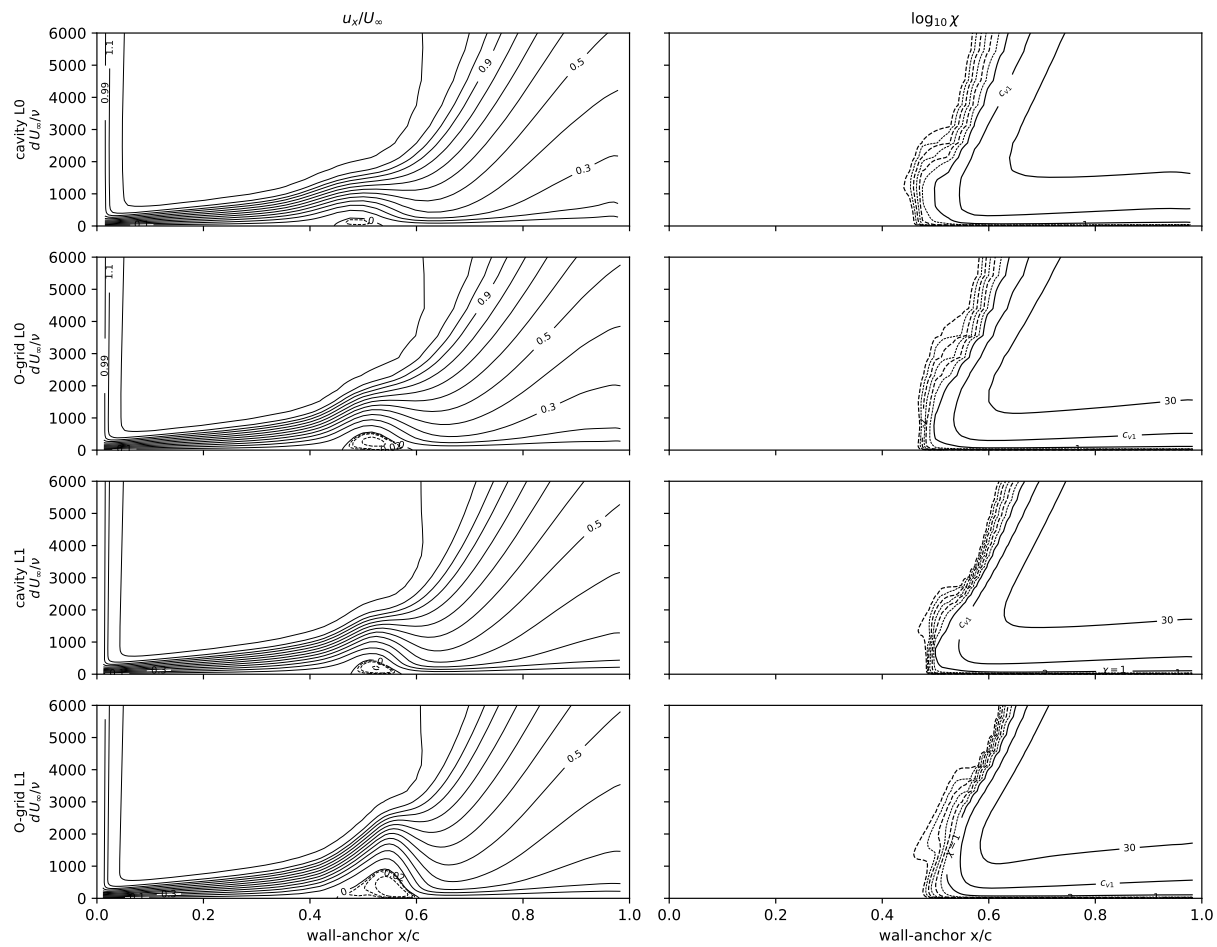


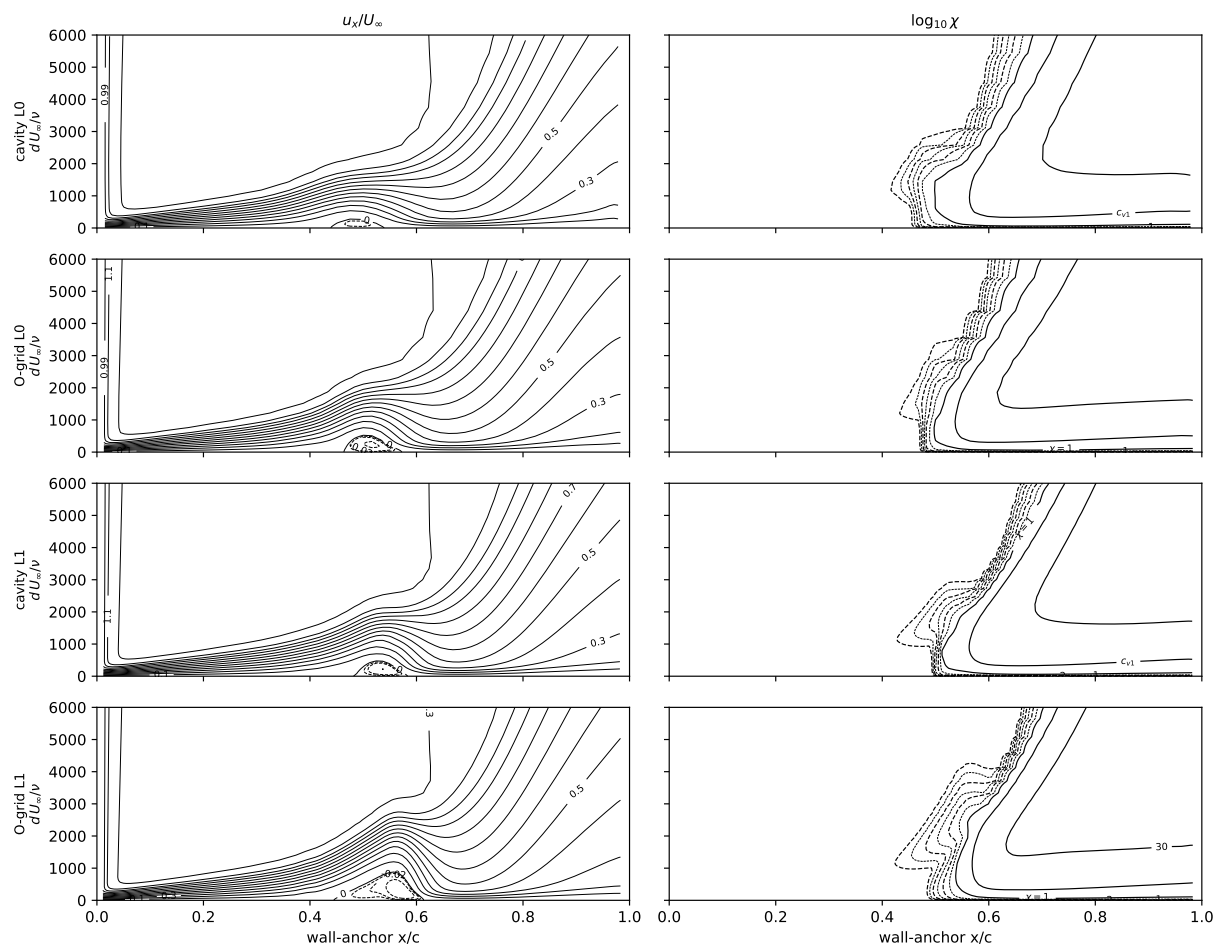


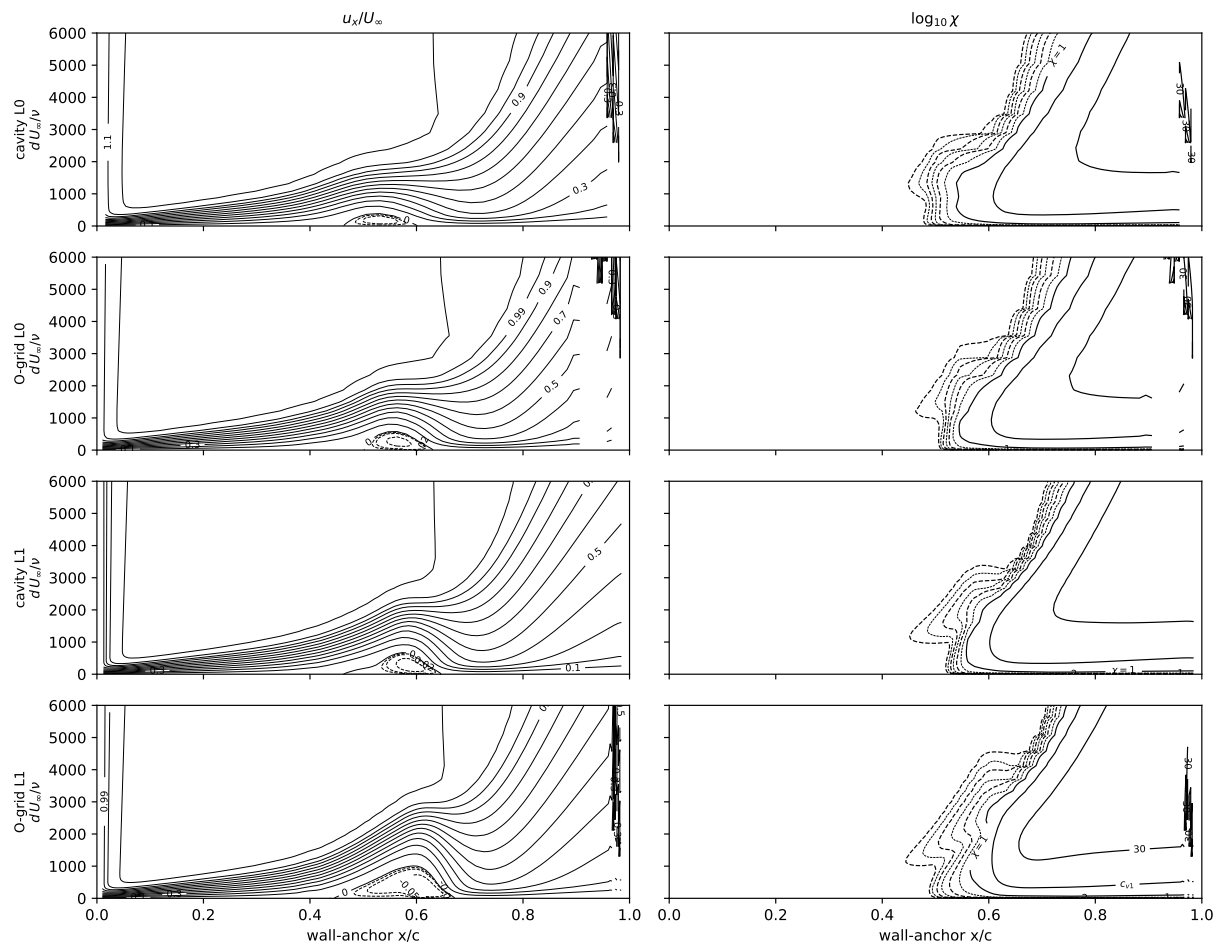












2.7 6:1 prolate spheroid, $\alpha = 0\text{--}10^\circ$

Analytic spheroid ($x^2/A^2 + r^2/B^2 = 1$, $A = L/2$, $B = L/12$, half-model) on the L1 structured O-grid (2.2M nodes, first wall spacing $1.5 \times 10^{-6}L$), campaign constants and kernel unchanged, at the DFVLR/ONERA measured conditions. Mach is the tunnel's: the Göttingen 3×3 m tunnel is atmospheric, so $M = 1.994 \times 10^{-8} Re_L$ (calibrated on its two published (Re_L, U_∞) pairs, 7.70×10^6 at 55.0 m/s and 6.54×10^6 at 45.0 m/s, reproduced to 2%), i.e. $M = 0.030$ at $Re_L = 1.52 \times 10^6$ to 0.144 at 7.2×10^6 ; the pressurised ONERA F1 tunnel does not have M fixed by Re_L , so its condition is run at its Göttingen twin's Mach. 45 000 cold-start pseudo-steps, seed applied in the freestream and the far-field boundary with a cold-start freestream initial condition. Each condition at the facility-calibrated seed (limiting $N_{TS} = 8.0$ Göttingen, 7.0 ONERA F1) and at the facility-measured hot-wire level (Göttingen 0.33% streamwise, ONERA F1 $< 0.1\%$).

Table 11: computed near-wall $\chi = 1$ front minus measured, azimuth by azimuth. $\alpha = 0$, $Re_L = 7.2 \times 10^6$: +0.077 and -0.097 at the calibrated and measured seed, uniform to ± 0.001 across seventeen azimuths. $\alpha = 2.5^\circ$: +0.016 leeward, +0.132 windward, against a measured azimuthal spread of 0.018. $\alpha = 5^\circ$: -0.088 to +0.246 at 6.49×10^6 . $\alpha = 10^\circ$: rms 0.027 at 1.52×10^6 and the measured seed; at 6.56×10^6 the windward residual is +0.375 to +0.549 and moves by ≤ 0.05 between seeds a factor 3.4 apart in Tu .

Table 11: 6:1 prolate spheroid at the tunnel conditions: computed near-wall $\chi=1$ front minus the measured front, at every azimuth carrying a measurement, on the L1 O-grid at the measured Mach number. Each condition is run at the facility-calibrated seed (the tunnel-calibrated limiting N_{TS}) and at the facility-measured hot-wire level. Δ lee / Δ wind are the residuals at the most leeward and most windward measured azimuth ($\phi=0$ windward). The n column is the number of measured azimuths at which the computed front was found, over the number measured: where they differ the near-wall χ never reaches 1 within $x/L \leq 0.97$, i.e. the computed layer is still laminar at the tail and the residual there is a bound, not a value.

α	Re_L [10^6]	tunnel	seed	Tu [%]	χ_∞	n	Δx_{tr}		Δ lee	Δ wind
							mean	rms		
0°	7.2	Göttingen	measured	0.330	0.036	17/17	-0.097	0.097	-0.097	-0.098
0°	7.2	Göttingen	calibrated	0.106	0.00238	17/17	+0.077	0.077	+0.078	+0.076
2.5°	7.2	Göttingen	calibrated	0.106	0.00238	3/3	+0.082	0.096	+0.016	+0.132
5°	1.52	Göttingen	calibrated	0.106	0.00238	4/8	-0.006	0.028	+0.026	-0.040
5°	6.49	Göttingen	calibrated	0.106	0.00238	4/4	+0.107	0.201	-0.088	+0.246
10°	1.52	Göttingen	measured	0.330	0.036	8/8	-0.013	0.027	+0.002	-0.041
10°	1.52	Göttingen	calibrated	0.106	0.00238	8/8	+0.019	0.082	+0.160	-0.053
10°	6.56	Göttingen	measured	0.330	0.036	5/5	+0.365	0.440	-0.105	+0.375
10°	6.56	Göttingen	calibrated	0.106	0.00238	5/5	+0.421	0.463	+0.068	+0.387
10°	6.56	ONERA F1	calibrated	0.161	0.00647	5/5	+0.399	0.475	+0.002	+0.549
10°	6.56	ONERA F1	measured	0.100	0.00205	5/5	+0.420	0.486	+0.051	+0.549